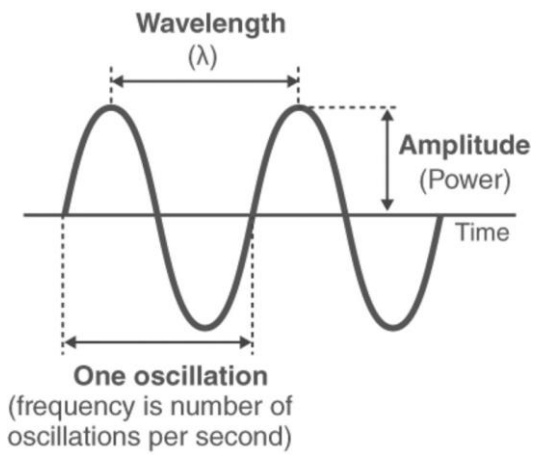
MODULE I - WAVE OPTICS

1. **Define amplitude (2marks)**

Amplitude – Wave is an energy transport phenomenon. Amplitude is the height of the wave, usually measured in meters. It is directly related to the amount of energy carried by a wave.

2. **Define wavelength (2marks)**

Wavelength – The distance between identical points in the adjacent cycles of crests of a wave is called a wavelength. It is also measured in meters.



3. **Define period and frequency (2marks)**

Period – The period of a wave is the time for a particle on a medium to make one complete vibrational cycle. As the period is time, hence is measured in units of time such as seconds or minutes.

Frequency – Frequency of a wave is the number of waves passing a point in a certain time. The unit of frequency is hertz (Hz) which is equal to one wave per second.

The period is the reciprocal of the frequency and vice versa.

$$\textit{Period} = \frac{1}{\textit{Frequency}} \quad \text{(or)}$$

$$\textit{Frequency} = \frac{1}{\textit{Period}}$$

4. **Write short notes about transverse and longitudinal wave. (2marks)**

Transverse Waves

Waves in which the medium moves at right angles to the direction of the wave.

Examples of transverse waves:

Water waves (ripples of gravity waves, not sound through water)
Light waves
S-wave earthquake waves
Stringed instruments
Torsion wave

The high point of a transverse wave is a crest. The low part is a trough.

Longitudinal Wave:

A longitudinal wave has the movement of the particles in the medium in the same dimension as the direction of movement of the wave.

Examples of longitudinal waves:

Sound waves
P-type earthquake waves
Compression wave

Parts of longitudinal waves:

Compression: where the particles are close together.

Rarefaction: where the particles are spread apart.

Write short notes on mechanical and matter waves. (2marks)

Mechanical waves:

A wave which needs a medium in order to propagate itself. Sound waves, waves in a Slinky, and water waves are all examples of this.

Matter Waves:

Any moving object can be described as a wave When a stone is dropped into a pond, the water is disturbed from its equilibrium positions as the wave passes; it returns to its equilibrium position after the wave has passed.

Write about electromagnetic wave. (2marks)

Electromagnetic Waves: These waves are the disturbance that does not need any object medium for propagation and can easily travel through the vacuum. They are produced due to various magnetic and electric fields. The periodic changes that take place in magnetic electric fields and therefore known as Electromagnetic Wave

Example: Radio signals, light rays, x-rays, and cosmic rays.

Write short notes on wavefront and wavelets.

Define phase speed.

Phase Speed (2marks)

Phase speed is the speed with which wave front moves and it is equal to wave speed.

The shape of wavefront due to a

(i) point source is spherical

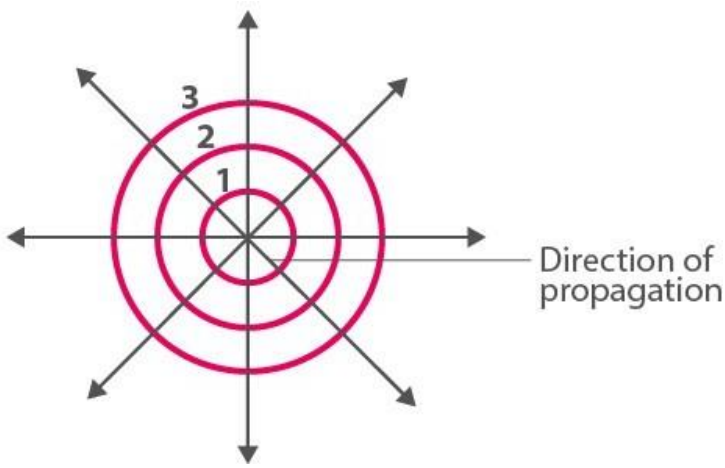
(ii) line source is cylindrical

(iii) source at infinity is a plane.

Write short notes on spherical wavefront (2marks)

Spherical Wave Front

When light is emerging from a point source, the wavefronts are spherical in shape.



In spherical wavefront,

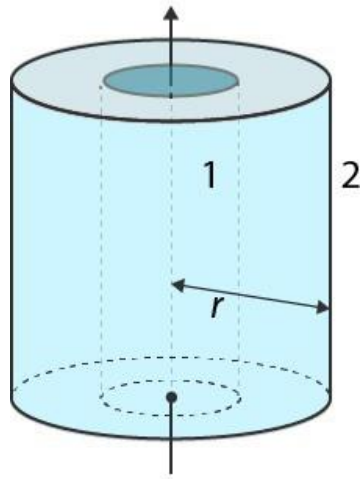
Amplitude of light waves, $A \propto \frac{1}{r}$

and, Intensity of light waves, $I \propto \frac{1}{r^2}$

Write short notes on cylindrical wavefront (2marks)

Cylindrical Wave Front

When the source of light is linear, the wavefronts are cylindrical in shape. All the points are equidistant from the source.



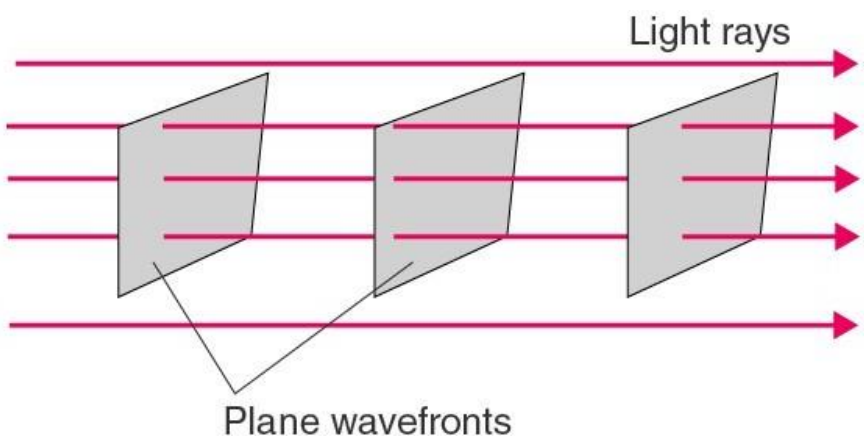
In this case, Amplitude of light waves, $A \propto \frac{1}{\sqrt{r^2}}$

Intensity of light waves $I \propto \frac{1}{r}$

Write short notes on plane wavefront (2marks)

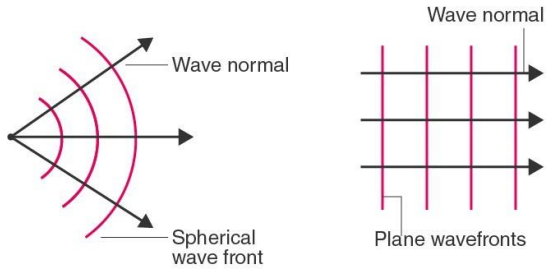
Plane Wave Front

when the light is coming from a very far-off source, the wavefronts are planar. For a plane wavefront, Amplitude remains constant therefore Intensity remains constant.



What is wave normal? (2marks)

A perpendicular drawn to the surface of a wavefront at any point, in the direction of propagation of light, is called “wave normal”. The direction in which light travels is called a ‘ray’ of light. Thus, a wave normal is the same as a ray of light.



What are the shapes of wavefront falls on different surfaces? (2marks)

Shape of wavefront in case light emerging out of a convex lens when illuminating point source is placed at its focus is

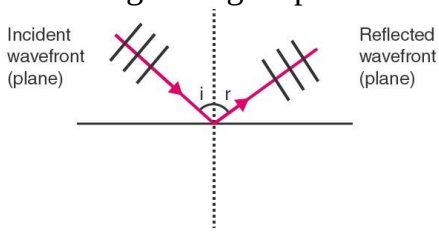
planar
spherical converging
spherical diverging
Cylindrical

Write briefly about shapes of wavefronts due to reflection of light. (2marks)

Wavefronts for Reflection

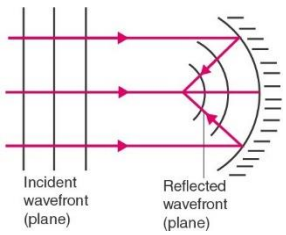
- **If light falls on a plane mirror:**

If the plane wavefronts are being reflected on the plane mirror, the shape of the wavefront of the reflected light is again planar.

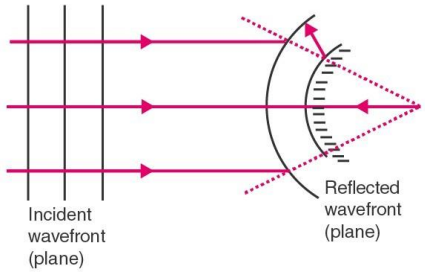


- **If light falls on a concave mirror; convex mirror:**

If a plane wavefront falls on a concave mirror, the shape of the reflected light is spherical.



If a plane wavefront falls on a convex mirror, the shape of the reflected light is spherical.

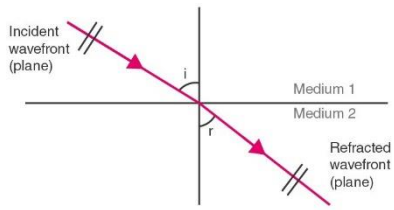


Write briefly about shapes of wavefronts due to refraction of light. (2marks)

Wavefronts for Refraction

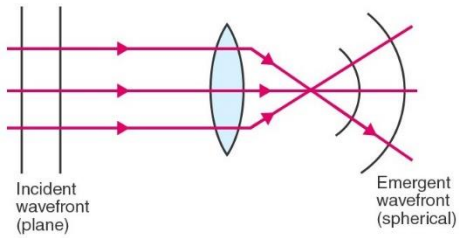
- **If light falls on plane surfaces:**

If a plane wavefront falls on a plane surface the refracted ray will also have a plane wavefront.



- **If light falls on curved surfaces:**

If a plane wavefront falls on a converging (or) diverging lens, the emergent light will have a spherical wavefront.



Write Huygen's principle. (2marks)

Huygens Principle

"Every point on a wavefront is in itself the source of spherical wavelets which spread out in the forward direction at the speed of light. The sum of these spherical wavelets forms the wavefront"

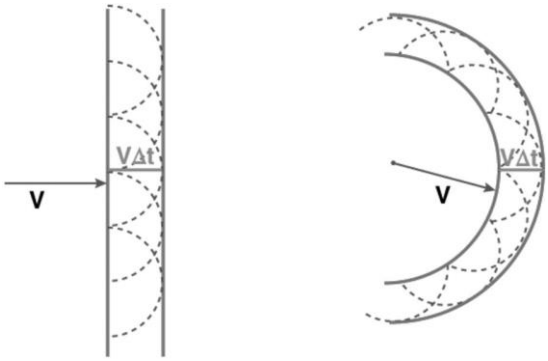
Define superposition. (2marks)

"The principle of superposition states that every charge in space creates an electric field at point independent of the presence of other charges in that medium. The resultant electric field is a vector sum of the electric field due to individual charges."

Write briefly about primary and secondary sources. (2marks)

Primary and Secondary Sources

Huygens stated that light is a wave that propagates through space much like ripples in water or sound in air. Hence, light spread out like a wave along with all directions from a source. The locus of points that travelled some distance during a fixed time interval is called a wavefront. Thus, from a point source of light, the locus of points that light has travelled during a fixed time period is a sphere (a circle if you consider a 2D source).



After the primary wavefront is created, a secondary wavefront is created from every primary wavefront. Secondly, every point on the wavefront acts as a secondary source of light that emits more wavefronts. The net effect is that the effective wavefront generated is tangential to all the secondary wavefronts generated by the secondary sources as shown in the figure. In this way, a light wave propagates through space by generating secondary sources and wavefronts. The direction of the traverse is always perpendicular to the wavefronts.

Write the advantages and disadvantages of Huygen's principle. (2marks)

Advantages and Disadvantages of Huygens Principle

Advantages:

- Huygens concept proved the reflection and refraction of light.
- The concepts like diffraction of light, as well as interference of light, were proved by Huygens.

Disadvantage:

- Concepts like emission of light, absorption of light and polarization of light were not explained by Huygens principle.
- Huygens principle failed to explain the photoelectric effect.
- A serious drawback is that the theory proposes an all-pervading medium required to propagate light called luminiferous ether. This was proved to be false in the 20th century.

Write about the applications of Michelson interferometer. (2marks)

APPLICATIONS OF MICHELSON INTERFEROMETER

Determination of wavelength of monochromatic light

Determination of difference in wavelength between two neighboring spectral lines

Write differences between Fresnel and Fraunhofer diffraction. (2 or 4 marks)

Fresnel's Diffraction	Fraunhoffer Diffraction
(1) The type of diffraction in which the distance of either source or screen or both from the obstacle is finite, such diffraction is known as Fresnel's diffraction. (2) No lenses are used to make the rays converge or parallel. (3) The incident wave front is either cylindrical or spherical. Ex: The diffraction at the straight edge.	(1) The type of diffraction in which the distance of either source or screen or both from the obstacle is infinite, such diffraction is known as Fraunhoffer diffraction. (2) Lenses are used to make the rays converge or parallel. (3) The incident wave front is plane. Ex:. The diffraction at the narrow.

Define interference (2marks)

The net effect of the combination of two or more wave trains moving on intersecting or coincident paths. The effect is that of the addition of the amplitudes of the individual waves at each point affected by more than one wave. If two of the components are of the same frequency and phase (i.e., they vibrate at the same rate and are maximum at the same time), the wave amplitudes are reinforced, producing constructive interference. But if the two waves are out of phase by 1/2 period (i.e., one is minimum when the other is maximum), the result is destructive interference.

Define diffraction (2marks)

Diffraction is the spreading out of waves as they pass through an aperture or around objects. It occurs when the size of the aperture or obstacle is of the same order of magnitude as the wavelength of the incident wave. For very small aperture sizes, the vast majority of the wave is blocked. For large apertures the wave passes by or through the obstacle without any significant diffraction.

Define polarization (2marks)

Polarisation is a phenomenon generated by the oscillation of light waves restricted to a single plane. In an ordinary beam of light with millions of waves, the light vectors of component waves will remain in all possible directions on a plane drawn at right angles to the direction of propagation.

Define refraction (2marks)

Refraction is the phenomenon of bending of light as it goes obliquely from one medium to another. It is caused by the change in speed of light as it goes from one medium to another.

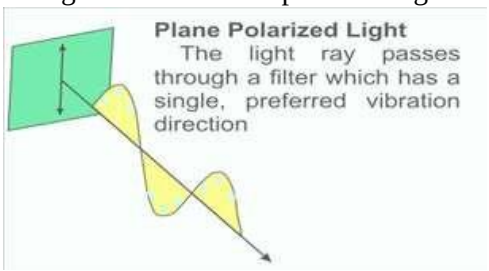
Write differences between ordinary light and polarized light. (2marks)

Polarised light	Ordinary light
1. The vibrations are confined in a particular direction.	1. The vibrations of light particle are not confined in a particular direction.
2. The probability of occurrence of vibration along the axis of crystal is not same in all position of crystal	2. The probability of occurrence of vibration along the axis of the crystal is not symmetries for all position of the crystal.
3. The intensity of light plate is not same in all position of the crystal plate.	3. The intensity of light plate is same in all position of the plate.

What is linearly polarized light? (2marks)

Linearly Polarised /Plane polarised:

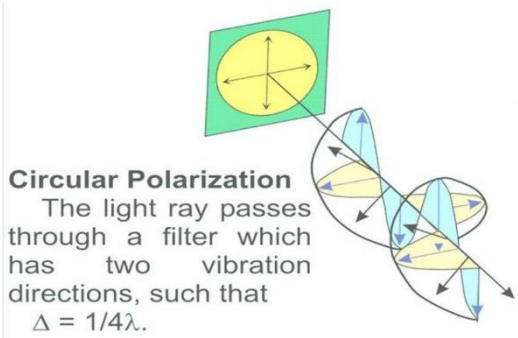
When the vibrations are confined to a single linear direction at right angles to the direction of propagation, such light is called Plane polarised light.



What is circularly polarized light? (2marks)

Circularly polarised light:

When the two plane polarised wave superpose under certain condition such that the resultant light vector rotate with a constant magnitude in a plane perpendicular to the direction of propagation and tip of light vector traces a circle around a fixed point such light is called circularly polarised light.

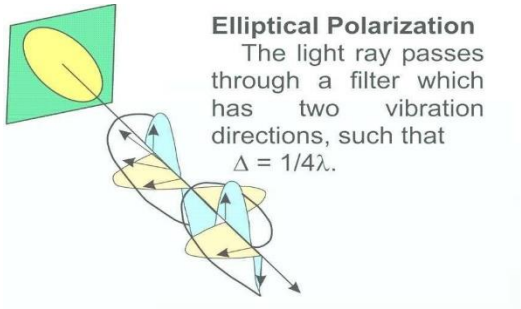


What is elliptically polarized light? (2marks)

Elliptically polarised light:

When two plane polarised light are superpose in such a way that the magnitude of the

resultant light vector varies periodically during its rotation then the tip of the vector traces an ellipse such light is called elliptically polarised light.



Define Brewster's law. (2marks)

Brewster's law is a statement that says that when unpolarized light falls on an interface, the reflected light is completely polarized if the angle of incidence is a specific angle called the Brewster's angle. In this case the angle made by the refracted ray and the reflected ray is 90 degree.

What are the applications of Brewster's law? (2marks)

Application of Brewster's Law

One general example for the application of Brewster's law is polarized sunglasses. These glasses use the principle of Brewster's angle. The polarized glasses reduce glare that is reflecting directly from the sun and also from horizontal surface like road and water. Photographers also use the same law to reduce the reflection from reflective surfaces by using polarizing filter for the lens.

Write about the Double refraction of light. (2marks)

Double Refraction

Bartholinus discovered that when a ray of unpolarized light is incident on a calcite crystal, two refracted rays are produced. This phenomenon is called **double refraction**. Hence, two images of a single object are formed. This phenomenon is exhibited by several other crystals like quartz, mica etc.

Write about Birefringence and its occurrence. (2marks)

Occurrence of Birefringence

Calcite crystals show some of the strongest birefringence effects. Light passing through a calcite crystal is split into two rays by refraction. Refraction of the light waves occurs in such a way that two refracted rays are mutually perpendicular and, also, perpendicular to the direction of wave propagation. The two rays are termed as slow and fast rays. The wave having greater refractive index is called the slow ray.

Write briefly about birefringence and its measurement. (2marks)

Birefringence is classified as intrinsic or stress-induced birefringence.

- Intrinsic Birefringence – This type of birefringence is caused by the anisotropy present in the crystals. The atomic arrangement of the crystal itself is the source of birefringence. Examples are calcite, tourmaline, etc
- Stress-Induced Birefringence – This type of birefringence is caused due to the stresses imposed on the material. Materials such as glass or plastics show strain birefringence

Birefringence can be quantified by measuring the changes in polarization of light waves. This method of measurement is called polarimetry. A special method, called dual polarization interferometry, is used to measure birefringence of lipid bilayers.

Give applications of birefringence. (2marks)

Applications of Birefringence

Birefringence finds use in the following applications:

- Polarizing prisms and retarder plates
- Liquid crystal displays
- Medical Diagnostics

Define optical axis in Nicol prism. (2marks)

Optic Axis

It is a direction inside the crystal such that the single ray of light does not split into O-ray and E-ray as their velocities are equal.

Double refraction or birefringence is not observed along optic axis.

What are principle plane and principle section in Nicol Prism? (2marks)

Principal plane and Principal section

A plane containing the optic axis and perpendicular to the opposite faces of the crystal called principal plane.

Section of a crystal along principal plane is called principal section of the crystal.

6, 8 AND 16 MARKS

Explain in detail about Huygen's Theory of Double Refraction - ---------- (6 marks)

Huygen's Theory of Double Refraction:

According to Huygen;s theory each point on wavefront act as a fresh source of disturbance and sends secondary wavelets. the envelope of all secondary wavelets at any instance represents a wavefront which is the locus of particles vibrating in phase. Huygen's theory cannot explain double refraction in uniaxial crystals. He extends the theory in secondary wavelets. According to this theory,

1. When any wavefront strikes a doubly refracted crystal, every point of crystal become a source of two wavefronts.

a. Ordinary wavefront corresponds to ordinary waves. Since ordinary waves have same velocity in all direction, the secondary wavefront is spherical.

b. Extraordinary wavefront corresponds to extra ordinary waves. Since extra ordinary waves have different velocities in different directions, the extra ordinary wavefront is ellipsoid of revolution, with optic axis as the axis of revolution.

2. The sphere and ellipsoid touch each other at points which lie on optic axis of the crystal, because the velocity of ordinary and extra ordinary ray is same along optic axis.

3. In certain crystals lie calcite and tourmaline called negative crystals, the ellipsoid lies outside the sphere as shown in figure below. This shows that in negative crystals, the extra ordinary wavefront travels faster than ordinary wavefront except along optic axis.

4. In certain crystals lie ice and quartz called the positive crystals, the sphere lies outside the ellipsoid as shown in figure below. This shows that velocity of ordinary wavefront is greater than extraordinary wavefront in positive crystals except along optic axis.

Explain in detail about Michelson interferometer experiment.

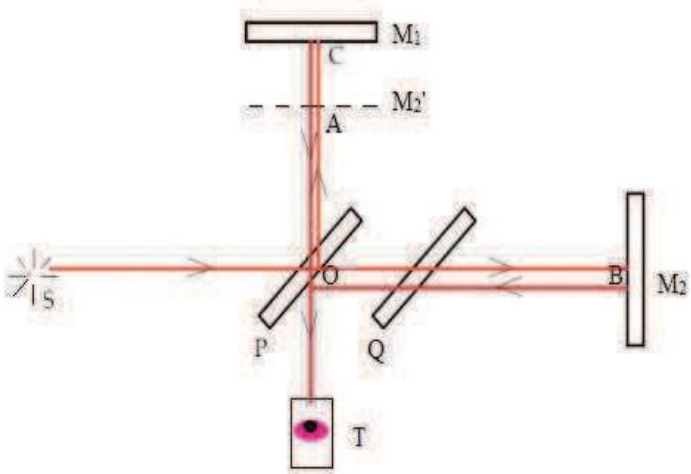
- ---------(16 marks)

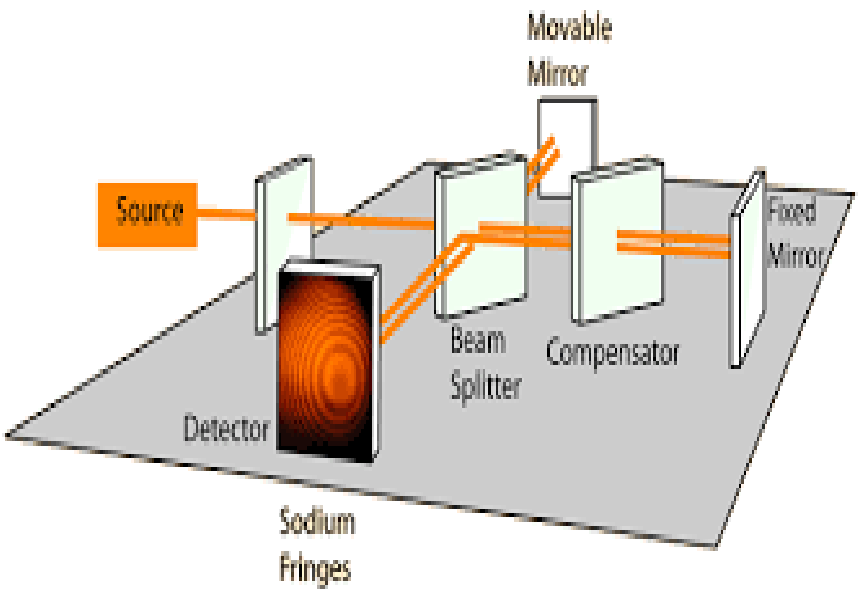
MICHELSON INTERFEROMETER

- Michelson interferometer is a device used for the formation and study of interference fringes by a monochromatic light.
- In this apparatus, a beam of light coming from an extended source of light is divided into two parts, one is reflected part and another is refracted part after passing through a partially polished glass plate.
- These two beams are brought together after reflected from plane mirrors, and finally interference fringes are produced in the field of view.

Construction

- The apparatus is shown in Figure below
- The main part of the apparatus is a half silvered glass plate P, on which a beam of monochromatic light is incident.
- The plate P inclined at an angle 45^0 with incident light as shown in figure





- The incident light then divided into two parts, one is reflected part and another is transmitted part.
- The transmitted light is then passes through another glass plate Q which is of equal thickness as of P, and parallel to plate P, this pate Q is called compensating plate.
- The transmitted and reflected parts of light are normally incident on two mirrors M_2 and M_1 respectively.
- The mirror M_1 and M_2 are perpendicular to each other as shown in figure.
- The mirror M_1 is fixed in a carriage and can be moved to and fro with help of a screw and micro scale.
- Therefore mirror M_1 is movable and the mirror M_2 is fixed.
- A telescope is also fixed as shown in figure.
- The light reflected from mirror M_1 and M_2 are superimposed and interference fringes are formed in the field of view.

Working

- S is a source of monochromatic light; the light coming from this source is rendered parallel by mean of a convex lens L, and after passing through Lens L the light falls on plate P.
- Since plate P is partially polished, some part of light reflected back from P and going toward direction AC and incident on mirror M_1.
- Similarly the light transmitted from plate P passing through compensating plate Q and then incident on mirror M_2.
- The compensating plate is used to compensate the optical path travelled by transmitted light.
- The beam of light reflected by P, crosses plate P two times, for transmitted light this optical path is compensated by using plate Q in which the transmitted light crosses Q two time.
- Thus by using compensating plate Q, the reflected and transmitted light travel equal optical path lengths.
- Now the reflected light is incident on mirror M_1 and reflected back towards the telescope T.
- Similarly the transmitted light incident normally on mirror M_2 and reflected back towards plate P, and at P some part of this light again reflected toward the telescope.
- Now in the direction of telescope we have two coherent beams of light reflected from mirror M_1 and M_2, and interference takes place and we observed interference pattern/beam in the field of view.

Formation of Fringes

- Since the fringes are form by the light reflected from mirror M_1 (movable) and M_2 (fixed) and we can consider a virtual image of M_2 called M_2' in the field of view as shown in figure 6.1.
- Further we can consider the interference fringes are now formed due to light reflected from the surface of air film formed between mirror M_1 and M_2'. Now it is clear that the shapes of fringes are depend upon the inclination of mirror M_1 and M_2

Explain in detail about diffraction due to single slit --------(16 marks)

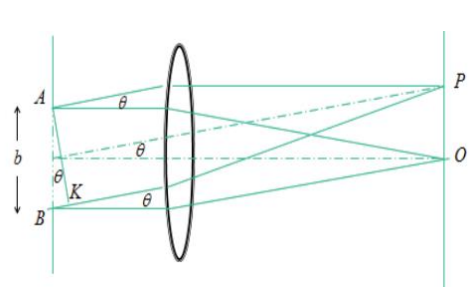
FRAUNHOFER DIFFRACTION

CLASSES OF DIFFRACTION

- Based on the distance between source, aperture and screen, and also on the shape of wavefront, diffraction pattern is classified into two classes
- Fresnel Diffraction-If the source of light and the screen are at finite distances from the diffracting aperture, then the wavefront falling on the aperture will not be plane (spherical or cylindrical).
- The diffraction obtained under this type of arrangement is called Fresnel Diffraction.
- This type of diffraction is also called near-field diffraction.
- No lenses are used to make the rays parallel or convergent.
- Fresnel Diffraction is obtained when light suffers diffraction at a straight edge, a thin wire, a narrow slit etc.
- Both the size and shape of the pattern depends on the distance between the diffracting aperture and the screen.
- Fraunhofer Diffraction-If both the source of light and the screen are effectively far enough from the aperture so that the wavefronts reaching the aperture and the screen can be considered plane.
- Then the source and the screen are said to be at infinite distances from the aperture.
- This kind of diffraction is called Fraunhofer Diffraction. This is also called far-field diffraction.
- Fraunhofer Diffraction is encountered in the case of gratings that contain number of slits.
- When the screen is moved, the size of the diffraction pattern changes uniformly while the shape of the pattern does not change.

FRAUNHOFER DIFFRACTION DUE TO A SINGLE SLIT

- Let AB is a slit of width b, the diffracted beam through the slit is tilted at an angle θ with respect to straight direction.



- Path difference between two rays diffracted from two extreme points of slit.

$$\Rightarrow BK = AB\sin\theta = b\sin\theta$$

$$\text{Phase difference} = \frac{2\pi}{\lambda} \times \text{path difference}$$

$$= \frac{2\pi}{\lambda}(b\sin\theta)$$

- Let the width AB of the slit be divide into n equal parts. The amplitude of vibration at P due to the waves from each part will be same, say a.
- The phase difference between the waves from any two consecutive parts is given by

$$\textbf{Let d} = \frac{Phase\ difference}{n}$$

$$\mathbf{d} = \frac{1}{n}\left(\frac{2\pi}{\lambda}\, b\sin\theta\right)$$

Then the resultant amplitude at P is given by

$$R = \frac{a\sin\left(\frac{nd}{2}\right)}{\sin\left(\frac{d}{2}\right)}$$

$$= \frac{a \sin\left(\frac{\pi\, b \sin\theta}{\lambda}\right)}{\sin\left(\frac{\pi\, b \sin\theta}{n\lambda}\right)}$$

Let us consider, $\frac{\pi\, b \sin\theta}{\lambda} = \alpha$

$$R = \frac{a \sin\alpha}{\sin\left(\frac{\alpha}{n}\right)}$$

Since $\left(\frac{\alpha}{n}\right)$ **is small value sin** $\left(\frac{\alpha}{n}\right)$ **is a smaller value, so that we can write directly** $\left(\frac{\alpha}{n}\right)$

Therefore, $R = \frac{a \sin\alpha}{\left(\frac{\alpha}{n}\right)} = \frac{na \sin\alpha}{\alpha}$

Let, A = na

Now, $R = A\,\frac{\sin\alpha}{\alpha}$

The resultant intensity at P, being proportional to the square of the amplitude, is

$$I = R^2 = A^2\left(\frac{\sin\alpha}{\alpha}\right)^2$$

Condition for Maxima

$$R = A\,\frac{\sin\alpha}{\alpha} = \frac{A}{\alpha}\sin\alpha$$

By binomial expansion

$$\sin\alpha = \left[\alpha - \frac{\alpha^3}{3!} + \frac{\alpha^5}{5!} - \frac{\alpha^7}{7!} + \cdots\right]$$

$$R = \frac{A}{\alpha}\left[\alpha - \frac{\alpha^3}{3!} + \frac{\alpha^5}{5!} - \frac{\alpha^7}{7!} + \cdots\right]$$

$$\text{Now, } R = A\left[1 - \frac{\alpha^2}{3!} + \frac{\alpha^4}{5!} - \frac{\alpha^6}{7!} + \cdots\right]$$

In the above expression if α = 0 ; R = A (This is intensity of central maxima)

According to the above condition $\alpha = \frac{\pi\, b \sin\theta}{\lambda} = 0$

Here π, b, λ are constants, and maximum intensity occurs only when θ = 0, so that sin 0° = 0.

Condition for Minima

Minima means there should minimum intensity (intensity is zero)

Now we consider α ≠ 0, but sin α = 0,

This occurs when α = ±mπ, where m = 1,2,3,.....

So $\alpha = \frac{\pi\, b \sin\theta}{\lambda} = \pm m\pi$

$$\Rightarrow \quad b \sin\theta = \pm m\lambda$$

Again we get π, b, λ are constants, this equation gives the position of first, second, third etc. minima for m = 1, 2, 3 etc.,

Secondary Maxima

To find secondary maxima substitute, $\frac{dI}{d\alpha} = 0$

$$\frac{d}{d\alpha}\left(A^2\left(\frac{\sin\alpha}{\alpha}\right)^2\right) = 0$$

$$A^2 \quad \frac{d}{d\alpha}\left(\frac{\sin\alpha}{\alpha}\right)^2 = 0$$

$$A^2 \quad \frac{d}{d\alpha}\left(\frac{\sin^2\alpha}{\alpha^2}\right) = 0$$

We can differentiate by quotient rule

$$\frac{u}{v} = \frac{u^1 v - v^1 u}{v^2}$$

Here we have u = $\sin^2\alpha$ **and v =** α^2

$$A^2\left(\frac{\alpha^2\,(2\sin\alpha\cos\alpha) - \sin^2\alpha\,(2\alpha)}{\alpha^4}\right) = 0$$

$$A^2\left(\frac{2\alpha \sin\alpha\,(\alpha\cos\alpha - \sin\alpha)}{\alpha^4}\right) = 0$$

$$\frac{A^2}{\alpha^4}\,(2\alpha \sin\alpha\,(\alpha\cos\alpha - \sin\alpha)) = 0$$

$$2\alpha \sin\alpha\,(\alpha\cos\alpha - \sin\alpha) = 0$$

$$(\alpha\cos\alpha - \sin\alpha) = 0$$

$$\alpha\cos\alpha = \sin\alpha$$

$$\alpha = \frac{\sin\alpha}{\cos\alpha} = \tan\alpha$$

Let y = α = tan α

Therefore y = ; y = tan α

The maxima will occur when

$$\alpha = \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \ldots\ldots$$

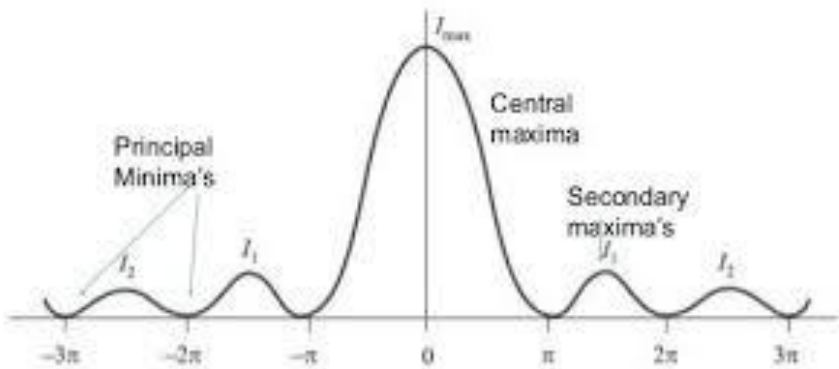
In general α = (2n+1)$\frac{\pi}{2}$ n = 1,2,3,……

These are points of secondary maxima

$$I = I_0\left(\frac{\sin\alpha}{\alpha}\right)^2$$

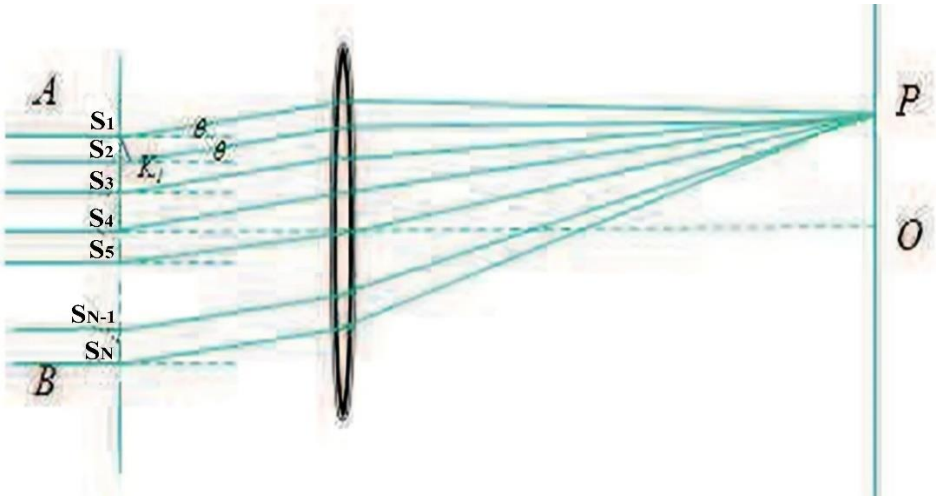
We have α = $\frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \ldots\ldots$

$$I_1 = \frac{4}{9\pi^2} I_0\ ,\ I_2 = \frac{4}{25\pi^2} I_0\ ,\ I_1 = \frac{4}{49\pi^2} I_0\ ,\ldots\ldots\ldots$$



Explain in detail about diffraction due to multi slit------(16 marks)

PLANE DIFFRACTION GRATING



- Here, S_1, S_2, S_3….. S_N are N narrow slits, in between points A and B. Let b= width of slit, d= width of opaque part between two slits.
- The amplitude from each slit in the direction θ is

$$R_0 = \frac{A \sin\alpha}{\alpha}$$

$$\text{Where } \alpha = \frac{\pi b}{\lambda} \sin\theta$$

(As derived, in case of Single slit Fraunhofer diffraction)

The path difference between the wavelets from S_1 and S_2 in the direction θ is

$$S_2 K_1 = (b + d) \sin\theta$$

Hence the phase difference between them

$$\frac{2\pi}{\lambda}(b+d)\sin\theta = 2\beta \text{, say}$$

If N be the total number of slits in the grating, the resultant amplitude in the direction of θ will be

$$R = R_0 \frac{\sin N\beta}{\sin\beta} = \left(\frac{A \sin\alpha}{\alpha}\right)\frac{\sin N\beta}{\sin\beta}$$

Thus, the resultant intensity at point P is

$$I = R^2 = A^2\left(\frac{\sin\alpha}{\alpha}\right)^2\left(\frac{\sin N\beta}{\sin\beta}\right)^2$$

$$\text{The factor } A^2\left(\frac{\sin\alpha}{\alpha}\right)^2$$

gives the intensity distribution due to single slit.

$$\text{while } \left(\frac{\sin N\beta}{\sin\beta}\right)^2$$

gives the distribution of intensity in the diffraction pattern due to the interference in the waves due to N slits.

Principal Maxima

$$I = R^2 = A^2\left(\frac{\sin\alpha}{\alpha}\right)^2\left(\frac{\sin N\beta}{\sin\beta}\right)^2$$

The intensity will be maximum when

$$\sin\beta = 0 \Rightarrow \beta = \pm n\pi$$

$$Where,\ n = 0,1,2,3.....$$

This result in

$$\frac{\sin N\beta}{\sin\beta} = \frac{0}{0} \text{ (Indeterminate)}$$

Applying L' Hospital rule

$$\lim_{\beta\to\pm n\pi}\frac{\sin N\beta}{\sin\beta} = \lim_{\beta\to\pm n\pi}\frac{\frac{d}{d\beta}(\sin N\beta)}{\frac{d}{d\beta}(\sin\beta)}$$

$$\lim_{\beta\to\pm n\pi}\frac{N\cos N\beta}{\cos\beta} \Rightarrow \pm N$$

This result in

$$I = A^2\left(\frac{\sin\alpha}{\alpha}\right)^2 N^2$$

The condition for principal maxima is

$$\sin\beta = 0 \qquad \text{or } \beta = \pm n\pi$$

$$\frac{\pi}{\lambda}(b+d)\,Sin\,\theta = \pm n\pi$$

$$(b+d)\,Sin\theta = \pm n\lambda$$

$$\sin\beta = 0 \qquad \text{or } \beta = \pm n\pi$$

$$\frac{\pi}{\lambda}(b+d)\,Sin\,\theta = \pm n\pi$$

$$(b+d)\,Sin\theta = \pm n\lambda$$

Minima

$$I = R^2 = A^2\left(\frac{\sin\alpha}{\alpha}\right)^2\left(\frac{\sin N\beta}{\sin\beta}\right)^2$$

The intensity will be minimum when

$$\sin N\beta = 0 \text{ but } \sin\beta \neq 0$$

$$N\beta = \pm m\pi$$

Missing Orders

As the resultant intensity due to N-parallel slits (plane diffraction grating) is given by

$$I = R^2 = A^2\left(\frac{\sin\alpha}{\alpha}\right)^2\left(\frac{\sin N\beta}{\sin\beta}\right)^2$$

Where $\alpha = \frac{\pi b}{\lambda}\sin\theta$

and $\beta = \frac{\pi}{\lambda}(b+d)\sin\theta$

Now the direction of principal maxima in grating spectrum is given as

$(b + d)\ Sin\ \theta = n\ \lambda$

Further the direction of minima of a single slit pattern is

$b\ Sin\ \theta = m\lambda$

Where $m = 1, 2, 3\ldots\ldots$

If both the conditions are simultaneously satisfied, a particular maximum of order n will be absent in the grating spectrum, these are known as absent spectra (or missing order spectrum).

Dividing equation

$$\frac{b+d}{b} = \frac{n}{m}$$

If b = d, then 2nd, 4th, 6th etc. orders maxima will be missing in the grating diffraction pattern.

If d = 2b, then 3rd, 6th, 9th etc. orders maxima will be missing in the grating diffraction pattern.

Explain elaborately about Brewster's law with its applications.

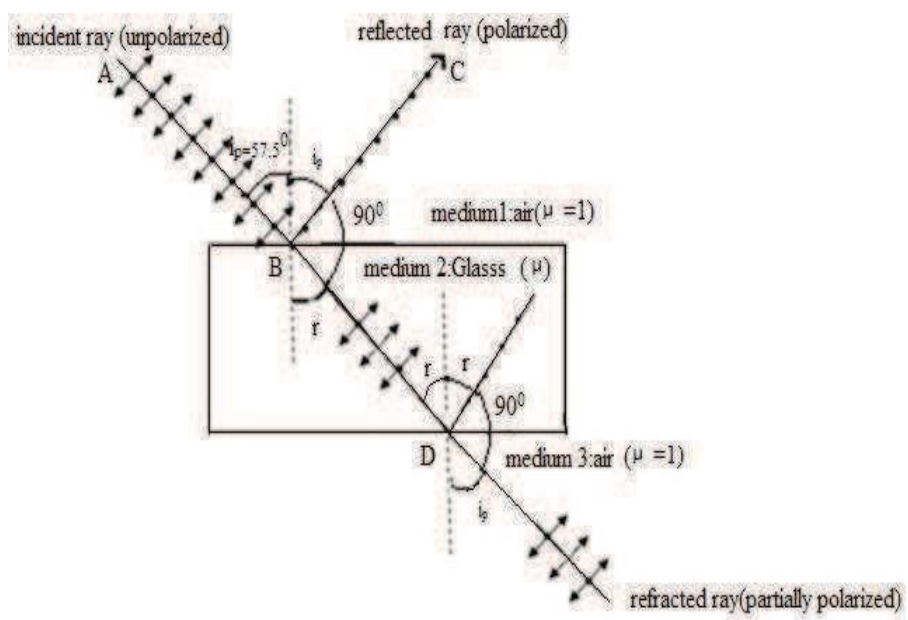
----------(8 marks)

Brewster's Law

- In 1811 A series of experiments on polarization of light by reflection from different reflecting surface was performed by Sir David Brewster.
- He observed that degree of polarization of ordinary light varies with angle of incidence and at a particular angle of incidence on the surface of a given transparent medium the reflected light is completely plane polarized.
- This particular angle of incidence is called polarizing angle or Brewster angle.
- However refracted light is still partially plane polarized.
- Further in case of reflected ray of light the plane of vibration is perpendicular to plane of incidence while it is parallel to plane of vibration in case of partially polarized transmitted light.
- Brewster found that polarizing angle depends upon the refractive index of the medium.
- As a result of experiments he proved that the tangent of angle of polarization is numerically equal to the refractive index of the reflecting medium.
- If i_p is angle of polarization and μ is the refractive index of the medium then according to Brewster's law

$$\mu = \tan i_p$$

- It must be noted here as refractive index depends upon the wavelength of incident light, angle of polarization varies with the wavelength hence complete polarization is possible with monochromatic light and not with white light.
- A direct consequence of Brewster's law is that when light is incident at the angle of polarization i_p, the reflected ray is at right angle to the refracted ray. This can be easily proved as follows



- Let unpolarized beam AB is incident on a smooth glass surface at the polarizing angle i_p. It is reflected along BC and refracted ray along BD as shown in figure.

Then from Brewster's law

$$\mu = \tan i_p = \frac{\sin i_p}{\cos i_p}$$

Also from Snell's law $\mu = \frac{\sin i_p}{\sin r}$

$$\frac{\sin i_p}{\cos i_p} = \frac{\sin i_p}{\sin r}$$

$$\frac{\sin i_p}{\sin\left(\frac{\pi}{2} - i_p\right)} = \frac{\sin i_p}{\sin r}$$

$$\frac{\pi}{2} - i_p = r$$

$$i_p + r = \frac{\pi}{2}$$

$$i_p + \angle CBD + r = \pi$$

$$\angle CBD = \pi - \left(i_p + r\right) = \pi - \frac{\pi}{2} = \frac{\pi}{2}$$

- That is, at polarizing angle of incidence, the reflected ray is perpendicular to refracted ray.
- Another important consequence of the law is that if unpolarized light is incident on the upper surface of a smooth parallel sided glass slab at the angle of polarization, the refracted ray also meets the lower surface of the slab at angle of polarization.
- As shown in the figure 10.8, the angle of incidence and refraction at the lower surface are r and i_p respectively.
- Hence at lower surface boundary between glass and air the refractive index of second medium (air) with respect to the first (glass).

$$\frac{1}{\mu} = \frac{\sin r}{\sin i_p}$$

Also, $\tan r = \frac{\sin r}{\cos r} = \frac{\sin r}{\cos\left(\frac{\pi}{2} - i_p\right)} = \frac{\sin r}{\sin i_p} = \frac{1}{\mu}$

As $i_p + r = \frac{\pi}{2}$ or $r = \frac{\pi}{2} - i_p$

$$\cos r = \cos\left(\frac{\pi}{2} - i_p\right) = \sin i_p$$

$$\frac{1}{\mu} = \tan r$$

- Therefore, r is the angle of polarization for the lower surface of the slab.
- Hence the light reflected from lower surface of glass slab is completely plane polarized while that refracted into the air is partially plane polarized.

Application of Brewster's Law

Here are some key applications of Brewster's Law:

- **Polarizing Sunglasses:**

  These glasses use a special filter aligned at the Brewster angle to block polarized light reflected from surfaces like roads and water, significantly reducing glare and improving visibility.

- **Photography:**
  Photographers use polarizing filters in their camera lenses to reduce glare from water and other reflective surfaces, allowing for clearer images with enhanced contrast.
- **Laser Technology:**
  Brewster windows are placed at the ends of laser tubes at the Brewster angle. This setup allows only the desired component of light to pass through, ensuring that the laser emits linearly polarized light by filtering out the non-polarized components.

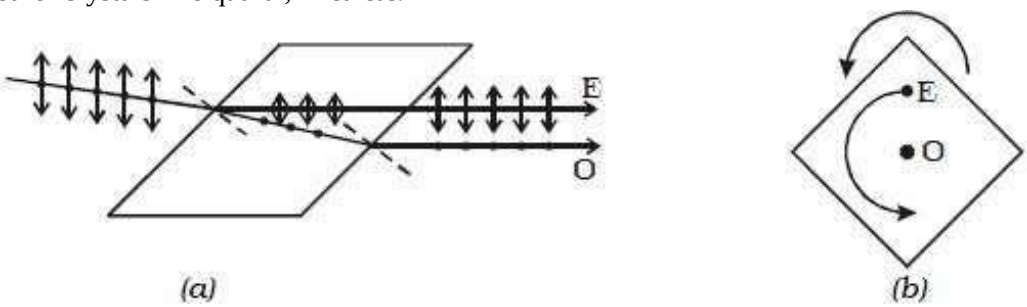
Scientific Instrumentation:

- **Brewster Angle Microscopy:** This technique allows for the direct observation and study of monolayers on liquid surfaces by analyzing the reflection of polarized light at the Brewster angle, providing insights into their structural changes.
- **Ellipsometry:** This non-destructive optical method uses the changes in the polarization state of light reflected or transmitted from a sample to determine properties of thin films, such as their thickness and refractive index

Explain in detail about double refraction, birefringence and its types with its applications. (8 marks)

Double Refraction

Bartholinus discovered that when a ray of unpolarized light is incident on a calcite crystal, two refracted rays are produced. This phenomenon is called **double refraction**. Hence, two images of a single object are formed. This phenomenon is exhibited by several other crystals like quartz, mica etc.



When an ink dot on a sheet of paper is viewed through a calcite crystal, two images will be seen (Fig: b). On rotating the crystal, one image remains stationary, while the other rotates around the first. The stationary image is known as the ordinary image (O), produced by the refracted rays which obey the laws of refraction. These rays are known as ordinary rays. The other image is extraordinary image (E), produced by the refracted rays which do not obey the laws of refraction. These rays are known as extraordinary rays.

Inside a double refracting crystal, the ordinary ray travels with same velocity in all directions and the extra ordinary ray travels with different velocities along different

directions.

A point source inside a refracting crystal produces spherical wavefront corresponding to ordinary ray and elliptical wavefront corresponding to extraordinary ray.

Inside the crystal there is a particular direction in which both the rays travel with same velocity. This direction is called optic axis. The refractive index is same for both rays and there is no double refraction along this direction.

Optical birefringence is an inherent property of a material, which causes double refraction of a ray of light resulting in polarization of the light wave. Birefringence is seen in crystallographic materials that have different refractive indices with respect to different crystallographic directions. Birefringence occurs when light passes through a transparent, molecularly ordered material, which exhibits orientation-dependent differences in refractive indices.

Birefringence is caused by the anisotropic forces that bind the atoms of a crystal. Anisotropy refers to the non-uniform distribution of properties in different directions. Some mineral crystals have two different refractive indices and, hence, show birefringence. Examples of minerals that exhibit birefringence are quartz, tourmaline, and calcite.

Occurrence of Birefringence

Calcite crystals show some of the strongest birefringence effects. Light passing through a calcite crystal is split into two rays by refraction. Refraction of the light waves occurs in such a way that two refracted rays are mutually perpendicular and, also, perpendicular to the direction of wave propagation. The two rays are termed as slow and fast rays. The wave having greater refractive index is called the slow ray.

Types of Birefringence and its measurement

Birefringence is classified as intrinsic or stress-induced birefringence.

- Intrinsic Birefringence – This type of birefringence is caused by the anisotropy present in the crystals. The atomic arrangement of the crystal itself is the source of birefringence. Examples are calcite, tourmaline, etc
- Stress-Induced Birefringence – This type of birefringence is caused due to the stresses imposed on the material. Materials such as glass or plastics show strain birefringence

Birefringence can be quantified by measuring the changes in polarization of light waves. This method of measurement is called polarimetry. A special method, called dual polarization interferometry, is used to measure birefringence of lipid bilayers.

Applications of Birefringence

Birefringence finds use in the following applications:

- Polarizing prisms and retarder plates
- Liquid crystal displays
- Medical Diagnostics

Write about the Huygen's theory of double refraction. -(6 marks)

Huygen's Theory of Double Refraction:

According to Huygen;s theory each point on wavefront act as a fresh source of disturbance and sends secondary wavelets. the envelope of all secondary wavelets at any instance represents a wavefront which is the locus of particles vibrating in phase.
Huygen's theory cannot explain double refraction in uniaxial crystals. He extends the theory in secondary wavelets. According to this theory,

1. When any wavefront strikes a doubly refracted crystal, every point of crystal become a source of two wavefronts.
   a. Ordinary wavefront corresponds to ordinary waves. Since ordinary waves have same velocity in all direction, the secondary wavefront is spherical.
   b. Extraordinary wavefront corresponds to extra ordinary waves. Since extra ordinary waves have different velocities in different directions, the extra ordinary wavefront is ellipsoid of revolution, with optic axis as the axis of revolution.
2. The sphere and ellipsoid touch each other at points which lie on optic axis of the

crystal, because the velocity of ordinary and extra ordinary ray is same along optic axis.

3. In certain crystals lie calcite and tourmaline called negative crystals, the ellipsoid lies outside the sphere as shown in figure below. This shows that in negative crystals, the extra ordinary wavefront travels faster than ordinary wavefront except along optic axis.

4. In certain crystals lie ice and quartz called the positive crystals, the sphere lies outside the ellipsoid as shown in figure below. This shows that velocity of ordinary wavefront is greater than extraordinary wavefront in positive crystals except along optic axis.

MODULE II - QUANTUM MECHANICS

What is the inadequacy of classical mechanics? (2 marks)

Inadequacy of Classical Mechanics

According to the classical mechanics, if we consider the case of an electron moving round the nucleus, its energy should decrease (because the accelerated charged particle loses energy in the form of electromagnetic waves) and therefore its velocity should decrease continuously. The ultimate result is that the electron comes closer and closer to the nucleus until it collapses. This shows the instability of the atom; it is in contradiction to the observed fact of the stability of an atom. Thus, the classical mechanics fails to explain the stability of an atom.

The classical mechanics also failed to explain the spectrum of the hydrogen atom. According to the classical theory, the excited atoms of hydrogen emit electromagnetic radiations of all wavelengths continuously, while it is observed that they emit the radiation of certain wavelengths only.

Explain briefly about origin of quantum theory of radiation (2 marks)

Difficulties with classical theories of Black Body Radiation and Origin of Quantum Theory of Radiation

We know that when bodies radiate energy, their temperature falls until the loss of energy is compensated by an external source. In case of heat radiation, we can obtain the thermal equilibrium by maintaining the body at a fixed temperature with the help of some heat-giving source. In this case the body gives as much radiation as it receives. If the body absorbs all the incident radiation, then it is called Black Body radiation. In actual practice, it is not possible to realize a perfectly black body, but an enclosure provided with a small opening serves the purpose because the radiation entering the enclosure will be reflected many times inside the enclosure and ultimately absorbed.

Define Kirchoff's law of radiation. (2 marks)

Kirchhoff's law:

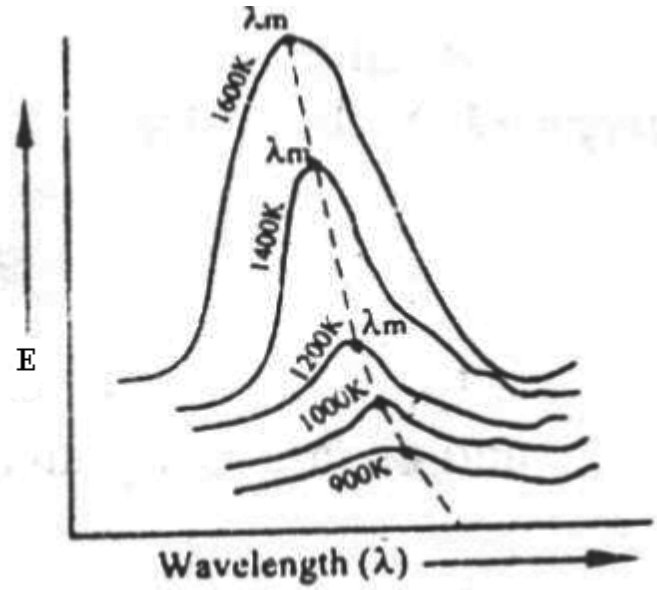
Ratio of emissive power to the coefficient of absorption of any given wavelength is the same for all bodies at a given temperature and is equal to the emissive power of the black body at that temperature.

$$\frac{e_\lambda}{e_\lambda} = \mathrm{E}$$

Discuss briefly about black body allowed to radiate at different temperatures (2 or 4 marks)

Energy spectrum:

When a perfect black body is allowed to emit radiations at different temperatures, then the distribution of the energy for different wavelengths at various temperatures is obtained as shown in the figure. From figure the following results are formulated.



i. The energy distribution is not uniform for a given temperature.
ii. The intensity of radiation (E) increases with respect to the increase in wavelength at particular wavelength in becomes maximum (λm) and after this it starts decreasing with respect to the increase in wavelength.
iii. When the temperature is increased, the maximum wavelength (λm) decreases.
iv. For all the wavelengths an increase in its temperature causes increase in energy.
v. The total energy emitted at any particular temperature can be calculated from the area under that particular curve.

Define Stefan Boltzmaan law (2 marks)

Stefan- Boltzmann Law

According to this law the radiant energy (E) of the body is directly proportional to the fourth power of the temperature (T) of the body.

$$E \propto T^4$$

$$E = \sigma T^4$$

Where σ = Stefan Constant, given by

$$\sigma = \frac{2\pi^4 K_B^4}{15\, h^3 c^2}$$

Define Wien's displacement law. (2 marks)

Wien's displacement law:

This law states that the product of then wavelength λ_m corresponding to the maximum energy and the absolute temperature (T) is a constant.

$$\lambda_m T = \text{constant}$$

This law shows that, as the temperature increases, the wavelength corresponding to maximum energy decreases.

Wien also showed that the maximum energy (Emax) is directly proportional to the fifth power of the absolute temperature.

$$E_{max} \propto T^5$$

$$E_{max} = \text{constant } T^5$$

By deducing this law, he obtained a law called Wien's law of distribution of energy (Eλ),

given by

$$E_\lambda = C_1\lambda^{-5} e^{-c_2/\lambda T}$$

Where C_1 and C_2 are constants given by

$$C_1 = 8\pi hc \text{ and } C_2 = \frac{hc}{K_B}$$

Define Raleigh jean law of radiation. (2 marks)

Raleigh Jeans law:

According to this law, the energy distribution is directly proportional to the absolute temperature and is inversely proportional to the fourth power of the wavelength.

It is governed by the equation

$$E_\lambda = \frac{8\pi K_B T}{\lambda^4}$$

Where K_B is Boltzmann Constant.

This law holds well only for longer wavelength regions and not for shorter wavelengths.

State Plank's hypothesis. (2 marks)

According to Planck's quantum theory, Different atoms and molecules can emit or absorb energy in discrete quantities only. The smallest amount of energy that can be emitted or absorbed in the form of electromagnetic radiation is known as quantum.

State the Plank's radiation law. (2 marks)

Statement:

The energy density of heat radiation emitted from an enclosure at temperature T in the wavelength ranges from λ to λ+dλ is given by

$$E_\lambda d\lambda = \frac{8\pi hc}{\lambda^5\left(e^{\frac{h\nu}{kT}} - 1\right)} d\lambda$$

Here h = Planck's constant
C= speed of the light
K= Boltzmann constant
T = temperature of the enclosure

Give appropriate examples for particle and wave nature of light. (2 marks)

The universe is made of radiation (light) and matter (particles). The light exhibits the dual nature (ie.) it can behave both a wave (interference, diffraction, phenomenon) and as a particle (Compton Effect, photo electric effect etc.)

Write about physical significance of wave function. (4 or 6 marks)

PHYSICAL SIGNIFICANCE OF WAVE FUNCTION ψ

1. The variable quantity that characterizes de-Broglie wave is called wave function ψ.

2. The wave function ψ represents the variations in the matter waves and it connects the particle nature and its associated wave nature statistically.

3. The wave function associated with a moving particle at a particular instant of time and at a particular point in space is related to the probability of finding the particle at that instant and at that point.

4. The probability 0 corresponds to the certainty of not finding the particle and probability 1 corresponds to the certainty of finding the particle.

$$\iiint \psi^* \psi d\tau \quad = 1 \; if \; particle \; is \; present$$

$$\iiint \psi^* \psi d\tau \quad = 0 \; if \; particle \; is \; not \; present$$

5. The wave function is a complex quantity that cannot be measured.

6. The probability of finding a particle at particular region must be real and positive, but the wave function is in general a complex quantity.

The probability density is given by

$$P(x,t) = |\psi(x,t)|^2 = \psi^* \psi$$

Define degeneracy and non degeneracy states. (2 or 4 marks)

DEGENERACY AND NON-DEGENERACY

Degeneracy:

For several combination of quantum numbers we have same energy Eigen value but different Eigen functions. Such states and energy levels are called Degenerate state.

The three combinations of quantum numbers (112), (121) and (211) which gives same Eigen value but different Eigen functions are 3-fold degenerate state.

Example: $n_x = 1$, $n_y = 1$, $n_z = 2$

Then $n_x^2 + n_y^2 + n_z^2 = 6$

Similarly, for the other two combinations also it is 6 only.

Since $E_{112} = E_{121} = E_{211} = \frac{6h^2}{8ma^2}$

The corresponding wave functions can be written as

$$\Psi_{112z} = \sqrt{\frac{8}{a^3}} \sin\frac{\pi x}{a} \sin\frac{\pi y}{a} \sin\frac{2\pi z}{a}$$

$$\Psi_{112z} = \sqrt{\frac{8}{a^3}} \sin\frac{\pi x}{a} \sin\frac{2\pi y}{a} \sin\frac{\pi z}{a}$$

$$\Psi_{112z} = \sqrt{\frac{8}{a^3}} \sin\frac{2\pi x}{a} \sin\frac{\pi y}{a} \sin\frac{\pi z}{a}$$

For various combinations of quantum number if we have same energy value and same (one) Eigen function then such states and energy levels are called Non – degenerate state.

Example

$$n_x = 2, n_y = 2, n_z = 2$$

$$E_{222} = \frac{12h^2}{8ma^2} \text{ and}$$

$$\Psi_{112} = \sqrt{\frac{8}{a^3}} \sin\frac{2\pi x}{a} \sin\frac{2\pi y}{a} \sin\frac{2\pi z}{a}$$

Write elaborately about Planck's hypothesis. – 4 or 6 marks

PLANCK'S HYPOTHESIS

Planck suggested that the correct results could be obtained if the energy of oscillating electrons is taken as discrete rather than continuous. He suggested quantum theory of radiation. Planck suggested in deriving the formula, which agrees extremely well with experimental results. He derived the radiation law by using the following assumptions.

a. A black body chamber is filled up not only with radiation, but also with simple harmonic oscillators or harmonic oscillators or resonators of molecular dimensions. They can vibrate with all possible frequencies.

b. The frequency of radiation emitted by an oscillator is the same as the frequency of its vibration.

c. An oscillator cannot emit energy in a continuous manner. It can emit energy in the multiples of a small unit called Quantum (Photon).

If an oscillator is vibrating with a frequency ν, it can radiate in quantas of magnitude h. The oscillator can have only discrete energy E_n given by

$$E_n = nh\nu = n\varepsilon \text{ where } h\nu = \varepsilon$$

Here n is an integer and h is the Planck's constant The energy of the single photon of the frequency is

$$\varepsilon = h\nu$$

d. The oscillators can emit or absorb radiation energy in packets of h

This implies that the exchange of energy between the radiation and matter cannot take place continuously but are limited to discrete set of values 0, hν, 2hν, 3hν, nhν.

The emission of radiation corresponds to a decrease and absorption to an increase in the energy and amplitude of an oscillator.

Define and derive the planck's law of radiation. -16 or 12 marks

PLANCK'S LAW OF RADIATION

Statement:

The energy density of heat radiation emitted from an enclosure at temperature T in the wavelength ranges from λ to λ+dλ is given by

$$E_\lambda d\lambda = \frac{8\pi hc}{\lambda^5\left(e^{\frac{h\nu}{kT}} - 1\right)} d\lambda$$

Here h = Planck's constant
C= speed of the light
K= Boltzmann constant
T = temperature of the enclosure

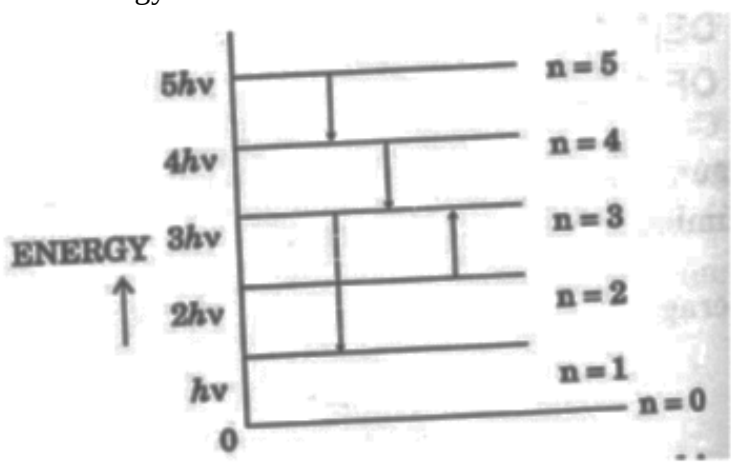
DERIVATION OF PLANCK'S LAW OF RADIATION

Assume that a black body consists of energy by a large number of atomic oscillators.

Average energy

$$\bar{E} = \frac{E}{N}$$

Where E is the total energy and N is the number of oscillators.



Number of atomic oscillators in ground state = N_0

From Maxwell's distribution law, the number of oscillators having energy ε_n in excess of ground state energy (ε_0) is given by

$$N_n = N_0 e^{\frac{-\varepsilon_n}{kT}}$$

Where T – Absolute temperature
K – Boltzmann's Constant

Let N be the total number of Planck's oscillators and N_1, N_2, N_3 etc be the number of oscillators with energies ε_0, ε_1, ε_2,…..then

$$N = N_0 e^{\frac{-\varepsilon_0}{kT}} + N_0 e^{\frac{-\varepsilon_1}{kT}} + \cdots$$

We know from Planck's quantum theory, ε can be a quantas of integral values of hν and so the possible values of ε are 0, 2hν, and 3hν etc.

$$N = N_0 e^0 + N_0 e^{\frac{-h\nu}{kT}} + N_0 e^{\frac{-2h\nu}{kT}} + \cdots$$

$$N = N_0 + N_0 e^{\frac{-h\nu}{kT}} + N_0 e^{\frac{-2h\nu}{kT}} + \cdots$$

Since $e^0 = 1$

$$put\ x = e^{\frac{-h\nu}{kT}}$$

$$N = N_0 + N_0 x + N_0 x^2 + N_0 x^3 + \cdots$$

$$N = N_0(1 + x + x^2 + x^3 + \cdots)$$

$$N = \frac{N_0}{1 - x}$$

By using binomial theorem, we get

$$\frac{1}{1 - x} = (-x) = 1 + x + x^2 + x^3 + \cdots$$

Total Energy $E = \varepsilon_0 N_0 + \varepsilon_1 N_1 + \varepsilon_2 N_2 + \cdots$

Substituting the values of $\varepsilon_0, \varepsilon_1, \varepsilon_2 \ldots\ldots$ We have

$$E = 0 \, x \, N_0 + hvN_0 e^{\frac{-hv}{kT}} + 2hvN_0 e^{\frac{-2hv}{kT}} + \ldots.$$

We know $x = e^{\frac{-hv}{kT}}$

$$E = hvN_0 x + 2hvN_0 x^2 + ..$$

$$E = hvN_0(x + 2x^2) + ..$$

$$E = hvN_0(x + 2x) + ..$$

$$E = \frac{hvN_0 x}{(-x^2)}$$

Since $\frac{1}{(1-x)^2} = (-x) = 1 + 2x + \cdots$

by using Binomial Series

$$\bar{E} = \frac{hvN_0 x}{\frac{(-x^2)}{\frac{N_0}{1 - x}}}$$

$$\bar{E} = \left[\frac{hvN_0 x}{(-x^2)}\right]\left[\frac{1 - x}{N_0}\right]$$

$$\bar{E} = \frac{hvx}{(-x)} = \frac{hvx}{x\left(\frac{1}{x} - 1\right)}$$

Substituting $x = e^{\frac{-hv}{kT}}$, *we have*

$$\bar{E} = \frac{hv}{\left(\frac{1}{e^{\frac{-hv}{kT}}} - 1\right)}$$

$$\bar{E} = \frac{hv}{\left(\frac{1}{e^{\frac{hv}{kT}}} - 1\right)}$$

Number of oscillators per unit volume in the wavelength range λ and λ + dλ is given by

$$\bar{E} = \frac{8\pi d\lambda}{\lambda^4}$$

The energy density of radiation between wavelengths λ and λ+dλ is given by

Eλdλ = (Number of oscillators per unit volume in the interval λ and λ+dλ)

× (Average energy per oscillator)

$$E_\lambda d\lambda = \left[\frac{8\pi d\lambda}{\lambda^4}\right]\left[\frac{hv}{\left(e^{\frac{hv}{kT}} - 1\right)}\right]$$

$$E_\lambda d\lambda = \left[\frac{8\pi d\lambda}{\lambda^4}\right]\left[\frac{\frac{hc}{\lambda}}{\left(e^{\frac{hv}{kT}} - 1\right)}\right] \quad since\ v = \frac{c}{\lambda}$$

$$E_\lambda d\lambda = \left[\frac{8\pi hcd\lambda}{\lambda^5}\right]\left[\frac{1}{\left(e^{\frac{hv}{kT}} - 1\right)}\right]$$

$$E_\lambda = \left[\frac{8\pi hc}{\lambda^5}\right]\left[\frac{1}{\left(e^{\frac{hv}{kT}} - 1\right)}\right]$$

This equation represents Planck's radiation law in terms of wavelength.

This law explained significantly the entire blackbody spectrum. It can also be expressed in

terms of frequencies by substituting $v = \frac{c}{\lambda}$ and dv = $\frac{cd\lambda}{\lambda^2}$ in the Planck law. Then

$$E_v = \left[\frac{8\pi h v^3}{c^3}\right]\left[\frac{1}{\left(e^{\frac{hv}{kT}}-1\right)}\right] dv$$

Deduce the Wein's displacement law and Raleigh Jean's law using Planck's radiation law. – 6 marks

DEDUCTION OF WIEN'S DISPLACEMENT LAW

When λ is small, is very large, hence hν/kT>>1 and $e^{\frac{hv}{kT}}$ is large when compared to 1. Hence, '1' is neglected in the denominator

Hence the equation reduces to

$$E_\lambda = \left[\frac{8\pi hc}{\lambda^5}\right]\left[\frac{1}{\left(e^{\frac{hv}{kT}}\right)}\right]$$

This is Wien's displacement law.
Therefore, Planck's law reduces to Wien's displacement for smaller wavelengths.
DEDUCTION OF RAYELIGH JEANS LAW:
When λ is very large, ν is very small, hν/kT << 1 and
$e^{\frac{hv}{kT}}=1+\frac{hv}{kT}$ (Using exponential series and neglecting higher powers)
Now equation reduces to

$$E_\lambda = \left[\frac{8\pi hc}{\lambda^5}\right]\left[\frac{1}{\left(1+\frac{hv}{kT}-1\right)}\right]$$

$$E_\lambda = \frac{8\pi hc}{\lambda^5 \frac{hv}{kT}}$$

$$E_\lambda = \frac{8\pi h\lambda}{\lambda^5 \frac{h}{kT}} \quad \text{i.e} \quad \lambda = \frac{c}{v}$$

$$E_\lambda = \frac{8\pi kT}{\lambda^4}$$

This is Rayleigh – Jeans law.

Therefore, Planck's law reduces to Rayleigh – Jeans law for longer wavelength.

Planck's formula is found to agree remarkably well with experimental observations of Lummer, Pringsheim, Kurlbaum etc. and is thus an established formula for all validity of quantum hypothesis.

Explain in detail about dual nature of radiation. – 6 marks

DUAL NATURE OF RADIATION (LIGHT) AND MATTER (PARTICLES) – MATTERWAVES

De- Broglie concept of dual nature:

The universe is made of radiation (light) and matter (particles). The light exhibits the dual nature (ie.) it can behave both a wave (interference, diffraction, phenomenon) and as a particle (Compton Effect, photo electric effect etc.)

Since the nature loves symmetry, in 1923 Louis debroglie suggested that an electron or any other material particle must exhibit wave like properties in addition to particle nature.

The waves associated with a material particle are called as matter waves.

De-Broglie wavelength:

From the theory of light, considering a photon as a particle the total energy of the photon is given by

$$E = mc^2 \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (1)$$

Where 'm' is the mass of the particle

'c' is the velocity of light

Considering the photon as a wave, the total energy is given by

$$E = h\nu \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (2)$$

Where h is the Planck's constant

ν is the frequency of the radiation.

From equations (1) and (2)

$$E = mc^2 = h\nu \quad \ldots\ldots\ldots\ldots\ldots\ldots (3)$$

We know Momentum = Mass × velocity

$$p = mc$$

$$hv = pc$$

i.e $p = \frac{hv}{c}$ Since $\frac{c}{v} = \lambda$

We can write $p = \frac{h}{\lambda}$

or the wavelength $\lambda = \frac{h}{p}$ …… (4)

de-Broglie suggested that the equation (4) can be applied both for photons and material particles.

If m is the mass of the particle and ν is the velocity the particle, then

Momentum p = mν

de-Broglie wavelength $\lambda = \frac{h}{mv}$ ……. (5)

Derive and explain about Schodinger time independent equation - 8 marks

SCHROEDINGER WAVE EQUATION

Schröedinger wave equation describes the wave nature of a particle in the mathematical form. It is the basic equation of motion of matter waves.

If the particle has wave properties, then there should be some sort of wave equation to describe the behavior of that particle.

Schrödinger connected the expression of de-Broglie's wavelength with the classical wave equation for a moving particle and he obtained a new wave equation

FORMS OF SCHROEDINGER WAVE EQUATION

There are two forms of Schröedinger wave equation. They are

a. Time independent wave equation

b. Time dependent wave equation

SCHROEDINGER TIME INDEPENDENT WAVE EQUATION

Consider a system of stationary waves associated with a particle.

Let x, y, z be the coordinates of the particle and ψ be the wave displacement for de-Broglie's waves at any time t.

The classical differential equation of a wave motion is given by

$$\frac{\partial^2\psi}{\partial x^2}+\frac{\partial^2\psi}{\partial y^2}+\frac{\partial^2\psi}{\partial z^2}=\frac{1}{v^2}\frac{\partial^2\psi}{\partial t^2} \quad \ldots\ldots\ldots (1)$$

Here v is the wave velocity

Equation (1) is written as

$$\nabla^2\psi=\frac{1}{v^2}\frac{\partial^2\psi}{\partial t^2} \quad \ldots\ldots\ldots \qquad (2)$$

$\nabla^2=\frac{\partial^2\psi}{\partial x^2}+\frac{\partial^2\psi}{\partial y^2}+\frac{\partial^2\psi}{\partial z^2}$ and ∇ is the Laplacian operator.

The solution of equation (2) gives as a periodic displacement in terms of time t,

$$\psi(x,y,z,t)=\psi_0(x,y,z,e^{-i\omega t}) \quad \ldots\ldots\ldots\ldots (3)$$

Here $\psi_0(x,y,z)$ is a function of x, y, z only and gives amplitude at the point considered.

Differentiating equation (3) with respect to t, we get

$$\frac{\partial\psi}{\partial t}=-i\omega\psi_0 e^{-i\omega t}$$

Again, differentiating with respect to t

$$\frac{\partial^2\psi}{\partial x^2}=(-\omega)(-i\omega\psi_0 e^{-i\omega t})$$

$$\frac{\partial^2\psi}{\partial x^2}=(i^2)(\omega^2\psi_0 e^{-i\omega t})$$

$$\frac{\partial^2\psi}{\partial x^2}=-\omega^2\psi \quad \ldots\ldots\ldots (3)$$

Substituting (3) in (2), we have

$$\nabla^2\psi=-\frac{\omega^2}{v^2}\psi \quad \ldots\ldots \qquad (4)$$

$$\nabla^2\psi+\frac{\omega^2}{v^2}\psi=0 \quad \ldots\ldots\ldots \qquad (5)$$

But, angular frequency ω = 2πν = 2π = $\frac{\nu}{\lambda}$

Here, ν is the frequency i.e ν = $\frac{\nu}{\lambda}$

$$\frac{\omega}{\nu} = \frac{2\pi}{\lambda} \quad \ldots\ldots\ldots\ldots\ldots \quad (6)$$

Squaring equation (6) both sides, we get

$$\frac{\omega^2}{\nu^2} = \frac{4\pi^2}{\lambda^2} \quad \ldots\ldots\ldots\ldots\ldots\ldots \quad (7)$$

Substituting equation (7) in equation (5), we have

$$\nabla^2\psi + \frac{4\pi^2}{\lambda^2}\psi = 0 \quad \ldots\ldots\ldots\ldots \quad (8)$$

We know that de-Broglie wavelength $= \frac{h}{mv}$, on substituting the value of λ in equation (8), we get

$$\nabla^2\psi + \frac{4\pi^2}{\frac{h^2}{m^2v^2}}\psi = 0$$

$$\nabla^2\psi + \frac{4\pi^2 m^2 v^2}{h^2}\psi = 0 \ldots\ldots\ldots\ldots (9)$$

If E is the total energy of the particle, V is the potential energy and $\frac{1}{2}mv^2$ is the Kinetic energy, then

Total energy (E) = potential Energy (V) + Kinetic Energy $\left(\frac{1}{2}mv^2\right)$

$$E = V + \frac{1}{2}mv^2$$

$$E - V = \frac{1}{2}mv^2$$

Multiplying by m on both sides

$$m^2v^2 = 2m(E - V) \ldots\ldots\ldots (10)$$

Substituting equation (10) in equation (9), we get

$$\nabla^2\psi + \frac{4\pi^2}{h^2}2m(E - V)\psi = 0$$

$$\nabla^2\psi + \frac{8\pi^2 m}{h^2}(E - V)\psi = 0 \ldots\ldots..(11)$$

This equation (11) is known as Schrödinger time independent wave equation

Let us now substitute $\hbar = \frac{\mathbf{h}}{\mathbf{2\pi}}$ in equation (11)

$$\hbar^2 = \frac{h^2}{4\pi^2}$$

Where ħ is a reduced Planck's constant
The equation (11) is modified as,

$$\nabla^2\psi + \frac{m}{\frac{h^2}{8\pi^2}}(E - V)\psi = 0$$

$$\nabla^2\psi + \frac{m}{\frac{h^2}{2 \times 2^2\pi^2}}(E - V)\psi = 0$$

$$\nabla^2\psi + \frac{2m}{\frac{h^2}{4\pi^2}}(E - V)\psi = 0$$

$$\nabla^2\psi + \frac{2m}{\hbar^2}(E - V)\psi = 0 \ldots\ldots\ldots(12)$$

This is the Schroedinger time independent wave equation

Derive and explain about Schodinger time dependent equation - 8 marks

SCHROEDINGER TIME DEPENDENT WAVE EQUATION

Schrödinger time dependent wave equation can be derived from Schrödinger time independent wave equation.

The solution of classical differential equation of the wave motion is given by

$$\psi(x, y, z, t) = \psi_0(x, y, z, e^{-i\omega t}) \ \ldots\ldots\ldots\ldots\ldots (1)$$

Differentiating equation (1) with respect to time t we get

$$\frac{\partial\psi}{\partial t} = -i\omega\psi_0 e^{-i\omega t} \ldots\ldots\ldots\ldots(2)$$

$$\frac{\partial\psi}{\partial t} = -i(2\pi\nu)\psi_0 e^{-i\omega t}$$

$$\therefore \omega = 2\pi\nu$$

$$\frac{\partial\psi}{\partial t} = -i(2\pi\nu)\psi \ldots\ldots\ldots (3)$$

$$\therefore \psi = \psi_0 e^{-i\omega t}$$

$$\frac{\partial\psi}{\partial t} = -2\pi i \frac{E}{h}\psi \qquad (\therefore \mathrm{E} = \mathrm{h}\nu = \nu = \frac{E}{h})$$

$$\frac{\partial\psi}{\partial t} = -i\frac{E}{\frac{h}{2\pi}}\psi = -i\frac{E}{\hbar}\psi \quad \left[\therefore \hbar = \frac{h}{2\pi}\right]$$

$$\frac{\partial\psi}{\partial t} = -i\frac{E}{\hbar}\psi$$

Multiplying by i on both the sides

$$i\frac{\partial\psi}{\partial t} = -i \times i\frac{E}{\hbar}\psi = -i^2\left(\frac{E}{\hbar}\right)\psi$$

$$i\frac{\partial\psi}{\partial t} = \frac{E}{\hbar}\psi = [\therefore i^2 = -1] \ldots\ldots..(4)$$

$$Or \;\; E\psi = i\hbar\frac{\partial\psi}{\partial t} \ldots\ldots\ldots\ldots\ldots.(5)$$

We know that Schrödinger time independent wave equation is given by

$$\nabla^2\psi + \frac{2m}{\hbar^2}(E - V)\psi = 0 \ldots\ldots.(6)$$

Substituting the value of Eψ from equation (5) in equation (6), we get

$$\nabla^2\psi + \frac{2m}{\hbar^2}\left[i\hbar\frac{\partial\psi}{\partial t} - V\psi\right] = 0$$

$$\nabla^2\psi = -\frac{2m}{\hbar^2}\left[i\hbar\frac{\partial\psi}{\partial t} - V\psi\right]$$

$$-\frac{\hbar^2}{2m}\nabla^2\psi = i\hbar\frac{\partial\psi}{\partial t} - V\psi$$

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V\psi = i\hbar\frac{\partial\psi}{\partial t}$$

$$\left[-\frac{\hbar^2}{2m}\nabla^2 + V\right]\psi = i\hbar\frac{\partial\psi}{\partial t} \ldots\ldots..(7)$$

$$H\psi = E\psi \ldots\ldots..(8)$$

This equation (8) is known as Schrödinger time dependent wave equation.

Where

$H = \left[-\frac{\hbar^2}{2m}\nabla^2 + V\right]$ is Hamiltonian Operator

$E = i\hbar\frac{\partial}{\partial t}$ is Energy Operator.

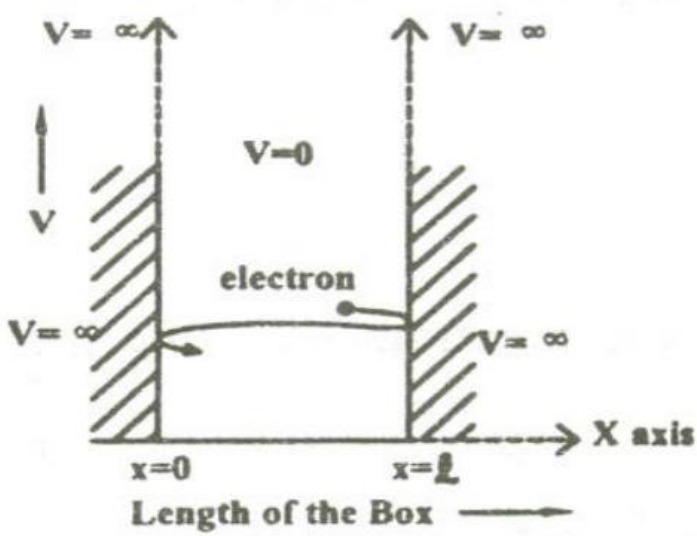
Explain in detail about particle in a one dimentional box condition. -16 marks

PARTICLE IN A ONE-DIMENSIONAL BOX

Let us consider particle (electron) of mass 'm' moving along the x-axis, enclosed in a one-dimensional potential box as shown in the figure.

Since the walls are of infinite potential the particle does not penetrate out from the box.

Also, the particle is confined between the length 'l' of the box and has elastic collisions with the walls. Therefore, the potential energy of the electron inside the box is constant and can be taken as zero for simplicity.



We can say that the Outside the box and on the wall of the box, the potential energy V of the electron is ∞.

Inside the box the potential energy (V) of the electron is zero.

In other words, we can write the boundary condition as

$$V(x) = 0 \; when \; 0 < x < l$$

$$V(x) = \infty \; when \; 0 \geq x \geq= l$$

Since the particle cannot exist outside the box the wave function ψ = 0 when 0 ≥ x ≥ l.

To find the wave function of the particle within the box of length 'l', let us consider the Schrödinger one dimensional time independent wave equation (i.e.,)

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - V)\psi = 0$$

Since the potential energy inside the box is zero [i.e. =0], the particle has kinetic energy alone and thus it is named as a free particle or free electron

For a free particle (electron), the Schrödinger wave equation is given by

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}(E - V)\psi = 0$$

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}\mathrm{E}\psi = 0 \ldots\ldots..(1)$$

$$Or \quad \frac{d^2\psi}{dx^2} + k^2\psi = 0 \quad where \; k^2 = \frac{2mE}{\hbar^2} \ldots\ldots..(2)$$

Equation (1) is a second order differential equation; therefore, it should have solution with two arbitrary constants.

The solution for equation (1) is given by

$$\psi(x) = A\sin kx + B\cos kx \ldots\ldots\ldots\ldots\ldots..(3)$$

Where A and B are called Arbitrary constants, which can be found by applying the boundary conditions.

$$V(x) = \infty \; when \; x = 0 \; and \; x = 1$$

Boundary condition (i) at x=0 Potential energy v = ∞. There is no chance for finding the particle at the walls of the box, ψ(x) = 0

Equation (3) becomes

$$0 = A\sin 0 + B\cos 0$$

$$0 = 0 + B$$

$$\therefore B = 0$$

Boundary condition (i) at x=1 Potential energy V = ∞. There is no chance for finding the particle at the walls of the box, ψ(x) = 0

Equation (3) becomes

$$0 = A \sin kl + B \cos kl$$

$$\therefore B = 0$$

$$0 = \text{Asin}\, kl$$

Since a ≠ 0; sinkl = 0

We know sin nπ = 0

Comparing two equations, we can write kl=nπ where n is an integer.

$$\text{Or k} = \frac{n\pi}{l} \quad \ldots\ldots\ldots\ldots\ldots..(4)$$

Substituting the value of B and k in equation 3 we can write the wave function associated with the free electron confined in a one-dimensional box as

$$\psi_n(\text{x}) = Asin\ \frac{n\pi x}{l} \quad \ldots\ldots\ldots\ldots.(5)$$

<u>Energy of the particle (Electron)</u>

We know from equation (2)

$$k^2 = \frac{2mE}{\hbar^2}$$

$$k^2 = \frac{2mE}{\frac{h^2}{4\pi^2}} \quad \therefore \left[\hbar^2 = \frac{h^2}{4\pi^2}\right]$$

$$k^2 = \frac{8\pi^2 mE}{h^2} \quad \ldots\ldots\ldots\ldots(6)$$

Squaring equation (3) we get

$$k^2 = \frac{n^2\pi^2}{l^2} \quad \ldots\ldots\ldots\ldots(7)$$

Equating (6) and (7), we can write

$$\frac{8\pi^2 mE}{h^2} = \frac{n^2\pi^2}{l^2}$$

$$Energy\ of\ the\ particle\ E_n = \frac{n^2 h^2}{8ml^2} \ldots\ldots (8)$$

From equations (8) and (5) we can say that, for each value of 'n', there is an energy level and the corresponding wave function. Thus, we can say that, each value of E_n is known as Eigen value and then corresponding value of ψ_n is called Eigen function.

ENERGY LEVELS OF AN ELECTRON

For various values of 'n' we get various energy values of the electron. The lowest energy value or ground state energy value can be got by substituting n = 1 in equation (8)

$$When\ n = 1\ we\ get\ E_1 = \frac{h^2}{8ml^2}$$

Similarly, we can get the other energy values

$$When\ n = 2\ we\ get\ E_2 = \frac{4h^2}{8ml^2} = 4E_1$$

$$When\ n = 3\ we\ get\ E_3 = \frac{9h^2}{8ml^2} = 9E_1$$

$$When\ n = 4\ we\ get\ E_4 = \frac{16h^2}{8ml^2} = 16E_1$$

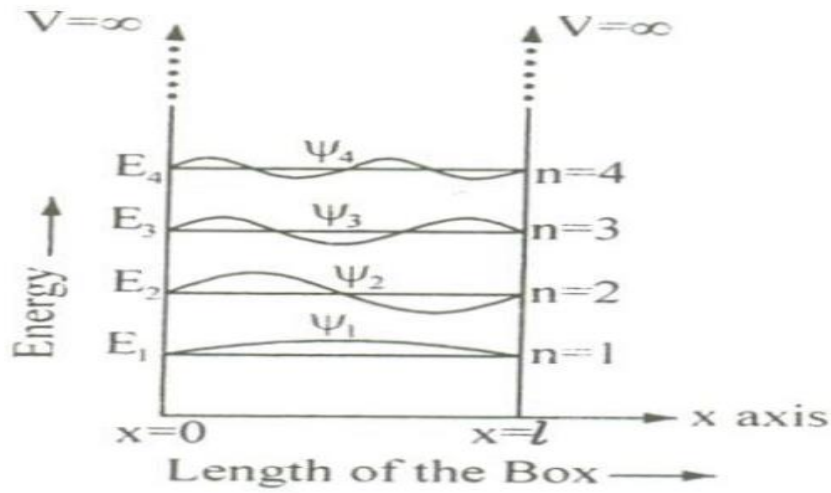
In general, we can write the energy Eigen function as

$$E_n = n^2 E_1 \ldots\ldots\ldots (9)$$

It is found that from the energy levels E_1,E_2,E_3……. etc the energy levels of an electron are discrete.

This is the great success which is achieved in Quantum Mechanics than classical mechanics, in which the energy levels are found to be continuous.

The various energy Eigen values and their corresponding Eigen functions of an electron enclosed in a one-dimensional box is as shown in the figure. Thus, we have discrete energy values.



Normalization of the wave function

Normalization:

It is the process by which the probability (P) of finding the particle (electron) inside the box can be done.

We know that the total probability (P) is equal to 1 means then there is a particle inside the box.

For a one-dimensional potential box of length 'l' the probability

$$P = \int_0^1 |\psi|^2 \, dx = 1 \ldots\ldots\ldots..(10)$$

Substituting equation (5) in equation (10), we get

$$P = \int_0^1 A^2 sin^2 \frac{n\pi x}{l} dx = 1$$

$$P = A^2 \int_0^l \left[\frac{1 - cos2n\pi x/l}{2}\right] dx = 1$$

$$A^2 \left[\frac{x}{2} - \frac{1}{2}\frac{sin2n\pi x/l}{2n\pi/l}\right]_0^l = 1$$

$$A^2 \left[\frac{l}{2} - \frac{1}{2}\frac{sin2n\pi l/l}{2n\pi/l}\right] = 1$$

$$A^2 \left[\frac{l}{2} - \frac{1}{2}\frac{sin2n\pi}{2n\pi/l}\right] = 1 \ldots\ldots..(11)$$

We know sin n π = 0 and

sin2nπ is also 0

Equation (11) can be written as

$$A^2 \frac{l}{2} = 1$$

$$A^2 = \frac{2}{l}$$

$$A = \sqrt{\frac{2}{l}}$$

Substituting the value of A in equation (5)

$$\psi_n(x) = \sqrt{\frac{2}{l}} sin \frac{n\pi x}{l}$$

The normalized wave function and their energy values are as shown in the figure.

The solution of one-dimensional potential box can be extended for a three-dimensional potential box. In a three-dimensional potential box, the particle (electron) can move in any direction in space. Therefore, instead of one quantum number 'n', we have to use three quantum number n_x, n_y, n_z corresponding to the three-coordinate axis x, y, z respectively

The energy of the particle = Ex + Ey+ Ez

$$E_{x,y,z} = \frac{n_x^2 h^2}{8ma^2} + \frac{n_y^2 h^2}{8mb^2} + \frac{n_z^2 h^2}{8mc^2}$$

If a=b=c for a cubical box

$$E_{x,y,z} = \frac{h^2}{8ma^2}\left[n_x^2 + n_y^2 + n_z^2\right]$$

The corresponding normalized wave function of an electron in a cubical box can be written as

$$\psi_{n_x n_y n_z} = \sqrt{\frac{2}{a} \times \frac{2}{a} \times \frac{2}{a}} \; sin \frac{n_x \pi x}{a} sin \frac{n_y \pi y}{a} sin \frac{n_z \pi z}{a}$$

$$\psi_{n_x n_y n_z} = \sqrt{\frac{8}{a^3}}\ sin\frac{n_x \pi x}{a} sin\frac{n_y \pi y}{a} sin\frac{n_z \pi z}{a}$$

Define Heisenberg Uncertainty Principle (2 or 4 marks)

HEISENBERG'S UNCERTAINTY PRINCIPLE

Statement: "It is impossible to determine both position and momentum of a particle simultaneously and accurately. The product of uncertainties involved in the determination of position and momentum simultaneously is greater or equal to $\frac{h}{4\pi}$

$$\Delta_{x\ X}\ \Delta_{px} = \frac{h}{4\pi}$$

This principle was formulated in 1927 by the German physicist Werner Heisenberg. It is also called the indeterminacy principle.

The Heisenberg's uncertainty principle can also be expressed in terms of the uncertainties involved in the simultaneous measurements of angular displacement and angular momentum and energy & time.

$$\Delta\theta\,.\,\Delta l\ \geq \frac{h}{4\pi}$$

$$\Delta t\ .\,\Delta E \geq \frac{h}{4\pi}$$

MODULE III - LASER AND FIBER OPTICS

Define LASER. (2 marks)

LASER stands for Light amplification by stimulated emission of radiation. Laser is a device which emits a powerful, monochromatic collimated beam of light. The emitted light waves are coherent in nature.

Write a note on directionality property of LASER (2 marks)

Directionality:

Ordinary light spreads in all directions and its angular spread is 1m/m. But it is found that laser is highly directional and is angular spread is 1mm/m. For example, the laser beam can be focused to very long distance with a few divergence or angular spread.

Divergence or angular spread is given by,

$$\phi = \frac{r_2 - r_1}{d_2 - d_1}$$

Where d_1, d_2 are any two distances for the laser source emitted and r_1, r_2 are the radii of the beam spots at a distance d_1 and d_2 respectively as shown.

Write a note on intensity property of LASER (2 marks)

Intensity:

Since an ordinary light spreads in all directions, the intensity reaching the target is very less. But in the case of laser, due to high directionality, the intensity of laser beam reaching the target is of high intense beam. For example, 1 mill watt power of He-Ne laser appears to be brighter than the sunlight.

Define monochromatic property of laser. (2 marks)

Monochromatic:

Laser beam is highly monochromatic; the wavelength is single, whereas in ordinary light like mercury vapor lamp, many wavelengths of light are emitted.

Define coherence property of laser. (2 marks)

Coherence:

It is an important characteristic of laser beam. In lasers the wave trains of same frequency are in phase, the radiation given out is in mutual agreement not only in phase but also in the direction of emission and polarization. Thus, it is a coherent beam. Due to high coherence it results in an extremely high power.

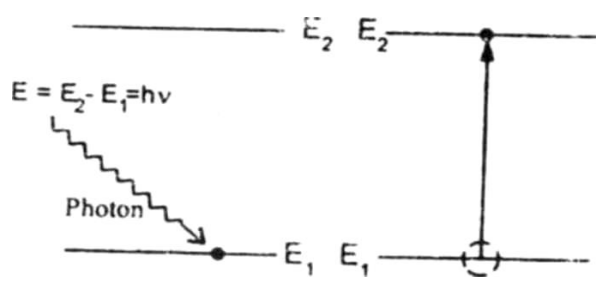
Difference between Ordinary light and LASER (2 marks or 4 marks)

S.No	Ordinary light	Laser beams
1	In ordinary light the angular spread is more.	In laser beam the angular spread is less.
2	They are not directional.	They are highly directional.
3	It is less intense	It is highly intense
4	It is not a coherent beam and is not in phase.	It is a coherent beam and is in phase
5	The radiations are polychromatic	The radiations are monochromatic
6	Example: Sun light, Mercury vapor lamp	He- Ne Laser, Co_2 laser

Define induced absorption. (2 marks)

Induced Absorption:

An atom in the lower energy level or ground state energy level E_1 absorbs the incident photon radiation of energy $h\ v$ and goes to the higher energy level or excited level E_2 as shown in figure.



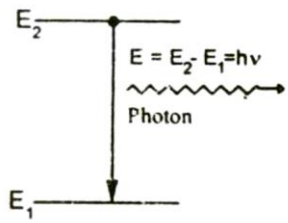
Before Absorption **After Absorption**

This process is called absorption.

Define spontaneous emission. (2 marks)

Spontaneous Emission:

The atom in the excited state returns to the ground state by emitting a photon of energy $E = (E_2 - E_1) = h\upsilon$ spontaneously without any external triggering as shown in the figure.

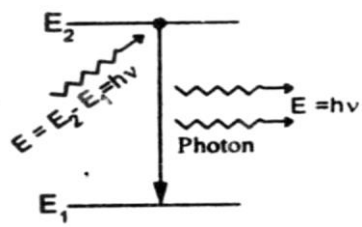


This process is known as spontaneous emission.

Define stimulated emission. (2 marks)

Stimulated Emission:

The atom in the excited state can also return to the ground state by external triggering or inducement of photon thereby emitting a photon of energy equal to the energy of the incident photon, known as stimulated emission. Thus, results in two photons of same energy, phase difference and of same directionality as shown.



Therefore, the rate of stimulated emission is given by

$$R_{21}\,(St) \propto \rho_v\, N_2$$

$$R_{21}\,(St) = B_{21}\, \rho_v\, N_2 \text{----------(3)}$$

Where B_{21} is constant which gives probability of stimulated emission transition per unit time.

State the difference between stimulated and spontaneous emission. (2 marks or 4 marks)

S.NO.	STIMULATED EMISSION	SPONTANEOUS EMISSION
1.	An atom in the excited state is induced to return to the ground state, thereby resulting in two photons of same frequency and energy is called Stimulated emission	The atom in the excited state returns to the ground state thereby emitting a photon, without any external inducement is called Spontaneous emission.
2.	The emitted photons move in the same direction and is highly directional	The emitted photons move in all directions and are random
3.	The radiation is highly intense, monochromatic and coherent	The radiation is less intense and is incoherent.
4.	The photons are in phase, there is a constant phase difference.	The photons are not in phase (i.e.) there is no phase relationship between them.
5.	The rate of transition is given by $R_{21}\,(St) = B_{21}\, \rho_v\, N_2$	The rate of transition is given by $R_{21}\,(Sp) = A_{21}\, N_2$

Define population inversion. (2 marks)

POPULATION INVERSION:

Population Inversion creates a situation in which the number of atoms in higher energy state is more than that in the lower energy state.

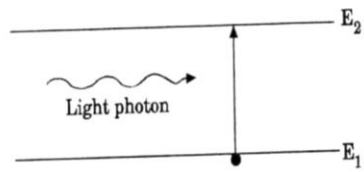
Usually at thermal equilibrium, the number of atoms N_2 i.e., the population of atoms at higher energy state is much lesser than the population of the atoms at lower energy state N_1 that is $N_1 > N_2$.

The Phenomenon of making $N_2 > N_1$ is known as Population Inversion.

Define optical pumping. (2 marks)

Optical pumping:

When the atoms are exposed to light radiations energy $h\ v$, atoms in the lower energy state absorb these radiations and they go to the excited state. This method is called Optical pumping. It is used in solid state lasers like ruby laser and Nd-YAG laser. In ruby laser, xenon flash lamp is used as pumping source.

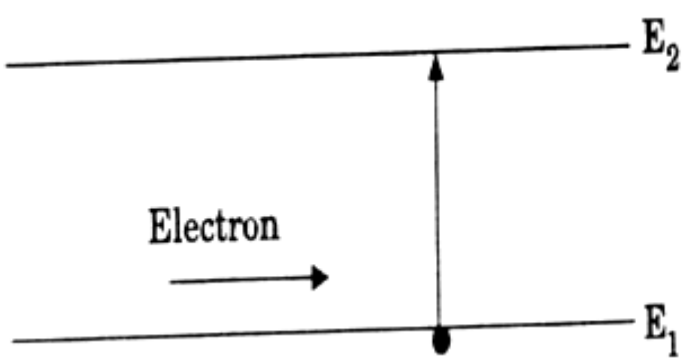


Define Electrical discharge method. (2 marks)

Electrical discharge method (Excitation by electrons):

In this method, the electrons are produced in an electrical discharge tube. These electrons are accelerated to high velocities by a strong electrical field. These accelerated electrons collide with the gas atoms.

By the process, energy from the electrons is transferred to gas atoms. Some atoms gain energy and they go to the excited state. This results in population inversion. This method is called Electrical discharge method.

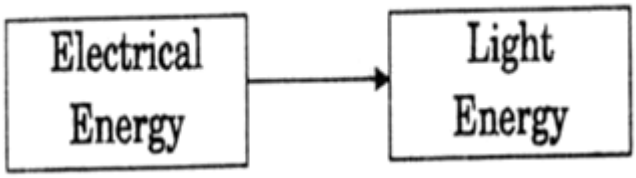


Define Direct Conversion. (2 marks)

Direct Conversion:

In this method, due to electrical energy applied in direct band gap semiconductor

like Ga As, recombination of electrons and holes takes place. During the recombination process, the electrical energy is directly is converted into light energy.



Define in-elastic atom - atom collision. (2 marks)

In elastic atom - atom collision:

In this method, a combination of two gases (Say A and B are used). The excited states of A and B nearly coincides in energy.

In the first step during the electrical discharge atoms of gas A are excited to their higher energy state A^* (metastable state) due to collision with the electrons.

$$A + e^* = A^* + e$$

Now A^* atoms at higher energy state collide with b atoms in the lower state. Due to inelastic atom - atom collision B atoms gain energy and they are excited to a higher state B^*. Hence, A atoms lose energy and return to lower state.

$$A^* + B = A + B^*$$

Categories with examples about the types of LASER. (2 marks)

S.NO	TYPE OF LASER	EXAMPLES
1.	Solid State laser	Ruby Laser Nd:YAG laser
2.	Gas laser	He-Ne Laser, CO_2 Laser, Argon - ion laser
3.	Liquid Laser	$SeOCL_2$ Laser Europium Chelate Laser
4.	Dye laser	Rhodamine 6G laser, Coumarin dye laser
5.	Semiconductor Laser	GaAs laser, GaAsP laser

Write about the principle of CO_2 LASER. (2 marks)

In a molecular gas laser, laser action is achieved by transitions between vibrational and rotational levels of molecules. Its construction is simple and the output of this laser is continuous.

In CO_2 molecular gas laser, transition takes place between the vibrational states of Carbon dioxide molecules.

Write about advantages and disadvantages of CO_2 LASER. (2 or 4 marks)

Advantages:

1. The construction of CO_2 laser is simple
2. The output of this laser is continuous.
3. It has high efficiency

4. It has very high output power.
5. The output power can be increased by extending the length of the gas tube.

Disadvantages:

1. The contamination of oxygen by carbon monoxide will have some effect on laser action
2. The operating temperature plays an important role in determining the output power of laser.
3. The corrosion may occur at the reflecting plates.
4. Accidental exposure may damage our eyes, since it is invisible (infra-red region) to our eyes.

What are all applications of CO_2 LASER? (2 marks)

Applications:

1. High power CO_2 laser finds applications in material processing, welding, drilling, cutting soldering etc.
2. The low atmospheric attenuation (10.6 μm makes CO_2 laser suitable for open air communication.
3. It is used for remote sensing
4. It is used for treatment of liver and lung diseases.
5. It is mostly used in neuro surgery and general surgery.
6. It is used to perform microsurgery and bloodless operations.

What are all the energy states can be obtained in CO_2 LASER production? (2 marks)

Energy states of CO_2 molecules:

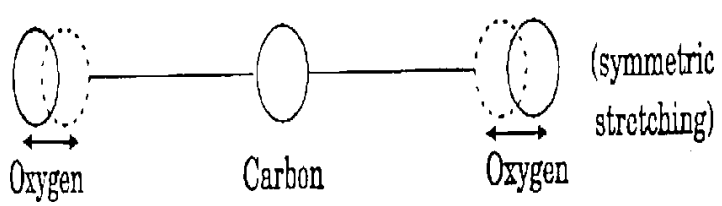
A carbon dioxide molecule has a carbon atom at the center with two oxygen atoms attached, one at both sides. Such a molecule exhibits three independent modes of vibrations. They are:

a) Symmetric stretching mode.
b) Bending mode
c) Asymmetric stretching mode.

Define symmetric stretching in CO_2 LASER (2 marks)

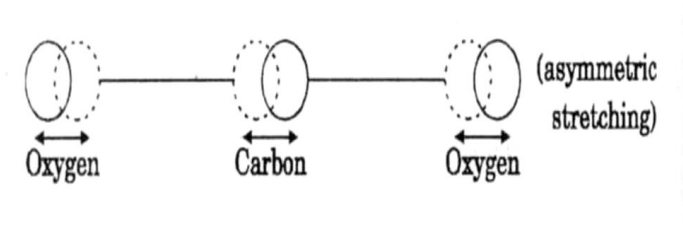
Symmetric Stretching Mode:

In this mode of vibration, carbon atoms are at rest and both oxygen atoms vibrate simultaneously along the axis of the molecule departing or approaching the fixed carbon atoms.



Define Asymmetric stretching in CO_2 LASER (2 marks)

Asymmetric Stretching Mode:

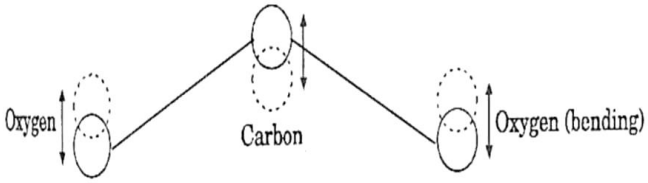


In this mode of vibration, oxygen atoms and carbon atoms vibrate asymmetrically, i.e., oxygen atoms move in one direction while carbon atoms in the other direction.

Define bending mode of vibration in CO_2 LASER (2 marks)

Bending Mode:

In this mode of vibration, oxygen atoms and carbon atoms vibrate perpendicular to molecular axis.



What are major categories of optic fibre? (2 marks)

Optical fibers are classified into three categories

1. Based on material used
2. The number of modes
3. The refractive index profiles

How can you classify optic fibre according to the material, number of modes, refractive index profile (4 or 6 marks)

Based on the type of material used, they are classified into two types

1. Glass fiber:

   Example:
   Core: SiO_2 Cladding: SiO_2
   Core: $GeO_2 – SiO_2$ Cladding: SiO_2
2. Plastic fiber:

   Example:

Core: Polymethyl methacrylate Cladding: Copolymer
Core: Polystyrene Cladding: Methyl methacrylate

Based on the number of modes, they are classified as

1. Single mode fiber
2. Multimode fiber

Based on the refractive index profile, they are classified as

1. Step- index fiber
2. Graded index fiber

Write elaborately about Einstein's theory of matter radiation interaction and A and B coefficients - 8 or 12 marks

MECHANISM OF LIGHT EMISSION:

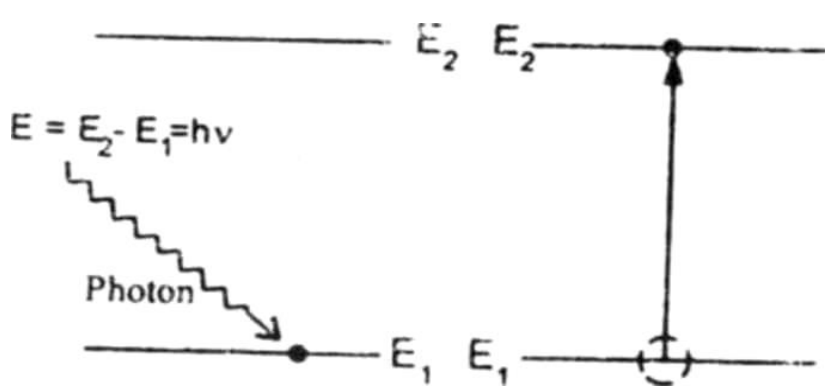
We know that, when light is absorbed by the atoms or molecules, then it goes from the lower energy level (E_1) to the higher energy level (E_2) and during the transition from higher energy level (E_2) to lower energy level (E_1) the light is emitted from the atoms or molecules.

Let us consider an atom exposed to light photons of energy $E_2 — E_1 = h\,v$, three distinct processes take place.

a. Induced Absorption
b. Spontaneous Emission
c. Stimulated Emission

(a) Induced Absorption:

An atom in the lower energy level or ground state energy level E_1 absorbs the incident photon radiation of energy $h\,v$ and goes to the higher energy level or excited level E_2 as shown in figure.



Before Absorption **After Absorption**

This process is called absorption.

If there are many numbers of atoms in the ground state then each atom will absorb the energy from the incident photon and goes to the excited state. Then, the rate of absorption (R_{12}) is proportional to the following factors.

R_{12} = Energy density of incident radiation (ρ_v)

= no of atoms in the ground state (N_1)

$R_{12} \propto \rho_v N_1$

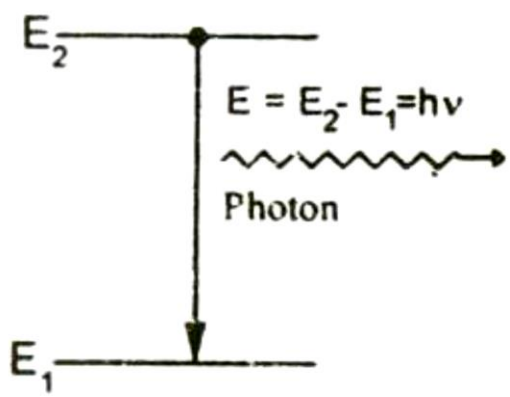
$R_{12} = B_{12} \rho_v N_1$ ----------(1)

Where, B_{12} is a constant which gives the probability of absorption transition per unit time.

Normally, the atoms in the excited state will not stay there for a long period of time, rather it comes to ground state by emitting a photon of energy $E = h\upsilon$ Such an emission takes place by one of the following two methods.

(b) Spontaneous Emission:

The atom in the excited state returns to the ground state by emitting a photon of energy $E = (E_2 - E_1) = h\upsilon$ spontaneously without any external triggering as shown in the figure.



This process is known as spontaneous emission.

Such an emission is random and is independent of incident radiation. If N_1 and N_2 are the numbers of atoms in the ground state (E_1) and excited state (E_2) respectively, then

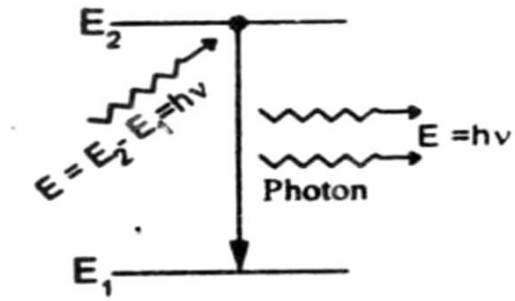
The rate of spontaneous emission R_{21} (Sp) $\propto N_2$

or R_{21} (Sp) = $A_{21} N_2$----------(2)

(c) Stimulated Emission:

The atom in the excited state can also return to the ground state by external triggering or inducement of photon thereby emitting a photon of energy equal to the

energy of the incident photon, known as stimulated emission. Thus, results in two photons of same energy, phase difference and of same directionality as shown.



Therefore, the rate of stimulated emission is given by

R_{21} (St) $\propto \rho_v N_2$

R_{21} (St) = $B_{21} \rho_v N_2$----------(3)

Where B_{21} is constant which gives probability of stimulated emission transition per unit time.

EINSTEIN'S THEORY OF MATTER RADIATION INTERACTION AND A AND B COEFFICIENTS

Einstein's theory of absorption and emission of light by an atom is based on Planck's theory of radiation. Also, under thermal equilibrium, the population of energy levels obeys the Boltzmann distribution law

Under thermal equilibrium
The rate of absorption = the rate of emission

Therefore, $B_{12} \rho_v N_1 = A_{21} N_2 + B_{21} \rho_v N_2$

$$B_{12} \rho_v N_1 - B_{21} \rho_v N_2 = A_{21} N_2$$

$$\rho_v (B_{12} N_1 - B_{21} N_2) = A_{21} N_2$$

$$\text{Therefore, } \rho_v = \frac{A_{21} N_2}{B_{12} N_1 - B_{21} N_2}$$

$$\rho_v = \frac{A_{21}}{B_{12} \frac{N_1}{N_2} - B_{21}} \quad \text{----------(4)}$$

From the Boltzmann distribution law, we get

$$N_1 = N_0 \, e^{-E_1/K_B T}$$

$$\text{Likewise, } N_2 = N_0 \, e^{-E_2/K_B T}$$

Where K_B is Boltzmann constant
T is absolute temperature
N_0 is number of atoms at absolute zero

At equilibrium, we write the ratio of population levels as follows

$$\frac{N_1}{N_2} = e^{(E_2 - E_1)/K_B T}$$

Since $E_2 - E_1 = h\nu$, we have

$$\frac{N_1}{N_2} = e^{h\nu/K_BT} \text{ ----------(5)}$$

Substituting equation (5) in (4) we get

$$\rho_v = \frac{A_{21}}{B_{12}e^{h\nu/K_BT} - 1}$$

$$\rho_v = \frac{A_{21}}{B_{21}} \frac{1}{\left(B_{12}/B_{21}\right)e^{h\nu/K_BT} - 1} \text{ --(6)}$$

This equation has a good agreement with Planck's energy distribution law.

That is $\rho_v = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{h\nu/K_BT} - 1}$ -------(7)

Therefore, comparing equations (6) and (7), we can write

$B_{12} = B_{21} = B$ and $\frac{A_{21}}{B_{21}} = \frac{8\pi h\nu^3}{c^3}$ ----------(8)

Taking $A_{21} = A$

The constants A and B are called as Einstein's coefficients, which accounts for spontaneous and stimulated emission probabilities.

Ratios of magnitudes of stimulated and spontaneous emission rates

From equations (2) and (3) we have

$$\frac{R_{21}(St)}{R_{21}(Sp)} = \frac{B_{21}\rho_v N_2}{A_{21} N_2}$$

$$\frac{R_{21}(St)}{R_{21}(Sp)} = \frac{B_{21}}{A_{21}} \rho_v \text{ ----------(9)}$$

Rearranging equation (6) we get

$$\frac{B_{21}}{A_{21}} \rho_v = \frac{1}{\left(B_{12}/B_{21}\right)e^{h\nu/K_BT} - 1}$$

Since $B_{12} = B_{21}$, we have

$$\frac{1}{e^{h\nu/K_BT} - 1} = \frac{B_{21}}{A_{21}} \rho_v \text{ ----------(10)}$$

Comparing (9) and (10) we get

$$\frac{R_{21}(St)}{R_{21}(Sp)} = \frac{1}{e^{h\nu/K_BT} - 1} = \frac{B_{21}}{A_{21}} \rho_v$$

In simpler way the ratio can be written as

$$R = \frac{B_{21}}{A_{21}} \rho_v$$

Generally spontaneous emission is more predominant in optical region (Ordinary light). To increase the number of coherent photons stimulated emission should dominate over spontaneous emission. To achieve this, an artificial condition called population inversion is necessary.

Write elaborately about CO_2 laser -16 or 12 marks

MOLECULAR GAS LASER

In a molecular gas laser, laser action is achieved by transitions between vibrational and rotational levels of molecules. Its construction is simple and the output of this laser is continuous.

In CO_2 molecular gas laser, transition takes place between the vibrational states of Carbon dioxide molecules.

CO_2 MOLECULAR GAS LASER:

It was the first molecular gas laser developed by Indian born American scientist Prof.C.K.N.Pillai. It is a four-level laser and it operates at 10.6 μm in the far IR region. It is a very efficient laser.

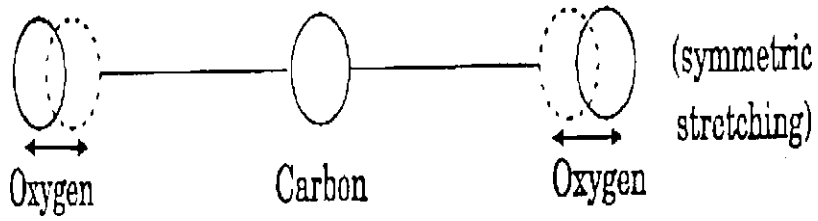
Energy states of CO_2 molecules:

A carbon dioxide molecule has a carbon atom at the center with two oxygen atoms attached, one at both sides. Such a molecule exhibits three independent modes of vibrations. They are:

a) Symmetric stretching mode.
b) Bending mode
c) Asymmetric stretching mode.

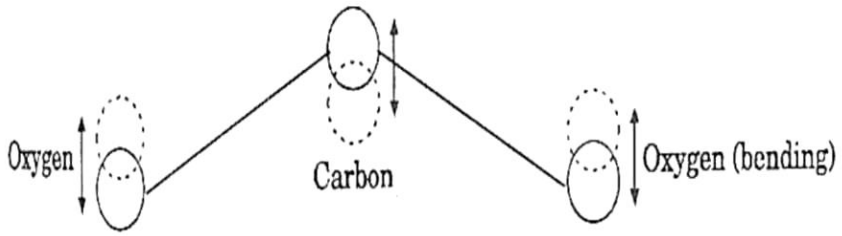
(a) Symmetric Stretching Mode:

In this mode of vibration, carbon atoms are at rest and both oxygen atoms vibrate simultaneously along the axis of the molecule departing or approaching the fixed carbon atoms.

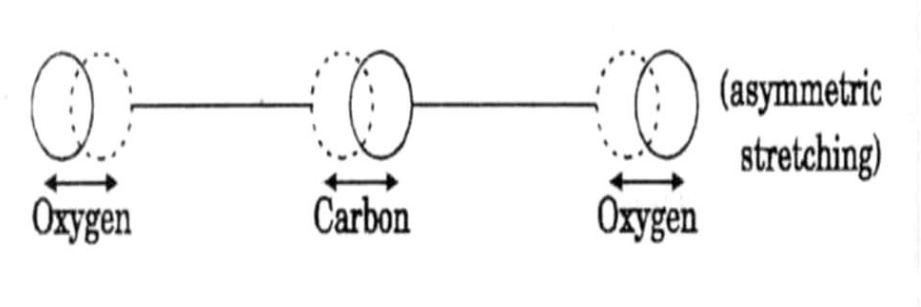


(b) Bending Mode:

In this mode of vibration, oxygen atoms and carbon atoms vibrate perpendicular to molecular axis.



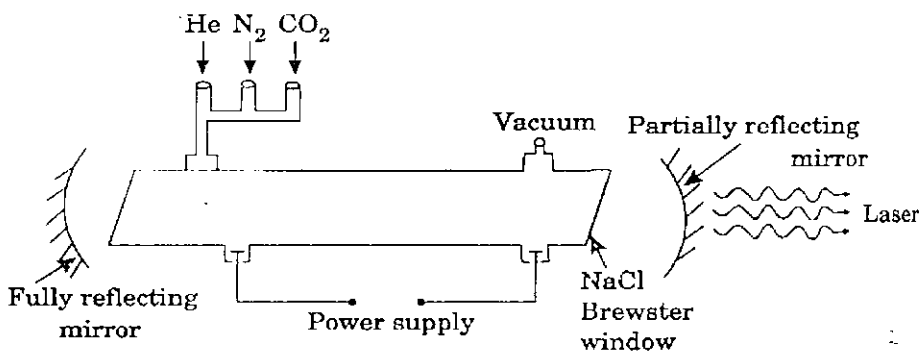
(c) Asymmetric Stretching Mode:



In this mode of vibration, oxygen atoms and carbon atoms vibrate asymmetrically, i.e., oxygen atoms move in one direction while carbon atoms in the other direction.

Principle:

The active medium is a gas mixture of CO_2, N_2 and He. The laser transition takes place between the vibrational states of CO_2 molecules.



Construction:

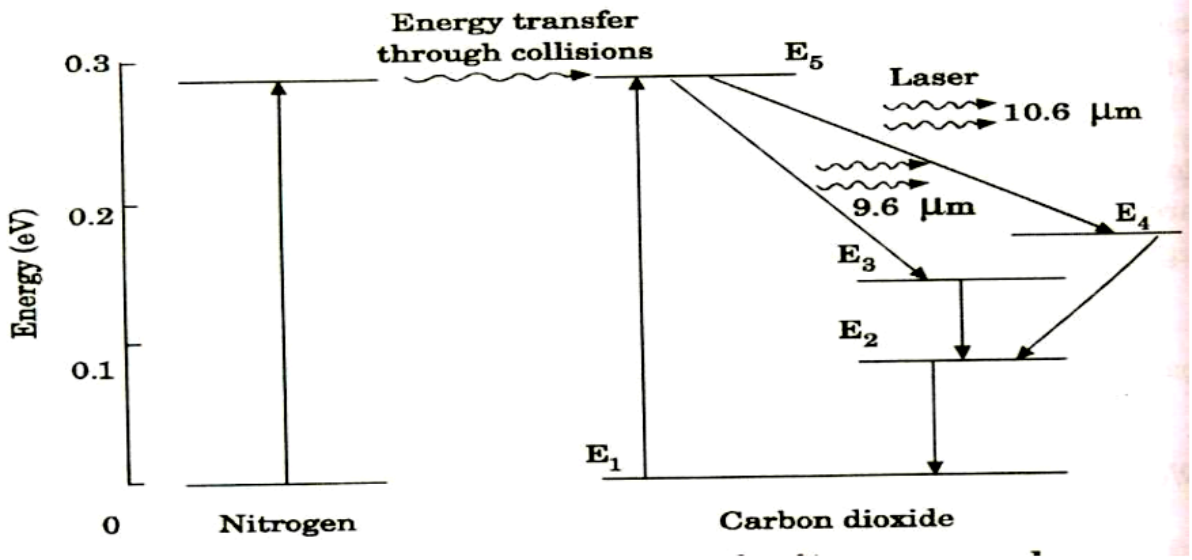
It consists of a quartz tube 5 m long and 2.5 cm in the diameter. This discharge tube is filled with gaseous mixture of CO_2 (active medium), helium and nitrogen with suitable partial pressures.

The terminals of the discharge tubes are connected to a D.C power supply. The ends of the discharge tube are fitted with NaCl Brewster windows so that the laser light generated will be polarized.

Two concave mirrors one fully reflecting and the other partially form an optical resonator.

Working:

Figure shows energy levels of nitrogen and carbon dioxide molecules.



When an electric discharge occurs in the gas, the electrons collide with nitrogen molecules and they are raised to excited states. This process is represented by the equation,

$$N_2 + e^* = N_2^* + e$$

N_2 = Nitrogen molecule in ground state
e^* = Electron with kinetic energy
N_2^* = Nitrogen molecule in excited state
e = Same electron with lesser energy

Now N_2 molecules in the excited state collide with CO_2 atoms in ground state and excite to higher electronic, vibrational and rotational levels. This process is represented by the equation

$$N_2^* + CO_2 = CO_2^* + N_2$$

N_2^* = Nitrogen molecule in excited state.
CO_2 = Carbon dioxide atoms in ground state
CO_2^* = Carbon dioxide atoms in excited state
N_2 = Nitrogen molecule in ground state.

Since the excited level of nitrogen is very close to the E_5 level of CO_2 atom, population in E_5 level increases.

As soon as population inversion is reached, any of the spontaneously emitted photon will trigger laser action in the tube. There are two types of laser transition possible.

Transition E_5 to E_4:

This will produce a laser beam of wavelength 10.6 μm.

Transition E_5 to E_3:

This transition will produce a laser beam of wavelength 9.6 μm. Normally 10.6 μm transition is more intense than 9.6 μm transition. The power output from this laser is 10kW.

Characteristics:

1. **Type:** It is a molecular gas laser.
2. **Active medium:** A mixture of CO_2, N2 and helium or water vapour is used as active medium

3. **Pumping method:** Electrical discharge method is used for Pumping action
4. **Optical resonator:** Two concave mirrors form a resonant cavity
5. **Power output:** The power output from this laser is about 10kW.
6. **Nature of output:** The nature of output may be continuous wave or pulsed wave.
7. **Wavelength of output:** The wavelength of output is 9.6 μm and 10.6 μm.

Advantages:

1. The construction of CO_2 laser is simple
2. The output of this laser is continuous.
3. It has high efficiency
4. It has very high output power.
5. The output power can be increased by extending the length of the gas tube.

Disadvantages:

1. The contamination of oxygen by carbon monoxide will have some effect on laser action
2. The operating temperature plays an important role in determining the output power of laser.
3. The corrosion may occur at the reflecting plates.
4. Accidental exposure may damage our eyes, since it is invisible (infra-red region) to our eyes.

Applications:

1. High power CO_2 laser finds applications in material processing, welding, drilling, cutting soldering etc.
2. The low atmospheric attenuation (10.6 μm makes CO_2 laser suitable for open air communication.
3. It is used for remote sensing
4. It is used for treatment of liver and lung diseases.
5. It is mostly used in neuro surgery and general surgery.
6. It is used to perform microsurgery and bloodless operations.

Write briefly about Fibre optics (2 or 4 marks)

The development of lasers and optical fiber has brought about a revolution in the field of communication systems. Experiments on the propagation of information – carrying light waves through an open atmosphere were conducted. The atmospheric conditions like rain, fog etc affected the efficiency of communication through light waves.

To have efficient communication systems, the information carried by light waves should need a guiding medium through which it can be transmitted safely.

This guiding mechanism is optical fiber. The communication through optical fiber is known as light wave communication or optical communication.

A light beam acting as a carrier wave is capable of carrying more information than that of radio waves and microwaves due to its larger bandwidth.

Currently in most part of the world, fiber optics is used to transmit voice, video and digital data signals using light waves from one place to other place.

Write about construction of fiber optic cable. (4 marks)

The optical fiber is a wave guide.

It is made up of transparent dielectrics (SiO_2), (glass or plastics).

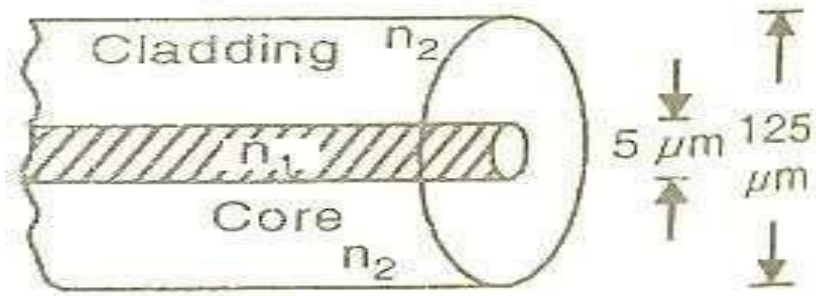
3.2.1Fiber Construction:

It consists of an inner cylinder made of glass or plastic called core. The core has high refractive index n_1.

This core is surrounded by cylindrical shell of glass or plastic called cladding.

The cladding has low refractive index n_2. This cladding is covered by a jacket which is made of polyurethane. It protects the layer from moisture and abrasion.

The light is transmitted through this fiber by total internal reflection. The fiber guides light waves to travel over longer distance without much loss of energy.



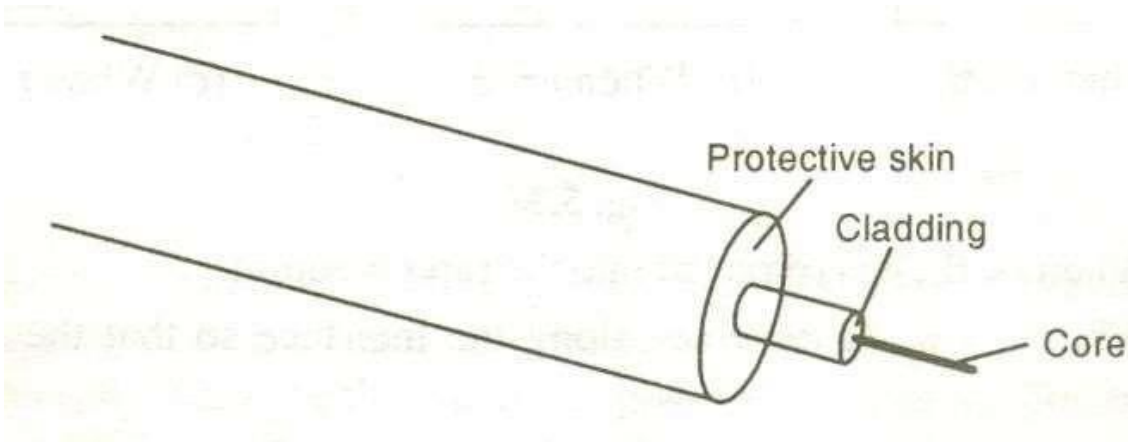
. Core diameters range from 5 to 600μm while cladding diameters vary from 125 to 750μm Core transmits the light waves. The cladding keeps the light waves within the core by ***total internal reflection.***

Define Refractive Index (2 marks)

Refractive index:

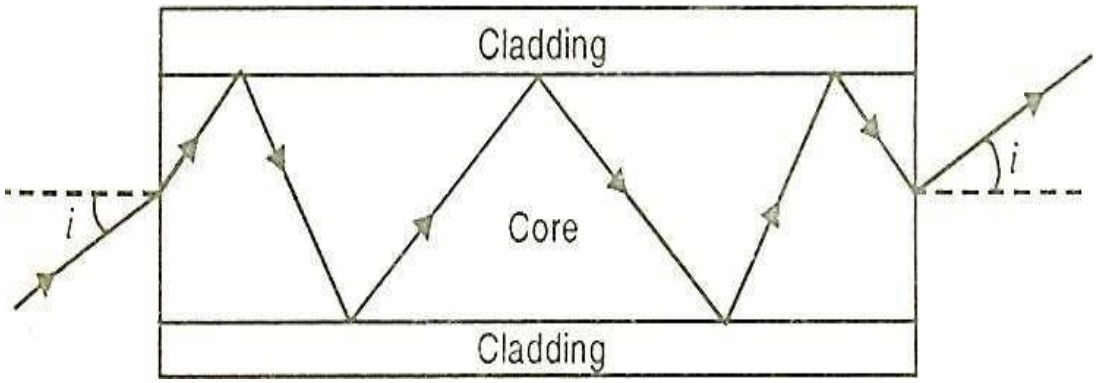
The ***refractive index*** *or* ***index of refraction*** of a substance is a measure of the speed of light in that substance. It is expressed as a ratio of the speed of light in vacuum relative to that in the considered medium.

Explain in detail about propagation of light in an Optical fiber (12 or 16 marks)



Principle of propagation of light in an optical fiber

The light launched inside the core at one end of the fiber propagates to the other end due to total internal reflection at the core and cladding interface.



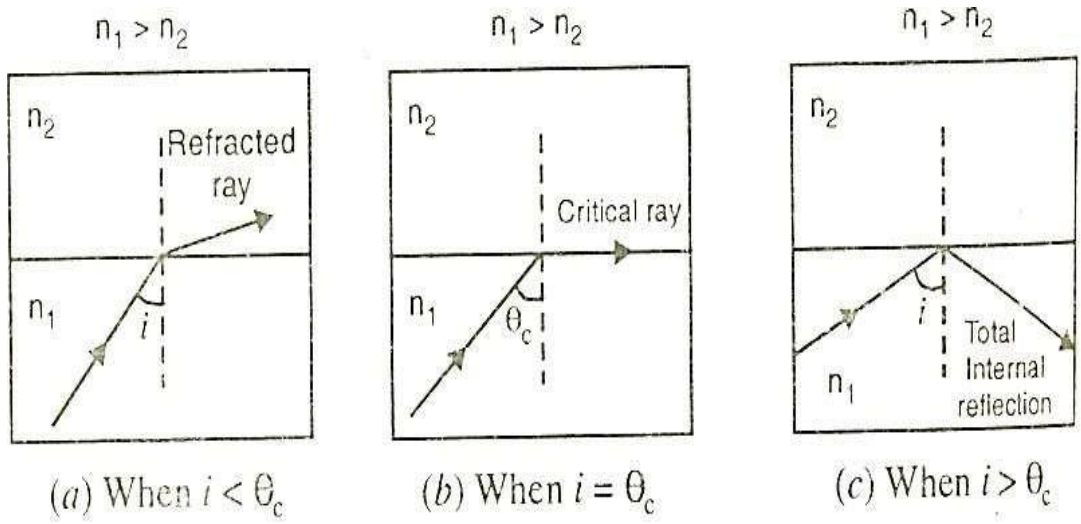
Total internal reflection at the fiber wall can occur only if two conditions are Satisfied.

1. The refractive index of the core material n_1 must be higher than that of the cladding n_2 surrounding it.
2. At the core – cladding interface, the angle of incidence (between the ray and normal to the interface) must be greater than the critical angle defined as

$$Sin\,\theta_c = \frac{n_2}{n_1}$$

Let the light ray travel from core of refractive index n_1 to cladding of refractive index n_2.

Where $n_1 > n_2$.



a) When $i < \theta_c$ it is refracted into rarer medium.

b) When $i = \theta_c$ it traverses along the interface so that angle of refraction is 90˚

c) When $i > \theta_c$ it is totally reflected back into denser medium itself.

When $i = \theta_c$ then by Snell's Law,

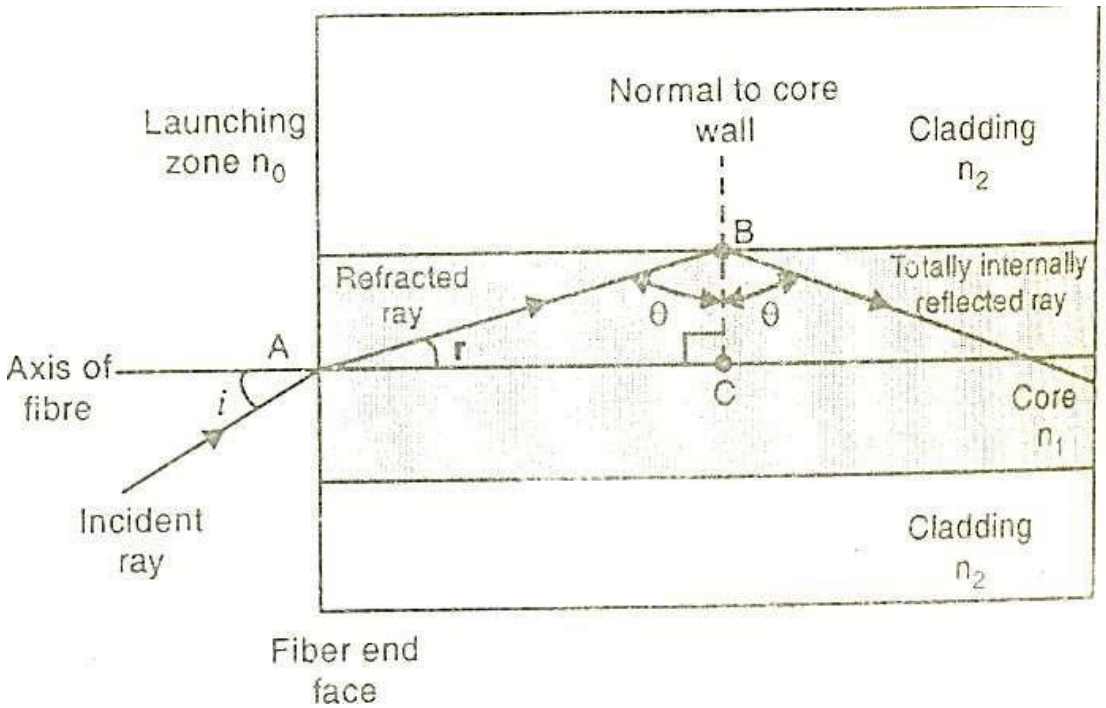
$$n_1 \sin \theta_c = n_2 \sin 90^\circ$$

$$\sin \theta_c = \frac{n_2}{n_1}$$

3.2.3 Propagation of light through fiber

Consider an optical fiber through which the light is being sent. The end at which light enters is called launching end. Let the refractive indices of the core and cladding be n_1 and n_2 respectively; $n_1 > n_2$. Let the refractive index of the medium from which the light is launched be n_0.

Let the light ray enter at an angle 'i' to the axis of the fiber



The ray refracts at an angle r.

The ray strikes the core – cladding interface at an angle θ. If θ is greater than the critical angle θ_c, the ray undergoes total internal reflection at the interface.

Let us now find out up to what maximum value of i at A total internal reflection at B is possible.

In triangle ABC, r = 90 – θ ----------(1)

From Snell's law $\frac{\sin i}{\sin r} = \frac{n_1}{n_0}$ ----------(2)

Sin i = $\frac{n_1}{n_0}$ sin r = $\frac{n_1}{n_0}$ sin (90-θ) = $\frac{n_1}{n_0}$ cos θ

If θ is less than the critical angle θ_c, the ray will be lost by refraction. Therefore, limiting value for containing the beam inside the core by total internal reflection is θ_c. Let i_m me the maximum possible angle of incidence at the fiber end face A for which $\theta = \theta_c$.

Sin i_m = $\frac{n_1}{n_0}$ cos θ_c ----------(3)

But sin θ_c = $\frac{n_2}{n_1}$

$$\cos\theta_c = \sqrt{1 - sin^2\theta_c} = \sqrt{1 - \left[\frac{n_2}{n_1}\right]^2}$$

$$= \sqrt{\frac{n_1^2 - n_2^2}{n_1^2}}$$

or $$i_m = \sin^{-1}\frac{n_1}{n_0}\left\{\frac{\sqrt{n_1^2 - n_2^2}}{n_1}\right\} = \sin^{-1}\left\{\frac{\sqrt{n_1^2 - n_2^2}}{n_0}\right\}$$

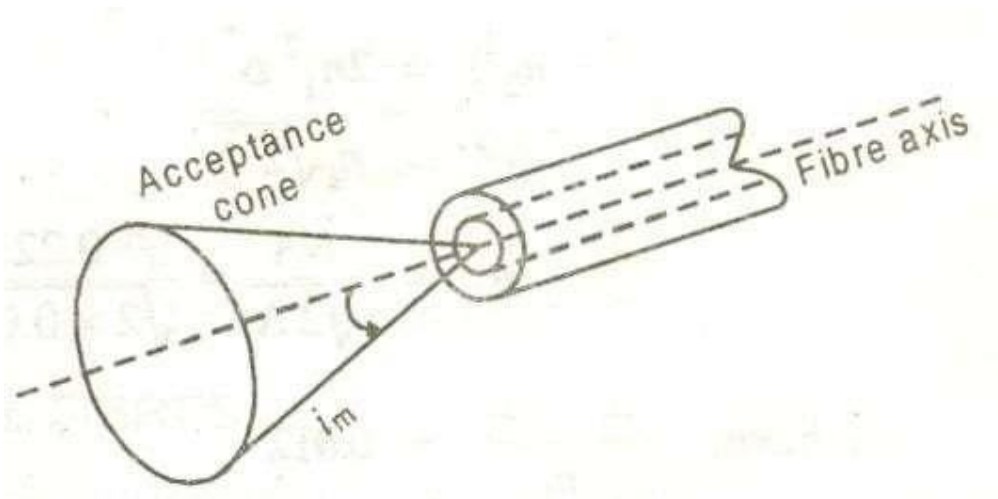
This angle i_m is called the acceptance angle of the fiber.

Definition: Acceptance angle is defined as the maximum angle that a light ray can have relative to the axis of the fiber and propagate down the fiber.

Or the maximum angle at or below which the light can suffer Total Internal Reflection is called acceptance angle.

Acceptance cone:

An optical fiber accepts only those rays which are incident within a cone having a semi angle i_m.



The light rays contained within the cone having a full angle $2i_m$ are accepted and transmitted along the fiber. Therefore, the cone is called the acceptance cone.

Light incident at an angle beyond i_m refracts through the cladding and the corresponding optical energy is lost. It is obvious that the larger the diameter of the core, the larger the acceptance angle.

Write about Numerical aperture (2 marks)

Numerical Aperture:

Definition:

The numerical aperture (NA) is defined as the sine of the acceptance angle.

$$NA = \sin i_m$$

$$\text{NA} = \left\{ \frac{\sqrt{n_1^2 - n_2^2}}{n_0} \right\}$$

Numerical aperture determines the light gathering ability of the fiber. It is a

measure of amount of light that can be accepted by a fiber. NA depends only on the refractive indices of the core and cladding materials. A large NA implies that a fiber will accept large amount of light from the source.

Write about Fractional index change (2 marks)

Fractional Index change:

It is the ratio of refractive index difference in core and cladding to the refractive index of the core.

$$\Delta = \frac{n_1 - n_2}{n_1}$$

Relation between NA and Δ

$n_1 \Delta = n_1 - n_2$

We know $NA = \sqrt{n_1^2 - n_2^2}$

NA = $\sqrt{(n_1 - n_2)(n_1 + n_2)}$

Substituting the value of $n_1 - n_2$ we have

$NA = \sqrt{\Delta n_1 (n_1 + n_2)}$

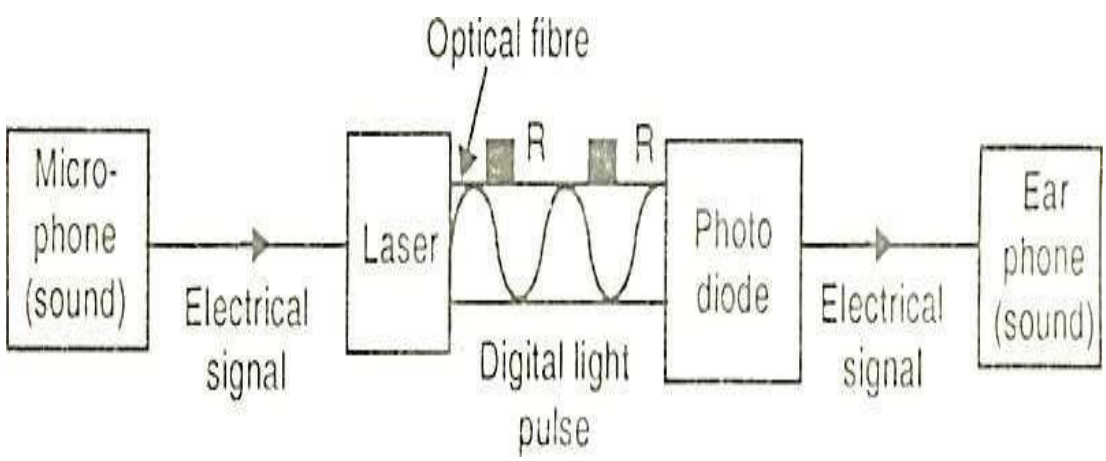
If $n_1 = n_2$ then NA $= \sqrt{2 n_1^2 \Delta}$

NA = $n_1 \sqrt{2\Delta}$

Explain working of optical fiber as an waveguide. (6 or 8 marks)

OPTICAL FIBER AS AN OPTICAL WAVEGUIDE

Optical fibers are used as dielectric waveguides for electromagnetic signals of optical frequencies. Figure shows the block diagram of transmission of sound along the optical fiber and conversion again to sound at the other end.



i. Sound is first converted into electrical signal by a microphone.
ii. The electrical signals modulate the intensity of light from laser.
iii. Then the information is carried along the fiber in a digital form.

Boosters or repeaters are placed at a distance of about 50km of cable to make

up the signal loses occurring due to scattering and absorption.

iv. At the receiving place, a photodiode converts the digital light pulses into corresponding electrical signals.

v. The electrical signals are then converted into sound by an earphone (receiver) Time division multiplexing system is used to transmit many thousands of telephone cells through a single optical fiber with the use of digital pulses.

Explain in detail about fiber optic communication system (8 or 12 marks)

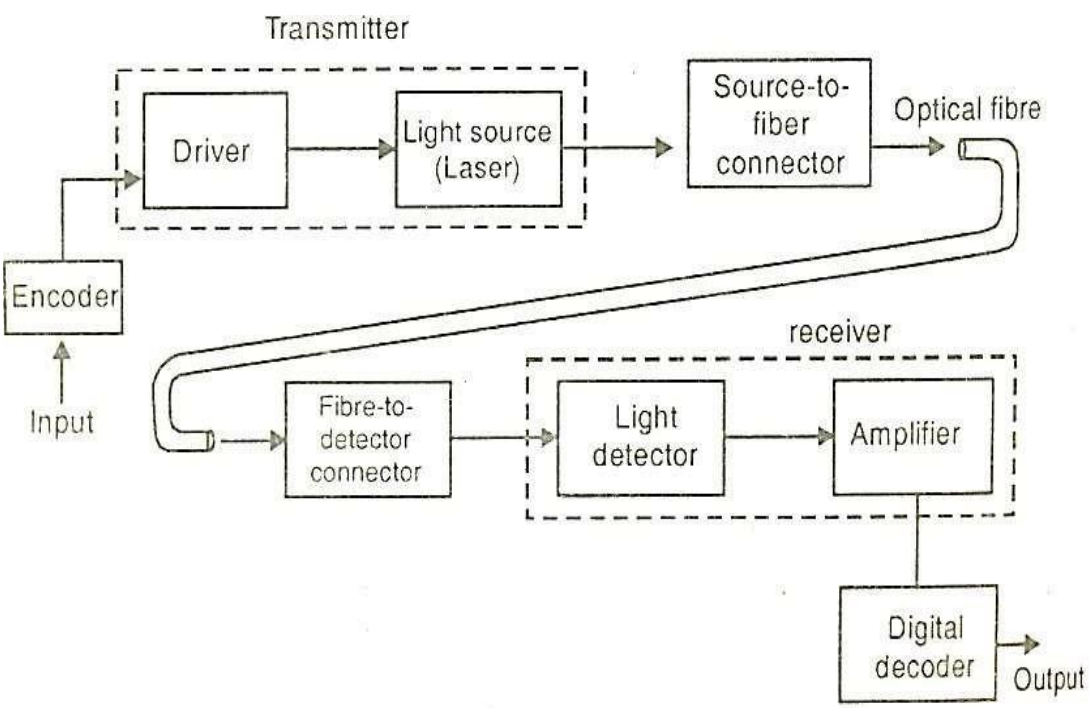
3.1.9 THE FIBER OPTIC COMMUNICATION SYSTEM

Figure shows the schematic diagram of a fiber optic communication system. The major components of an optical fiber communication system are

1. The optical transmitter
2. The optical fiber
3. The optical receiver

PRINCIPLE:

Basically, a fiber optic system converts an electrical signal to an infrared light signal. This signal is transmitted through an optical fiber. At the end of the optical fiber, it is reconverted into an electric signal.



Working:

1. Encoder encodes the information in the binary sequence zeros and ones.

a. Encoder is an electric circuit where in the information is encoded into binary sequences of zeros and one. In the light wave transmitter each 'one' corresponds to an electrical pulse and 'zero' corresponds to an absence of a pulse. These electrical pulses are used to turn a light

source on and off very rapidly. The driver converts the incoming electrical signal into a form that will operate with the light source.

2. These electrical pulses are used to turn a light source on and off rapidly.

3. The optical fiber acts as a wave guide and transmits the optical pulses towards the receiver, by the principle of total internal reflection.

4. The light detector receives the optical pulses and converts them into electrical pulses. These signals are amplified by the amplifier.

5. The amplified signals are decoded by the decoder.

APPLICATIONS:

1. Extremely wide bandwidth.
   a. Optical frequencies are very large (10^{15} Hz) as compared to radio frequencies (10^{6}Hz) and microwave frequencies (10^{10} Hz). The rate at which information can be transmitted is directly related to signal frequency. Therefore, a transmission system that operates at the frequency of light can transmit information at a higher rate than systems that operate at radio frequencies or microwave frequencies.

2. Lack of cross talk between parallel fibers.

   a. There is virtually no signal leakage from fibers. Hence, cross-talks between neighboring fibers are almost absent. This is quite frequent in conventional metallic system

3. Immunity to inductive interference

   a. Since optical fibers are not metallic, they do not pick up electromagnetic waves. The result is noise free transmission i.e., fiber optic cables are immune to interference caused by lighting or other electromagnetic equipment

4. Smaller diameter and light weight cable

   a. Optical fibers, because of their light weight and flexibility, can be handled more easily than copper cables.

5. Signal security

   a. The transmitted signal through the fibers does not radiate. Further the signal cannot be tapped from a fiber in an easy manner. Therefore, optical fiber communication provides a hundred percent signal security hence this system is highly suited to secure communications in defense communication networks.

MODULE IV - ELECTRONICS AND OPTOELECTRONIC DEVICES

Write the properties of semiconductors. (4 or 6 marks)

Properties of semiconductor

1. The resistivity of semiconductors lies between a conductor and an Insulator. (It various from 10^{-4} to 0.5 Ω m).

2. At 0 K it behaves as insulator.

3. They have negative temperature Coefficient of resistance. (when the temperature is increased large number of charge carriers are produced due to breaking of covalent bonds and hence these electrons move freely and gives rise to conductivity)

4. In semiconductors, both electrons and holes are charge carriers.

$$\sigma = \sigma_e + \sigma_h$$

Where σ_e and σ_h is the conductivities due to electron and holes.

5. If we increase the temperature of semiconductor, its electrical conductivity also increases.

6. They have an empty conduction band and almost filled valence band 0 K.

7. They are formed by a covalent bond.

8. They have small energy gap (or) band gap.

Explain elaborately about the intrinsic and extrinsic semiconductors (8 or 12 marks)

CLASSIFICATION OF SEMICONDUCTORS

Based on the purity, semiconductor is classified into two types.

1. Intrinsic semiconductors or Elemental semiconductors

2. Extrinsic semiconductors or Compound semiconductors

Intrinsic semiconductors

A pure semiconductor without any impurities is known as an intrinsic semiconductor. Example: Ge, Si (In the form of pure)

These are made from single element. They also known as indirect band gap semiconductors. In which the recombination of free electron from the conduction band with the hole in the valence band takes place via traps. During recombination phonons [lattice vibrations] are produced and they heat the crystal lattice (position of the atom). These are the IV group element in the periodic table.

Compound Semiconductors

The Compound Semiconductor is a semiconductor compound composed of elements from two or more different groups of the periodic table. They also known as direct band gap semiconductors.

i.e., **III – V group, II – VI group and IV – VI group.**

Here the recombination electron and hole take place directly, during recombination photons are emitted.

Example: GaAs, GaP,

Based on the type of impurity they are classified into

i) N-type semiconductor

ii) P-type semiconductor

N-type Semiconductor (Donor impurity)

1. Donor impurity means it donates the electron to the semiconductor materials.
2. Pentavalent atoms (five valence electrons in their outer most orbit) are called as donor impurities. ***Example :*** *Phosphorous, Arsenic and Antimony.*
3. When a pentavalent atom is added with tetravalent atoms (Ge and Si), the covalent bond is formed and one element is left free. Thus one impurity atom is surrounded by four Ge or Si atoms.
4. The remaining electron is loosely bound to the parent impurityatom is detached from the impurity atom by supplying ionization energy.
5. Each impurityatom donates one free electron. Thus this type of semiconductor is called as N-type semiconductor.
6. The donor atoms form the new energy level called donor energy level (E_D) very near and below the conduction band.
7. At room temperature, almost all the excess electrons donated by the donor atoms are raised to the conduction band as majority charge carriers (free electrons) in N-type semiconductor.

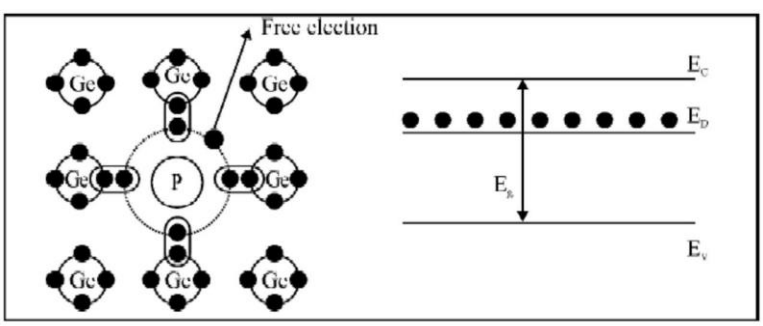


Fig. 2.6 N type semiconductor

2.10.1 P – type Semiconductor (Acceptor Impurities)

1. Acceptor impurity means it ready to accept an electron to form the covalent bond in semiconductormaterials.
2. Trivalent atoms (three valence electrons in their outer most orbits) are called as acceptor impurities. ***Example:*** *Aluminum, Gallium, Boron and Indium.*
3. When a trivalent atom is added with tetravalent atoms (Ge or Si), the covalent bond is formed and there is one vacancy (hole) for one electron in one of the covalent bonds, thereby one impurity atom is surrounded by four Ge or Si atoms.
4. Thus each impurity atom hole is ready to accept an electron. Thus this type of semiconductor is called P-type semiconductor.
5. The Acceptor atoms form the new energy level called acceptor energy level (E_A) very near and above the valence band.
6. When a small amount of energy is applied, the electrons from valence band are moved to the acceptor level and creating holes in the valence band.
7. These valence band holes are the majority charge carriers in the P-type semiconductor material.

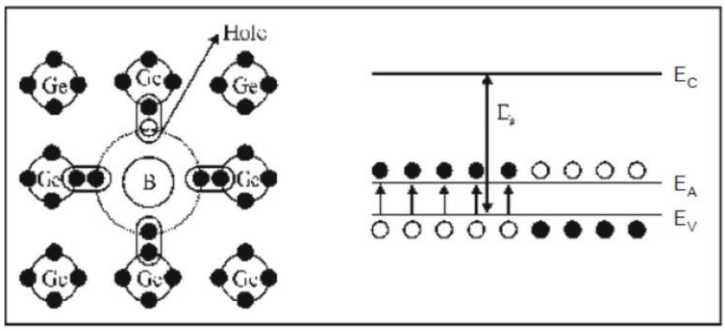


Fig. 2.7 P type semiconductor

Differenciate between N-type and P-type semiconductor (2 marks)

Difference between N-type and P-type semiconductor

S.No	N-type	P-type
1	It is donor type	It is acceptor type
2	Impurity atom is pentavalent	Impurity atom is trivalent
3	Donor level lies close to the bottom of the conduction band	Acceptor level lies close to the top of the valence band.

Difference between Elemental and Compound Semiconductors (8 or 12 marks)

Difference between Elemental and Compound Semiconductors

S.No	Elemental semiconductors	Compound semiconductors
1	These are made from single element.	These are made from compound (mixed) element.
2	These are made from IV group and VI group elements	These are made from III and V [or] II and VI elements.

3	These are called as indirect band gap semiconductor (electron-hole recombination takes place through traps)	These are called as direct band gap semiconductor (electron-hole recombination takes place directly)
4	Heat is produced in the recombination	Photons are emitted during Recombination
5	Life time of charge carriers is more due to indirect recombination	Life time of charge carriers is less due to direct recombination.
6	Current amplification is more	Current amplification is less.
7	These are used for making diodes, transistor, etc.	These are used for making LED, laser diodes, etc.
8	Example: Ge, Si	Example: GaAs, GaP, CdS, MgO

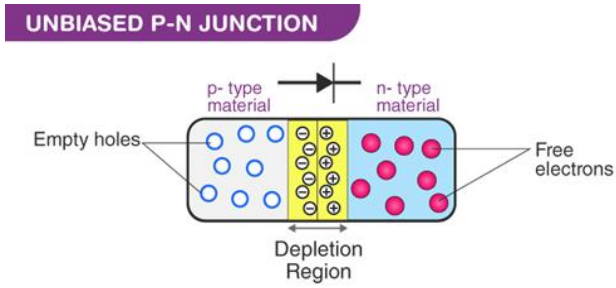
What is P-N Junction? (2 Marks)

Definition: *A P-N junction is an interface or a boundary between two semiconductor material types, namely the p-type and the n-type, inside a semiconductor.*

In a semiconductor, the P-N junction is created by the method of doping. The p-side or the positive side of the semiconductor has an excess of holes, and the n-side or the negative side has an excess of electrons. The process of doping is explained in further detail in the next section.

Write in detail characteristics of P-N Junction diode. (16 MARKS)

As we know, if we use different semiconductor materials to make a P-N junction, there will be a grain boundary that would inhibit the movement of electrons from one side to the other by scattering the electrons and holes and thus, we use the process of doping. We will understand the process of doping with the help of this example. Let us consider a thin p-type silicon semiconductor sheet. If we add a small amount of pentavalent impurity to this, a part of the p-type Si will get converted to n-type silicon. This sheet will now contain both the p-type region and the n-type region and a junction between these two regions. The processes that follow after forming a P-N junction are of two types – diffusion and drift. There is a difference in the concentration of holes and electrons at the two sides of a junction. The holes from the p-side diffuse to the n-side, and the electrons from the n-side diffuse to the p-side. These give rise to a diffusion current across the junction.



Also, when an electron diffuses from the n-side to the p-side, an ionised donor is left behind on the n-side, which is immobile. As the process goes on, a layer of positive charge is developed on the n-side of the junction. Similarly, when a hole goes from the p-side to the n-side, an ionized acceptor is

left behind on the p-side, resulting in the formation of a layer of negative charges in the p-side of the junction. This region of positive charge and negative charge on either side of the junction is termed as the depletion region. Due to this positive space charge region on either side of the junction, an electric field with the direction from a positive charge towards the negative charge is developed. Due to this electric field, an electron on the p-side of the junction moves to the n-side of the junction. This motion is termed the drift. Here, we see that the direction of the drift current is opposite to that of the diffusion current.

Biasing Conditions for the P-N Junction Diode

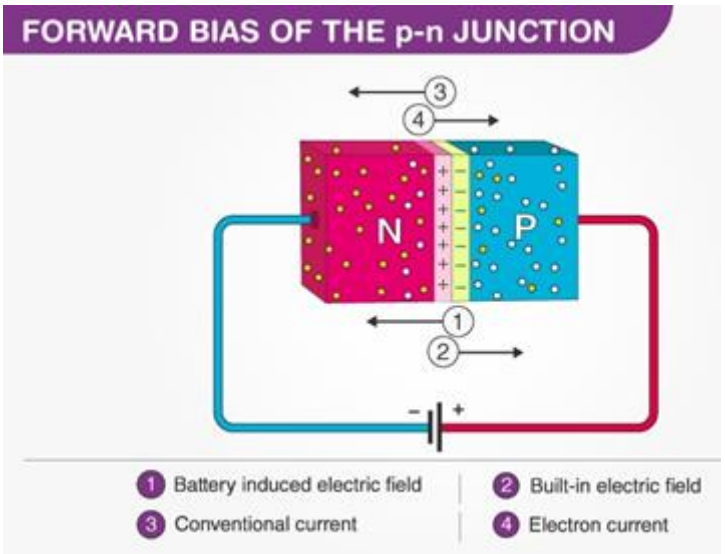
There are two operating regions in the P-N junction diode:

- P-type
- N-type

There are three biasing conditions for the P-N junction diode, and this is based on the voltage applied:

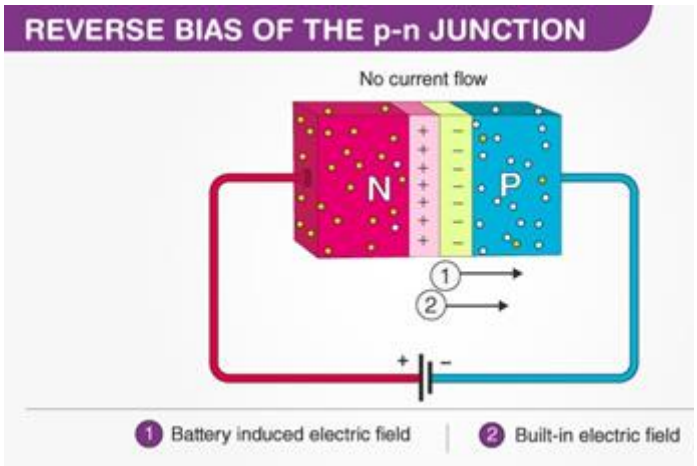
- Zero bias: No external voltage is applied to the P-N junction diode.
- Forward bias: The positive terminal of the voltage potential is connected to the p-type while the negative terminal is connected to the n-type.
- Reverse bias: The negative terminal of the voltage potential is connected to the p-type and the positive is connected to the n-type.

Forward Bias



When the p-type is connected to the battery's positive terminal and the n-type to the negative terminal, then the P-N junction is said to be forward-biased. When the P-N junction is forward biased, the built-in electric field at the P-N junction and the applied electric field are in opposite directions. When both the electric fields add up, the resultant electric field has a magnitude lesser than the built-in electric field. This results in a less resistive and thinner depletion region. The depletion region's resistance becomes negligible when the applied voltage is large. In silicon, at the voltage of 0.6 V, the resistance of the depletion region becomes completely negligible, and the current flows across it unimpeded.

Reverse Bias



When the p-type is connected to the battery's negative terminal and the n-type is connected to the positive side, the P-N junction is reverse biased. In this case, the built-in electric field and the applied electric field are in the same direction. When the two fields are added, the resultant electric field is in the same direction as the built-in electric field, creating a more resistive, thicker depletion region. The depletion region becomes more resistive and thicker if the applied voltage becomes larger.

P-N Junction Formula

The formula used in the P-N junction depends upon the built-in potential difference created by the electric field is given as:

$$E_o = V_T \ln \left[\frac{N_d \cdot N_A}{n_i^2}\right]$$

Where,

- E_0 is the zero bias junction voltage
- V_T is the thermal voltage of 26mV at room temperature
- N_D and N_A are the impurity concentrations
- n_i is the intrinsic concentration.

Explain in detail about Biasing in diode (8 or 12 marks)

Biasing:

The way of connecting emf source to PN junction is known as biasing.

Types of biasing:

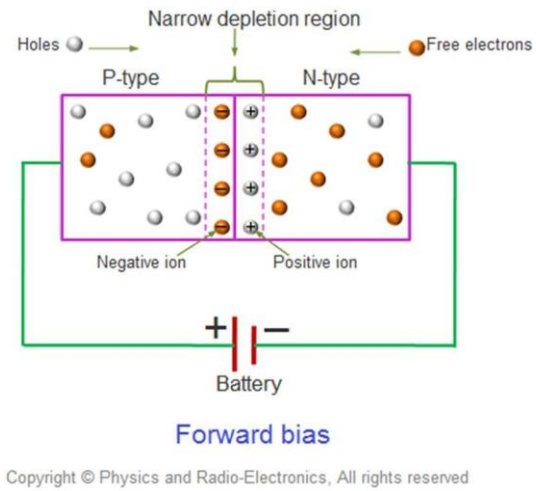
(i) Forward biasing

(i) Reverse biasing

(i) Forward biasing

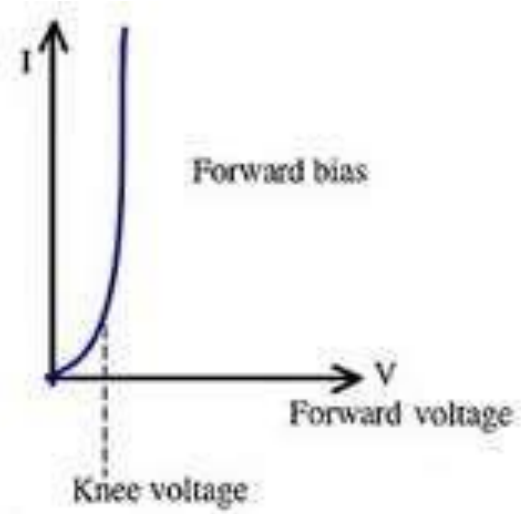
Positive terminal of battery is connected to p-type semiconductor

Negative terminal of battery is connected to n-type semiconductor



In forward biasing the thickness of depletion layer decreases

- More majority charge carriers will be allowed to flow across the junction.
- It means current flow primarily due to majority charge carriers
- The strength of current is in order of mA
- The resistance offered is 10-25 Ω



Knee voltage or Cut in voltage

The voltage from which the current starts to increase rapidly.

The current is given by $I = I_S [e^{eV/kT} - 1]$

Where I_S – saturation current

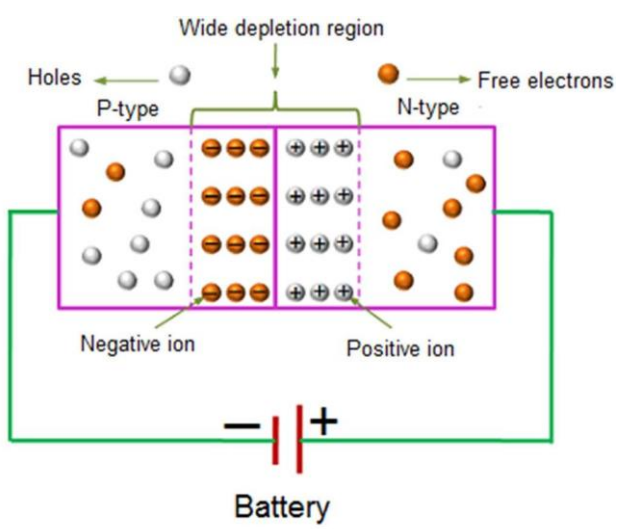
e – Charge of electron

k – Boltzmann's constant

Reverse Biasing:

Positive terminal of the battery is connected to N type semiconductor

Negative terminal of the battery is connected to P type semiconductor.



Reverse bias

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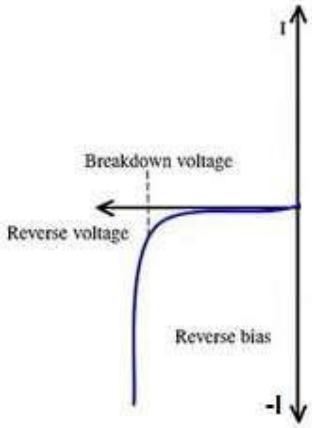
In reverse biasing, the thickness of depletion layer increases.

The reverse bias supports the potential barrier. No current flow across the junction due to the diffusion of majority carriers.

A very small reverse current may exist in the circuit due to movement of minority carriers across the junction.

The strength of current is in order of **μ**A

The resistance offered is approximately = $10^5\ \Omega$



Breakdown Voltage:

Reverse voltage at which breakdown of semiconductor

For Ge = 25V

For Si = 35 V

Applications:

1. It is used in clipping circuits as wave shaping circuits in computers, radios, radars etc.
2. It is used as switches in digital logic designs.
3. It is used in detector and demodulator circuits.
4. It is used in clamping circuits in TV receivers as well as voltage multipliers.
5. It is used as rectifiers in DC power supply manufacturing.

The device having same P-N material but different doping concentrations is known as homo junction. This will have many applications as mentioned below.

1. Used as zener diode in peak clippers, voltage regulators and for switching.
2. Used as varactor diode for tuning applications in TV and radio receivers.
3. Used as PIN photo diode in optical communication system.
4. Used as tunnel diode as oscillator at microwave frequencies.
5. Used as Laser diodes for optical communication.

6. Used as LEDs in various display panels.

Explain in detail about LED. (12 or 16 marks)

LIGHT EMITTING DIODES:

It is a semiconductor p-n junction diode which emits light when it is forward biased.

Principle:

The electrons injected into the p- region make a direct downward transition from the conduction band into valence band and they recombine with holes and emit photons of energy E_g.

We know that the forbidden gap energy is given by

$$E_g = h\nu \quad \text{----------} \quad (1)$$

Where h = Planck's constant

ν = frequency of the emitted radiation.

But we know $\nu = \frac{c}{\lambda}$ ---------- (2)

Substituting (2) in (1)

$$E_g = h\frac{c}{\lambda}$$

Hence, the wavelength of the emitted photon is given by relation

$$\lambda = h\frac{c}{E_g}$$

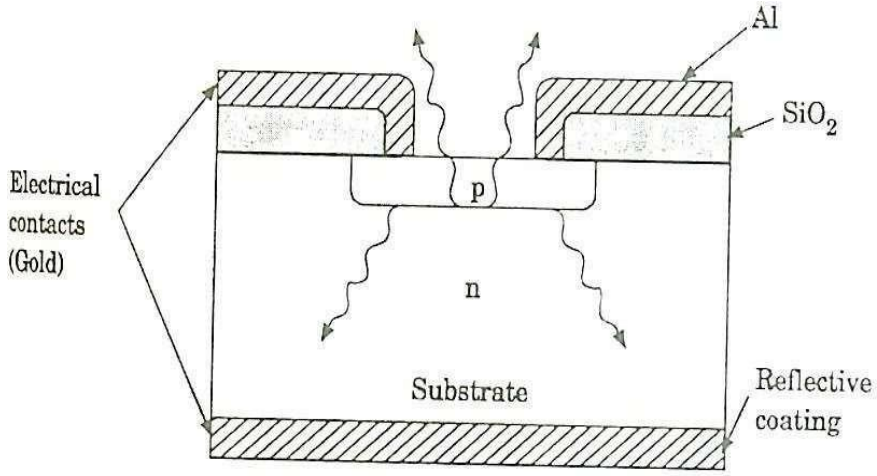
Therefore, the wavelength of the light emitted purely depends on the band gap energy.

Construction:

Figure shows crossectional view of a LED.

A n- type layer is grown on a substrate and a p- type layer is deposited on it by diffusion. Since carrier recombination takes place in the p-layer, it is deposited upper most.

For maximum light emission, a metal film anode is deposited at the outer edges of the p-layer. The bottom of the substrate is coated with metal (gold) film for reflecting most of the light surface of the device and also to provide connection with n- type layer.

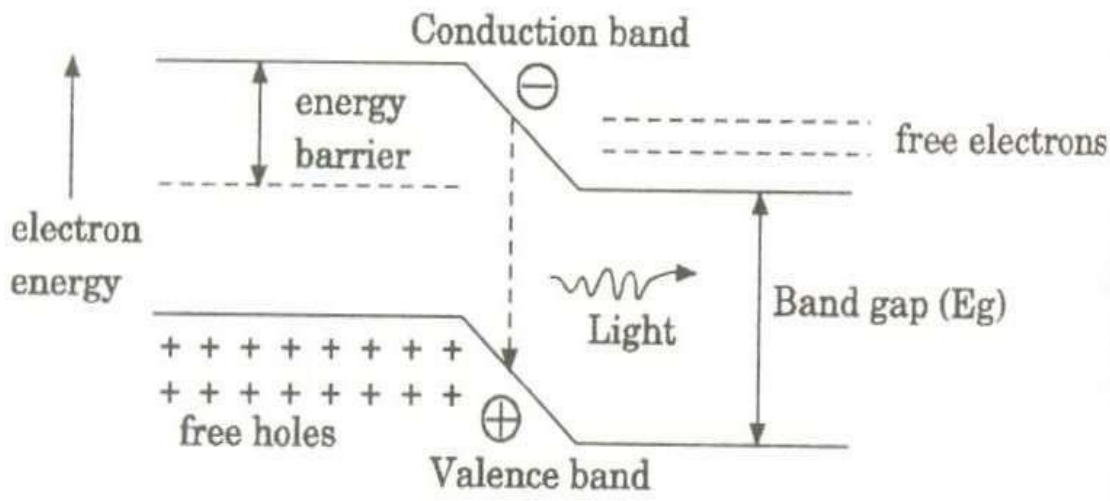


Working:

When the p –n junction diode is forward biased, the barrier width is reduced, raising the potential energy on the n-side and lowering that on the p-side.

The free electrons and holes have sufficient energy to move into the junction region. If a free electron meets a hole, it recombines with each other resulting in the release of a photon

Thus light radiation of the LED is caused by the recombination of holes and electrons that are injected into the junction by forward bias voltage.



Advantages of LED:

1. Light output is proportional to the current. Hence, the light intensity of LEDs can be controlled easily by varying the current flow.
2. LEDs are rugged and therefore withstand shocks and vibrations.
3. Varieties of LEDs are available which emit in different colours like red, green, yellow etc.
4. It has long life time and high degree of reliability.
5. It has low drive voltage and low noise.

6. It is easily interfaced to digital logic circuits.
7. It can be operated over a wide range of temperatures.

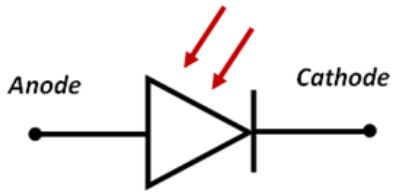
Disadvantages of LED:

1. It requires high power.
2. Its preparation cost is high.
3. LED is not suitable for large area display because of its high cost.
4. It cannot be used for illumination purposes.

Explain in detail about Photo diode. (12 or 16 marks)

Photo diode is a semiconductor device which converts light energy into electric current. When it is exposed to light , it absorbs the light and forms the electron hole pairs, thus the current is generated. It is also called as photo detector or light detector and photo sensor or light sensor.

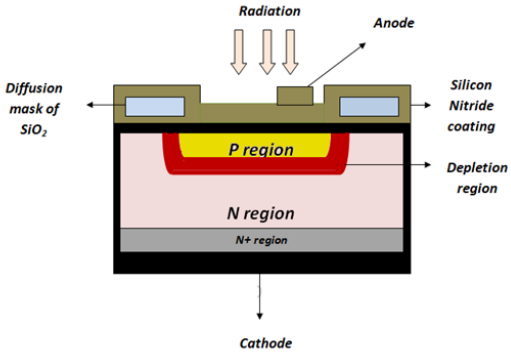
Symbol of photo diode:



Symbol of photo diode

The symbol of photo diode is similar to the PN junction diode. The difference is the arrow pointing towards the diode.

Construction of photo diode:



Construction of photo diode

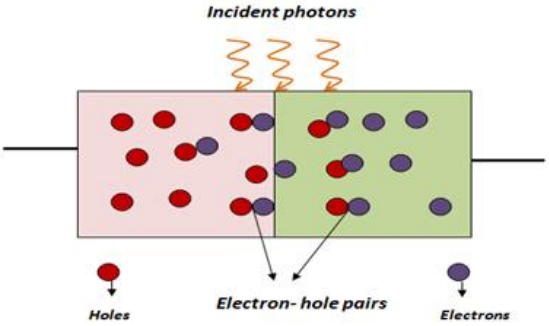
Photodiodes are formed from single crystal silicon wafers. It consists of two layers. P type layer is formed above the N type layer. Depletion layer is formed at the junction of P region and N region. It has a small transparent window which allows the light to strike the PN Junction.

The non active surface is coated with silicon di oxide SiO2 .The active region where the device is exposed to radiation is coated with anti reflection material so that maximum light energy is converted into current.

Working of photo diode:

When the photo diode is reversed biased the depletion layer near the PN junction increases.

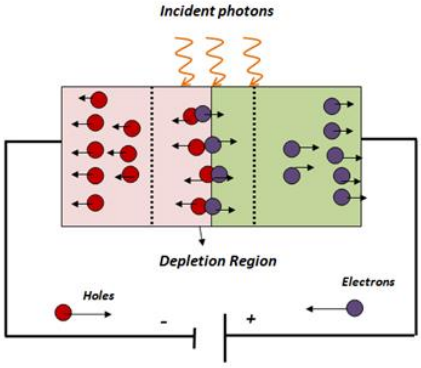
- **Photons create electron-hole pairs:**



Photons create electron-hole pairs

When the device is exposed to radiation or illuminated the photons are absorbed by the diode. The electrons in the valence band get energized and they jump to the conduction band leaving hole in the valence band. Thus electron hole pair is produced at the junction.

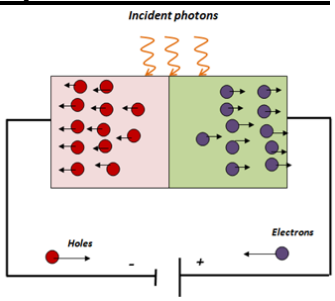
- **Holes and electrons attracted by reverse bias:**



Holes and electrons attracted by reverse bias

When the photo diode is reverse biased, the electrons are attracted towards the positive terminal and the holes are attracted towards negative terminal.

- **Holes and electrons forms the photo electric current:**



Holes and electrons forms the photo current

When the reverse biased voltage is increased the electron hole pair breaks and holes move toward the negative terminal and the electrons move towards positive terminal which results in high reverse current. This is called photo current.

Modes of operation of photo diode:

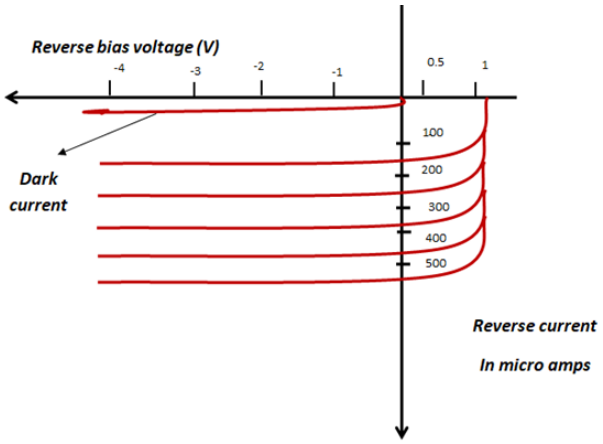
- **Photo conductive**

When the reverse bias is applied the electron hole combination takes place thus the depletion layer is formed. Thus the diodes become conductive. The width of the depletion layer is varied by changing the reverse voltage.

- **Photo- voltaic mode:**

This is called Zero bias mode. No external voltage is applied to the device. When the device is exposed to the radiation current is produced due to minority carriers.

V-I characteristics of photo diode:



V-I Characteristics of photo diode

The dark current is due to minority carriers when the device is not exposed to light. In the above figure we can observe that between each curve there is equal spacing. This is because increase in current which is directly proportional to the luminous flux.

Advantages:

- Low noise
- Highly sensitive to noise
- It operates at high frequencies
- Long lifetime
- High voltage is not required
- It operates very fast

Disadvantages:

- The active area is very small
- Temperature stability is less, when the temperature increases the dark current also increases
- Change in current is very small
- Should not exceed the working temperature limit

Application of Photodiode:

- Used in counters and switching circuits.
- Used widely in optical communication system.
- Used in detection of both visible and invisible light rays.
- Used in safety electronics like fire and smoke detector .
- Used as photo sensors in camera.
- Used in medical applications like pulsed oximeters, in instruments which analyze sample.

Explain about transistor working (12 or 16 marks)

TRANSISTOR

Transistor is known as Bipolar junction transistor. It consists of three main regions.

Emitter (E):

It provides majority carriers by which current flows in the transistor.

Hence the emitter semiconductor is heavily doped.

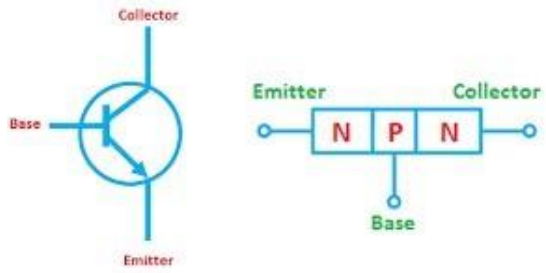
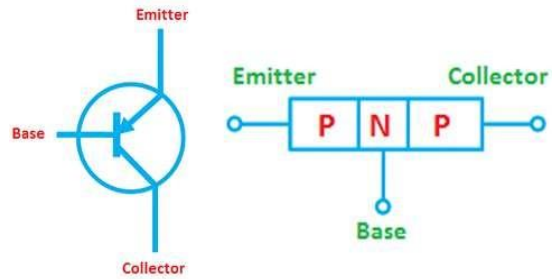
Base (B):

The base semiconductor is lightly doped and very thin

Collector (C):

The size of collector region is larger than emitter and base.

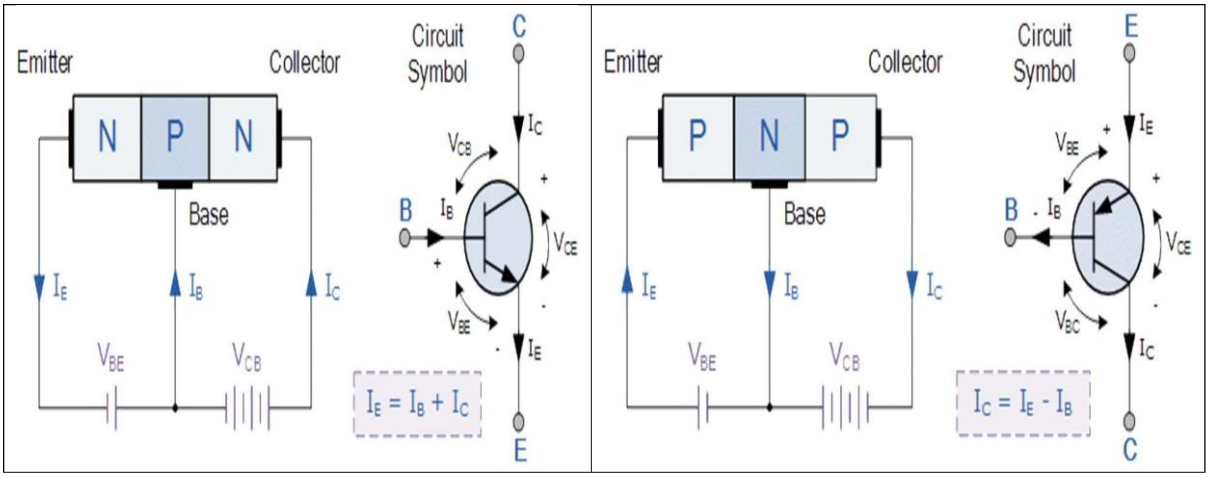
Types of Transistor

NPN	PNP
By sandwiching a thin layer of P-type semiconductor between two N-type semiconductors.	By sandwiching a thin layer of N-type semiconductor between two P-type semiconductors.
 NPN transistor symbol	 PNP transistor symbol
Electrons are majority carriers flow from emitter to base	Holes are majority carriers flow from base to emitter

Biasing:

Emitter – Base should be forward bias

Base – Collector should be reverse bias



Using Kirchoff's law

$$I_E = I_B + I_C \quad (1)$$

α and β parameters:

α and β parameters of a transistors are defined as

$$\alpha = \frac{I_C}{I_E} \text{ and } \beta = \frac{I_C}{I_B} \quad \text{.....(2)}$$

From (1) we get

$$I_E = I_B + I_C$$

Divide both side by I_C We get

$$\frac{I_E}{I_C} = \frac{I_B}{I_C} + \frac{I_C}{I_C}$$

Sub (2) in above equation we get

$$\frac{1}{\alpha} = \frac{1}{\beta} + 1$$

$$\beta = \frac{\alpha}{1-\alpha} \quad \text{and} \quad \alpha = \frac{\beta}{1-\beta}$$

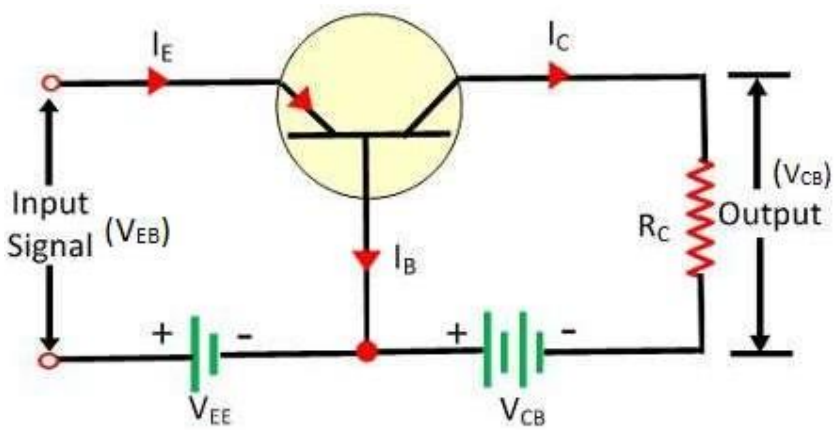
TRANSISTOR CONFIGURATIONS

A transistor can be connected in a circuit in the following three different configurations.

(i) Common base
(ii) Common emitter
(iii) Common collector

COMMON BASE CONFIGURATION

Here base is common to both emitter and collector



Common Base Connection of PNP Transistor

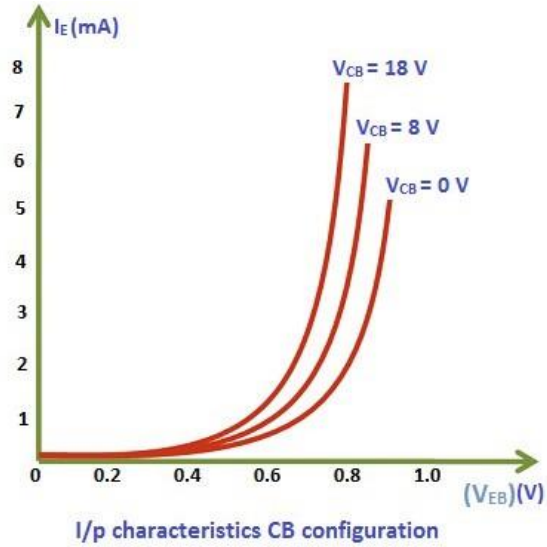
Let I_E be the input current
Let V_{EB} be the input voltage
Let I_C be the output current
Let V_{CB} be the output voltage

With small increase in emitter base voltage V_{EB}, the emitter current I_E increases rapidly due to small input resistance.

- The emitter base junction is forward bias
- Holes are injected by emitter to base
- Hence the input current I_E is created
- The thickness of base region is so small
- Most of the holes cross over into the collector region.
- Hence base current I_B is created.

- Holes in the collector go through the battery V_{CC}

Input Characteristics:



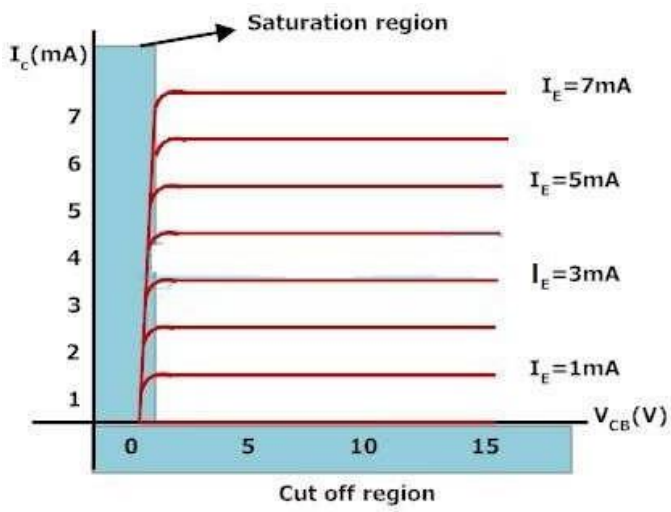
V_{CB} is kept constant.

- The input current I_E is measured by increasing the V_{EB}
- Now V_{CB} is increased by step of 10V and kept constant.
- The same procedure is repeated and graph is plotted between I_E and V_{EB}
- The slope measured from the graph gives us dynamic input resistance of a transistor

$$R_i = \left[\frac{\Delta V_{EB}}{\Delta I_E}\right] \text{ When } V_{CB} = \text{constant.}$$

- Input characteristics of NPN transistor are also similar to PNP transistor.

Output Characteristics:



- I_E is kept constant
- The output current I_C is measured by increasing CB
- Now I_E is increased in the step of 5mA
- The same procedure is repeated and the graph is plotted between I_C and V_{CB}
- The slope is measured from the graph

$$\text{Slope} = \left[\frac{\Delta I_C}{\Delta V_{CB}}\right] I_E = \text{constant}$$

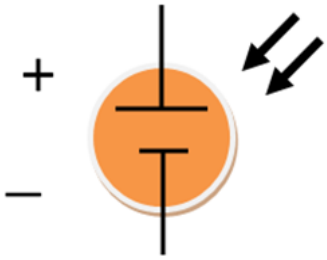
$$R_o = \left[\frac{\Delta V_{CB}}{\Delta I_C}\right] I_E = \text{constant}$$

Where R_o - dynamic output resistance

Explain in detail about Solar cell - construction, working, V-I characteristics and Applications. (16 marks)

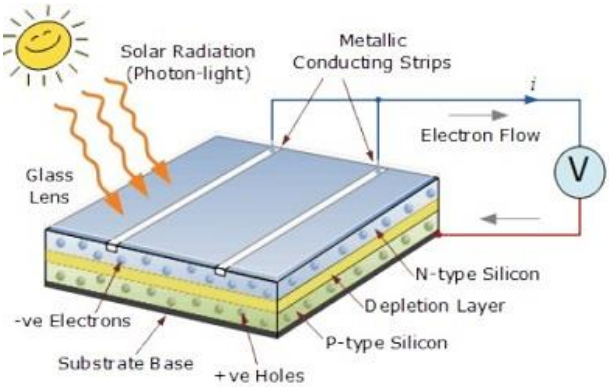
Solar cell is also called as photovoltaic cell and this is a device which converts light energy into electrical energy by using photovoltaic effect. Solar cell is basically a normal PN Junction diode.

Symbol of Solar cell:



Symbol of Solar Cell

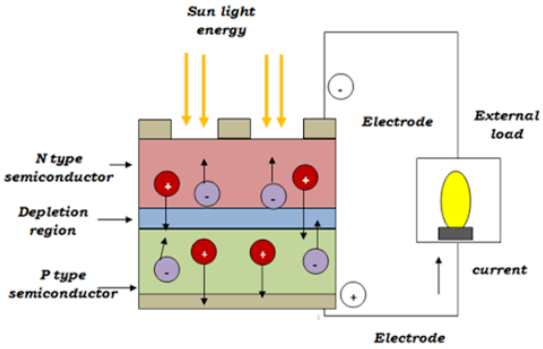
Construction of Solar cell:



Construction of Solar cell

It consists of N type and P type semiconductor material. N type is highly doped and P type is lightly doped. Top and bottom is of conducting electrode to collect the current. The bottom is fully covered with the conductive layer and top layer is not fully covered because the sun rays should not be fully blocked. Since semiconductors are reflective in nature, antireflective coating is used. The whole arrangement is kept inside a thin glass to avoid mechanical shock.

Working of Solar cell:



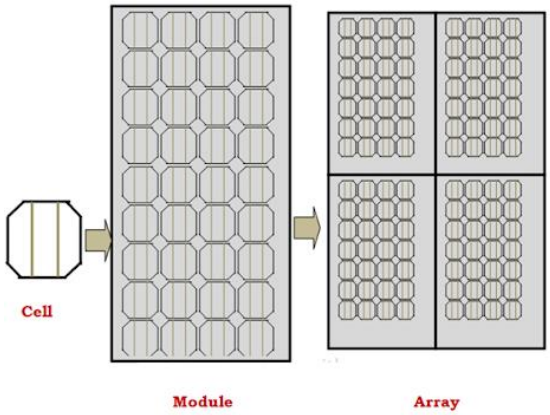
Working of Solar cell

The working of solar cell is based on photovoltaic effect. It is a effect in which current or voltage is generated when exposed to light. Through this effect solar cells convert sunlight into electrical energy.

A depletion layer is formed at the junction of the N type and P type semiconductor material. When light energy of the sun rays falls on the solar panel, the photons which is the small bundle of

energy whose energy is higher than the energy gap gives energy to the electrons and holes in the depletion region. The electrons and holes move to the higher level which is the conduction band. The electrons move towards N type and holes move towards P type and they act as a battery. So this movement of electrons and holes forms the electric current.

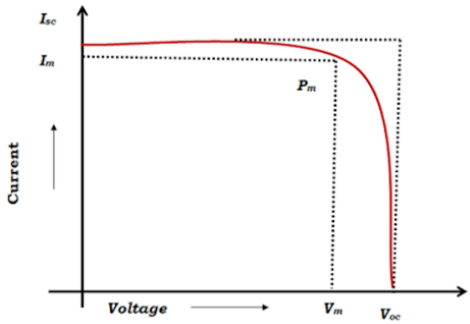
Solar cell to Solar farm:



Solar cell to Solar farm

Solar cell is the basic building module and it is in octagonal shape and in bluish black colour. Each cell produces 0.5 voltage. 36 to 60 solar cells in 9 to 10 rows of solar cells are joined together to form a solar panel. For commercial use upto 72 cells are connected. By increasing the number of cells the wattage and voltage can be increased. The thickness of solar panel is in the range 2.5 to 4cm. Many modules together form the solar array.

V-I characteristics of Solar cell:



V-I characteristics of Solar cell

Isc is the short circuit current and it is measured by short circuiting the terminals. Voc is the open circuit voltage and it is measured when no load is connected. Pm is maximum power, Im is maximum current, Vm is maximum voltage and it occurs at the bend of the characteristic curve.

Advantages of Solar cell:

- It uses renewable energy
- No pollution so it is environment friendly
- It lasts for many years
- No maintenance cost

Disadvantages of Solar cell:

- Energy is not produced during rainy, cloudy days and during night times.
- Cost of installation is high.

Applications of Solar cell:

- It is used in calculators and in wrist watches
- Used in storage batteries
- Street lights
- Portable power supplies
- Satellites

MODULE V - NANO MATERIALS

Write about the types of nanomaterials (2 marks)

Based on dimensions, nano materials are classified in to four types.

1. Nanoparticles
2. Nanoclusters
3. Nanowire
4. Nanorods
5. Nanotubes

Co-ordinates : Examples

1. 0 – dimension : Nanoclusters
2. 1 - dimension : Thin Films, surface coatings
3. 2 - dimension : Nanotubes, nanowires
4. 3 - dimension : Precipitates, colloids

Write a note on Nanoparticles. (2 Marks)

Nanoparticles

Nanoparticles are the particles, the size of which ranges from 1 to 100 nm. These are tiny aggregates of atoms but smaller than their crystals but bigger than molecules. They have three dimensional structures.

Examples : TiO_2, gold, silver, ZnO, etc.,

Applications of Nanoparticles

1. TiO_2 is used in cosmetics as they are very good UV - absorber.
2. Nano silver particles are used as a catalyst in industries.
3. Nanoparticles are used in medicine.

Nano silver particles are used in making bone cement, surgical instruments, etc.,

Write a note on Nanoclusters. (4 Marks)

Nanoclusters

Nanoclusters are fine aggregates of atoms or molecules. The size of which ranges from 0.1 to 10 nm. Of all the nano materials, nanoclusters are the smallest sized nano materials because of their close packing arrangement of atoms.

Examples : CdS, ZnO, etc.,

All the atoms, in nanocluster, are bound by forces like metallic, covalent, ionic, hydrogen bond or Vander Waals forces of attraction. Clusters of certain critical size are more stable than others. Nanoclusters consisting of up to a couple of hundred atoms, but larger aggregates, containing 10° or more atoms, are called nanoparticles.

Magic number

Magic number is the number of atoms present in the clusters of criticle sizes with higher stability.

Different types of nanoclusters can be distinguished from the nature of forces present between atoms. Generally clusters containing transition metal atom have unique chemical, electronic and magnetic properties. These properties vary with the number of constituent atoms, the type of element and the net charge on the cluster.

Properties of nanoclusters

1. Atomic clusters or molecular clusters are formed by the nucleation of atoms or molecules respectively.
2. The reactivity of nanoclusters are decreased due to their decrease in size.
3. The melting point of nanoclusters are lower than the bulk materials due to high surface to volume ratio.
4. The electronic structure of the nanocluster is more confined than the bulk materials.

Applications of nanocluster

1. Nanoclusters are used as catalysts in many reactions.

2. It is used in nano based chemical sensors.
3. It is also used as a light emitting diode in quantum computers.

Write a note on Nanorods. (4 Marks)

Nanorods

Nanorod is two dimensional cylindrical solid material having an aspect ratio i.e., length to width ratio less than 20.

Examples : Zinc oxide, Cadmium sulphide, Gallium nitride nano rods.

Synthesis of nanorods

Nano-rods are produced by direct chemical synthesis. A combination of ligands act as shape control agents and bond to different facets of the nano-rods with different strength.

This allows different nanorods to grow at different rates producing an elongated objects. Many of the above nanorods are not manufactured due to lack of commercial demand.

Properties of nanorods

1. Nanorods are two-dimensional materials.
2. It exhibits optical and electrical properties.

Applications of nanorods

1. Nanorods find application in display technologies.
2. It is also used in the manufacturing of micro mechanical switches.
3. Nanorods are used in an applied electric field, micro electro mechanical systems, etc.,
4. Nanorods along with noble metal nanoparticles function as theragnostic agents.
5. They are used in energy harvesting and light emitting devices.
6. Nanorods have used as cancer therapeutics.

Write a note on Nanowires. (6 Marks)

Nanowires

Nanowire is two dimensional cylindrical solid material having an aspect ratio ie., length to width ratio greater than 20. Diameter of the nanowire ranges from 10 - 100 nm.

Examples : Different types of nanowires

Types of nanowires : Examples

1. Metallic nanowires : Au, Ni, Pt
2. Nanowires of semiconductors : InP, Si, GaN
3. Nanowires of insulators : SiO_2, TiO_2
4. Molecular nanowires : DNA

Synthesis of nanowires

1. Template-assisted synthesis

Template assisted synthesis of nanowires is a simple way to fabricate nanostructures. These templates contain very small cylindrical pores or voids within the host material and the empty spaces are filled with the chosen material to form nanowires.

2. VLS (Vapour - Liquid - Solid) method

It involves the absorption of the source material from the gas phase into a liquid phase of catalyst. Upon supersaturation of the liquid alloy, a nucleation event generates a solid precipitate of the source material. This seed serves as a preferred site for further deposition of material at the interface of the liquid droplet, promoting the elongation of the seed into a nanowire.

Properties of nanowires

1. Nanowires are two-dimensional material.
2. Conductivity of a nanowire is less than that of the corresponding bulk materials.
3. It exhibits distinct optical, chemical, thermal and electrical properties due to this large surface area.
4. Silicon nanowires show strong photoluminescence characteristics.

Uses of nanowires

1. Nanowires are used for enhancing mechanical properties of composites.

2. It is also used to prepare active electronic components such as p-n junction and logic gates.
3. Semiconductor nanowire crossings are expected to play a important role in future of digital computing
4. Nanowires find applications in high-density data storage either as magnetic read heads or as patterned storage media.
5. Nanowires replace conventional copper wires used in computers, televisions.
6. It is also used to link tiny components into very small circuits.

Write a note on Nanotubes. (2 Marks)

Nanotubes

Nanotubes are tube like structures with diameter of 1-100 nm and a length of few nm to microns. Nanotubes consist of tiny cylinders of carbon and other materials like boron nitride. Nanotubes may be organic (or) inorganic.

Examples

1. Carbon nanotube
2. Silicon nanotube
3. DNA nanotube
4. Boron nitride nanotube

Write elaborately about Sol gel method (8 marks)

WET CHEMICAL SYNTHESIS OF NANOPARTICLES

In principle we can classify the wet chemical synthesis of nanomaterials into two broad groups:

1. The top down method: where single crystals are etched in an aqueous solution for producing nanomaterials, For example, the synthesis of porous silicon by electrochemical etching.

2. The bottom up method: consisting of sol-gel method, precipitation etc. where materials containing the desired precursors are mixed in a controlled fashion to form a colloidal solution.

SOL – GEL PROCESS:

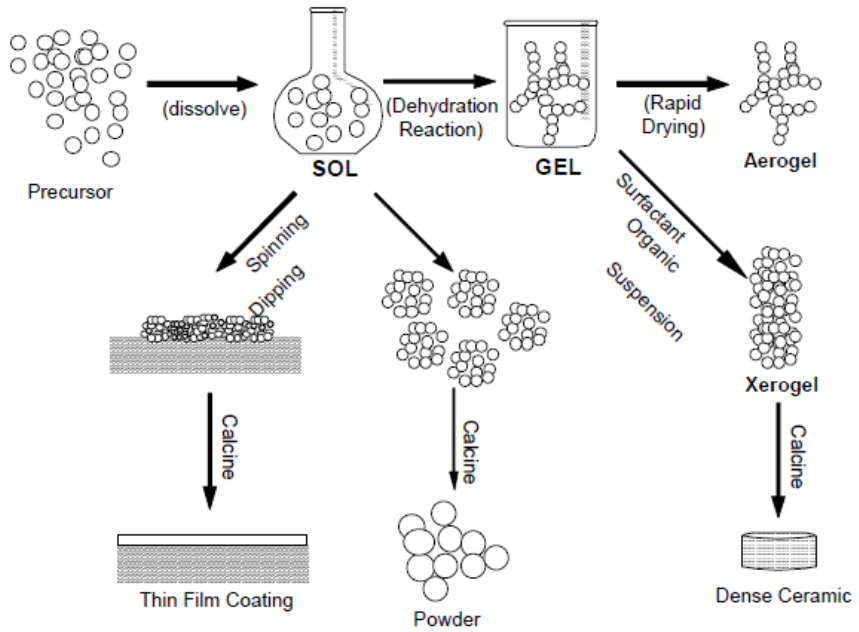
The sol-gel process, involves the evolution of inorganic networks through the formation of a colloidal suspension (sol) and gelation of the sol to form a network in a continuous liquid phase (gel). The precursors for synthesizing these colloids consist usually of a metal or metalloid element surrounded by various reactive ligands. The starting material is processed to form a dispersible oxide and forms a sol in contact with water or dilute acid. Removal of the liquid from the sol yields the gel, and the sol/gel transition controls the particle size and shape. Calcination of the gel produces the oxide.

Sol-gel processing refers to the hydrolysis and condensation of alkoxide-based precursors such as Si(OEt) 4 (tetraethyl orthosilicate, or TEOS). The reactions involved in the sol-gel chemistry based on the hydrolysis and condensation of metal alkoxides M(OR)z can be described as follows:

$$MOR + H_2O \rightarrow MOH + ROH \text{ (hydrolysis)}$$

$$MOH + ROM \rightarrow M\text{-}O\text{-}M + ROH \text{ (condensation)}$$

Sol-gel method of synthesizing nanomaterials is very popular amongst chemists and is widely employed to prepare oxide materials. The sol-gel process can be characterized by a series of distinct steps.



Schematic representation of sol-gel process of synthesis of nanomaterials

1. Formation of different stable solutions of the alkoxide or solvated metal precursor.

2. Gelation resulting from the formation of an oxide- or alcohol- bridged network (the gel) by a polycondensation reaction that results in a dramatic increase in the viscosity of the solution.

3. Aging of the gel (Syneresis), during which the polycondensation reactions continue until the gel transforms into a solid mass, accompanied by contraction of the gel network and expulsion of solvent from gel pores. Ostwald ripening (also referred to as coarsening, is the phenomenon by which smaller particles are consumed by larger particles during the growth process) and phase transformations may occur concurrently with syneresis. The aging process of gels can exceed 7 days and is critical to the prevention of cracks in gels that have been cast.

4. Drying of the gel, when water and other volatile liquids are removed from the gel network. This process is complicated due to fundamental changes in the structure of the gel. The drying process has itself been broken into four distinct steps:

1. The constant rate period,
2. The critical point,
3. The falling rate period,
4. The second falling rate period.

If isolated by thermal evaporation, the resulting monolith is termed a *xerogel*. If the solvent (such as water) is extracted under supercritical or near super critical conditions, the product is an *aerogel*.

5. Dehydration, during which surface bound M-OH groups are removed, there by stabilizing the gel against rehydration. This is normally achieved by calcining the monolith at temperatures up to 800°C.

6. Densification and decomposition of the gels at high temperatures (T>800°C). The pores of the gel network are collapsed, and remaining organic species are volatilized. The typical steps that are involved in sol-gel processing are shown in the schematic diagram below.

The interest in this synthesis method arises due to the possibility of synthesizing nonmetallic inorganic materials like glasses, glass ceramics or ceramic materials at very low temperatures compared to the high temperature process required by melting glass or firing ceramics.

The major difficulties to overcome in developing a successful bottom-up approach is controlling the growth of the particles and then stopping the newly formed particles from agglomerating. Other technical issues are ensuring the reactions are complete so that no unwanted reactant is left on the product and completely removing any growth aids that may have been used in the process. Also production rates of nano powders are very low by this process. The main advantage

is one can get monosized nano particles by any bottom up approach.

Advantages:
The following are the advantages of the sol-gel process:

1. It is economical.
2. It generates highly pure, well controlled ceramics compared to other techniques.
3. It is low temperature process i.e.. it has less energy consumption and low pollution.
4. The instrument processing is easy and synthesis almost all the materials.
5. It is possible to get the unique materials like zerogel, zeolites, ordered porous solids by organic – inorganic hybridization.
6. It is also possible to synthesis nanoparticle, nanorods, nanotubes etc using this technique.

Application:

1. The following are some of the applications of sol-gel process:
2. The nanomaterials produces by this technique are used for electronic thin film devices, for wear, chemical or oxidation protection, as well as for their optical properties (eg: anti reflection).
3. This method is used to produce the lightest materials.
4. Optical, protective and decorative coating can be done by this method.
5. Sol – gel routed ceramic fibers are used for fiber optic sensors and thermal insulation.
6. The ultra fine particle prepared by this technique is used for dental and biomedical applications.
7. Silicate sol is formed by this method.

What is the solvo-thermal method of synthesis? (2 Marks)

It is a method of preparing nanomaterials by carrying out a chemical reaction in a solvent at a temperature higher than its boiling point in a sealed vessel (usually an autoclave). The high temperature and pressure promote the crystallization of nanoparticles.

Why is a sealed autoclave used in the solvo-thermal method? (2 Marks)

The sealed vessel allows the solvent to be heated above its boiling point without evaporating, thus creating high pressure, which promotes better dissolution, nucleation, and controlled growth of nanocrystals.

How does the solvo-thermal method differ from the hydrothermal method? (2 Marks)

Hydrothermal: Uses **water** as the solvent.
Solvo-thermal: Uses **non-aqueous organic solvents** (e.g., ethanol, methanol, ethylene glycol) to allow different reaction environments and better control over particle size and morphology.

State two advantages of the solvo-thermal method. (2 Marks)

1) Produces highly crystalline nanoparticles with controlled size and shape.
2) Can synthesize materials that are not stable in water (hydrothermally unstable).

Name one application of nanomaterials prepared by solvo-thermal method. (2 Marks)

Solvo-thermally synthesized nanoparticles are used in catalysts, photocatalysts, batteries, sensors, and drug delivery systems due to their controlled morphology and surface properties.

How does the choice of solvent affect the product in the solvo-thermal method? (2 Marks)

The solvent affects the dielectric constant, viscosity, and polarity, which control the solubility of precursors, nucleation rate, and growth process — thus determining particle size and morphology.

What is one limitation of the solvo-thermal method? (2 Marks)

Requires high-pressure equipment (autoclave), which can be expensive and may limit large-scale production.

Explain in detail about Solvo-thermal method. (8 marks)

The solvo-thermal method is a versatile and cost-effective technique for synthesizing solid-state materials and nanomaterials in a closed system at temperatures above the solvent's boiling point and under high pressure. This technique utilizes a range of solvents, such as water (hydrothermal synthesis), organic solvents, or even liquid ammonia (ammothermal process), to facilitate chemical reactions and control the size, shape, and structural properties of the resulting materials, including zeolites, Metal-Organic Frameworks (MOFs), and various carbon nanomaterials.

Key Aspects of the Solvo-thermal Method

- Closed System and Elevated Pressure/Temperature: Reactions occur in a closed vessel, such as an autoclave, leading to high pressure from the solvent's vapor. This high-pressure, high-temperature environment allows for the dissolution of precursors and promotes the formation of crystalline materials that might not be achievable under ambient conditions.

Choice of Solvent is Crucial: The solvent is not just a reaction medium but also a key factor influencing the properties of the final product. The choice of solvent affects:

- **Solubility:** Different solvents are chosen to dissolve specific precursors, such as inorganic salts in polar solvents like water or organic molecules in nonpolar solvents.
- **Kinetics:** Solvent viscosity impacts diffusion rates, with higher viscosity slowing diffusion and promoting the formation of smaller particles.
- **Reactivity:** Some solvents can act as reactants or stabilizing agents (ligands) that control crystal growth and stabilize certain facets of the nanoparticles.

Control Over Material Properties: By carefully adjusting processing parameters such as temperature, reaction time, pressure, and solvent selection, researchers can precisely control:

- **Size and Morphology:** The size and shape of the resulting particles.
- **Crystallinity:** The degree of structural perfection and crystallinity.
- **Defects:** The number and type of structural and chemical defects in the material.
- **Functionalization:** The degree of functionalization, which is important for applications in photocatalysis and energy.

Materials Synthesized

The solvo-thermal method is used to prepare a broad range of materials, including:

Zeolites: Porous crystalline aluminosilicates with applications in catalysis and adsorption.

Metal-Organic Frameworks (MOFs): Hybrid porous materials formed from metal ions and organic ligands.

Nanomaterials: A wide variety of nanoparticles, nanorods, and carbon-based nanomaterials like carbon spheres, dots, and nanotubes.

Advantages

Cost-Effective: The method is generally cost-efficient for producing nanomaterials.

Environmentally Friendly: The use of water as a solvent in hydrothermal synthesis provides an environmentally benign approach.

Versatile: It can be used to synthesize a wide variety of materials by selecting appropriate precursors and solvents.

Precise Control: Offers precise control over the size, shape, and crystallinity of the synthesized products.

Subcategories of Solvo-thermal Synthesis

- Hydrothermal Synthesis: A specific type of solvo-thermal synthesis where water is used as the solvent.
- Ammothermal Synthesis: Uses liquid ammonia as the solvent for reactions like the synthesis of nitrides.
- Alcoholothermal Synthesis: Utilizes alcohols as the solvent.

Explain in detail about Chemical vapour deposition (12 marks)

CHEMICAL VAPOUR DEPOSITION:

Chemical vapour deposition involves the reaction and decomposition of vapour phase species at elevated temperatures and subsequent deposition from a vapour by a chemical reaction occurring on a heated substrate surface.

1. Principle:

Precursor gases are delivered into the reaction chamber at elevated temperature usually under vacuum. As they pass over or come into contact with the heated substrate, they react or decompose forming a solid phase. They are deposited onto the substrate. The substrate temperature is critical and can influence what reaction will take place.

2. Different types:

1. Atmospheric Pressure Chemical Vapour Deposition (APCVD)
2. Low Pressure Chemical Vapour Deposition (LPCVD)
3. Metal Organic Chemical Vapour Deposition (MOCVD)
4. Plasma Assisted Chemical Vapour Deposition (PACVD) or Plasma Enhanced Chemical Vapour Deposition (PECVD).
5. Laser Chemical Vapour Deposition (LCVD)
6. Photo Chemical Vapour Deposition (PCVD)

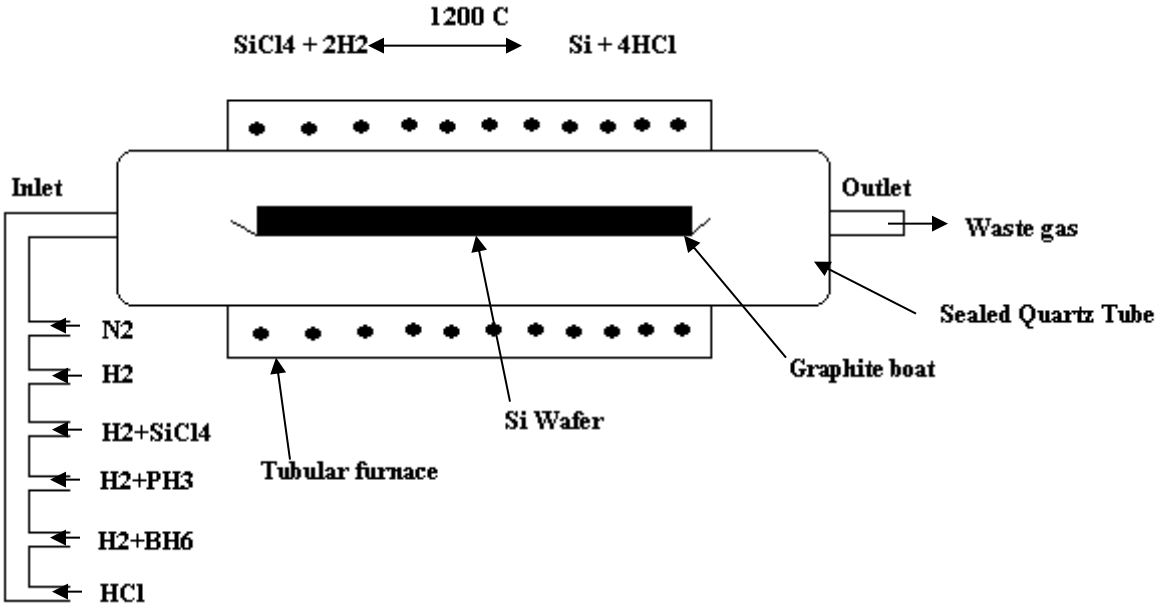
3. Construction and working:

Chemical Vapour Deposition processes are distinguished from PVD by the fact that growth proceeds as a result of series of chemical reactions above the heated substrate. Further CVD process can be considered as near equilibrium growth phenomenon while PVD is clearly very far from an equilibrium process.

The below fig shows the apparatus used in the CVD process for a thin film single crystal growth on the silicon wafer. The basic chemical reaction used to describe the epitaxial growth of pure silicon is hydrogen reduction of silicon tetrachloride.

1. **Gas delivery system:** For the supply of precursors to the reactor chamber.
2. **Reactor chamber:** Chamber within which deposition takes place.

3. **Substrate loading mechanism:** A system for introducing and removing substrates.
4. **Energy source:** Provides the energy / heat that is required to get the precursors to react/decompose.
5. **Vacuum system:** A system for removal of all other gaseous species other than those required for the reaction / deposition.
6. **Exhaust system:** System for removal of volatile by products from the reaction chamber.



Schematic design of a thermal CVD system with a tube furnace

The following steps are involved in the chemical vapour deposition:

1. Vapourisation and transport of precursor molecules into reactor.
2. Diffusion of precursor molecules to the surface of particle.
3. Adsorption of precursor molecules to the surface of particle.
4. Decomposition of precursor molecules on the surface and incorporation into solid films.

4. Advantages:

1. CVD coatings are typically fine grained, high purity and harder than other materials produced using conventional ceramic fabrication processes.
2. CVD coatings are usually only a few microns thick and are generally deposited at fairly slow rates, usually of the order of a few hundred microns per hour.
3. CVD is an extremely versatile process that can be used to process almost any metallic or ceramic compounds.

5. Applications:

1. It is used to produce the nanomaterials like metals and alloys, carbides, nitrites, borides, oxides and inter metallic compounds.
2. CVD has variety of applications across a wide range of industries.
3. Coatings for a variety of applications such as wear resistance, corrosion resistence, high temperature protection, erosion protection and combinations thereof.
4. It can be used to produce the semiconductor and relate devices like integrated circuits, sensors, opto-electronic devices.
5. CVD can be used to produce components that are difficult or uneconomical to produce using conventional fabrication techniques.

6. Optical fibers - for telecommunications.
7. Powder production – production of novel powder and fibers, catalyst, nanomachines.

Explain in detail about Electro deposition (12 marks)

ELECTRO DEPOSITION:

1. Definition:

The process that exploits the creation of solid materials directly from electro chemical reactions in liquid compositions with substrate materials. This is one of the chemical methods. Electrodeposition is also known as electroplating.

2. Principle:

Substrate is placed in electrolyte. When an electrical potential is applied between a conducting area on the substrate and a counter electrode in the liquid, a chemical redox process takes place resulting in the formation of a layer of material on the substrate and usually some gas generation at the counter electrode

3. Theory:

It may be classified as either anodic or cathodic processes. In addition to the simple deposition of metallic thin films, it includes the deposition of metal oxides and chalcogenides.

(i) Anodic Process:

In an anodic process, a metal anode is electrochemically oxidized in the presence of other ions in solution, which then react together and deposit on the anoide

$$Cd \longrightarrow Cd^{2+} + 2e^-$$
$$Cd^{2+} + 2e^- \longrightarrow CdTe$$

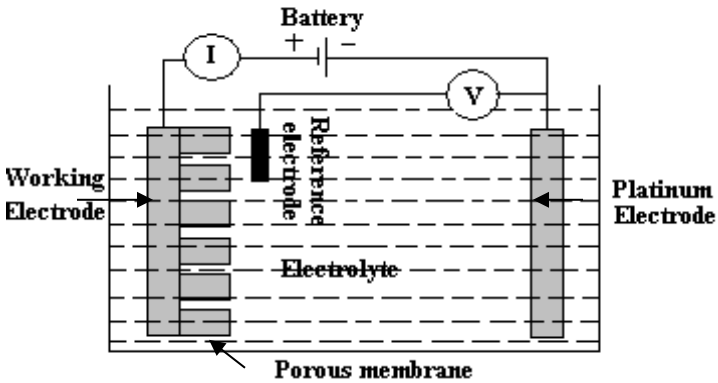
(i) Cathodic Process:

Meanwhile in a cathodic process, components are deposited onto the cathode fron solution precursors. In the case of metals this process is known as electroplating.

$$Cd^{2+} + 2e^- \longrightarrow Cd$$
$$HTeO_2^+ + 4e^- + 3H^+ \longrightarrow Te + 2\,H_2O$$
$$Cd^{2+} + 2e^- \longrightarrow CdTe$$

If a metal is oxidized and then undergoes hydrolysis or the local pH at the electrode is changed by liberating hydrogen gas electrochemically and producing OH^- ions, then metal oxide materials may be deposited.

The potential during the pulse will determine the species deposited while the thickness of individual layers is determined by the charge passed. The final film can range in thickness from a few nanometer to tens of microns and can be deposited onto large specimen areas of complex shape, making the process highly suitable for industrial use.



Experimental Arrangement – Electro Deposition

4. Experimental Setup and Working:

The experimental arrangement of the electro deposition consists of a three electrode electro chemical cell(a reference electrode, a specially designed cathode and an anode or counter electrode), porous membranes and AC/DC power supply as shown in the above fig.

The power supply is used to apply controlled current at a certain voltage. The template can be made of either non metallic or metallic materials. The surface morphology of the deposits depends on the surface structure and chemical composition of the cathode substrate as well as other electro chemical parameters.

The nucleation of nanostructures on the electrode substrate is influenced by (i) the crystal structure of the substrate (ii) specific free surface energy (iii) adhesion energy (iv) Lattice orientation of the electrode surface (v) crystallographic lattice mismatch at the nucleus substrate interface boundary.

The final size distribution of the electro deposits strongly depends on the kinetics of the nucleation and growth.

- ***Instantaneous nucleation:***

  All the nuclei form instantaneously on the electrode substrate and subsequently grow with the time of electro deposition.

- ***Progressive nucleation:***

  The number of nuclei that are formed is a function of time of electro deposition. These nuclei gradually grow and overlap, and therefore the progressive nucleation process exhibits zones of reduced nucleation rate around the growth stable nuclei.

  The formation of nano structures results from growth of the nuclei that invariably nucleate at the holes and defects of the electrode substrate. Subsequent growth of these nuclei at the template yields the desired surface morphology of the nano structures, which can therefore be synthesized by choosing the appropriate surface of the electrode.

Example: nanoporous polymers, porous alumina etc

5. Advantages:

1. Electro deposition is relatively cheap.
2. It can be performed at low temperatures which will minimize inter diffusion in the preparation of multilayered thin film.
3. The film thickness can be controlled by monitoring the amount of charge delivered and the deposition rate can be followed by the variation of the current with time.
4. The composition and defect chemistry can be controlled by the magnitude of the applied potential, which can be used to deposit non-equilibrium phases.
5. Plusing or cycling the applied current or potential in a solution containing a mixture of precursors allows the production of a multi layered material.
6. It is an important technique for synthesizing metallic nano materials with controlled shape and size.
7. It is used to prepare nanometer size particles, fibers, wires, rods and tubes.

6. Disadvantages:

1. The electro deposition process obviously requires electrically conducting substrate materials.

2. If a growth is epitaxial, then any strain due to lattice mismatch between the nanocrystal and the substrate can be growth limiting.
3. Difficulties in the preparation of desired templates.
4. Additional high temperature annealing steps are expensive and un suitable for polymer substrates.

7. Applications:

1. This technique is used to produce the nano intended holes, meso-porous silica, nano porous polymers, porous alumina.
2. This is a versatile technique to synthesize various kinds of nano materials with desired surface morphologies (nano rods, nano particles and nanowires).

Explain in detail about Applications of nanomaterials (12 marks)

APPLICATIONS OF NANOMATERIALS:

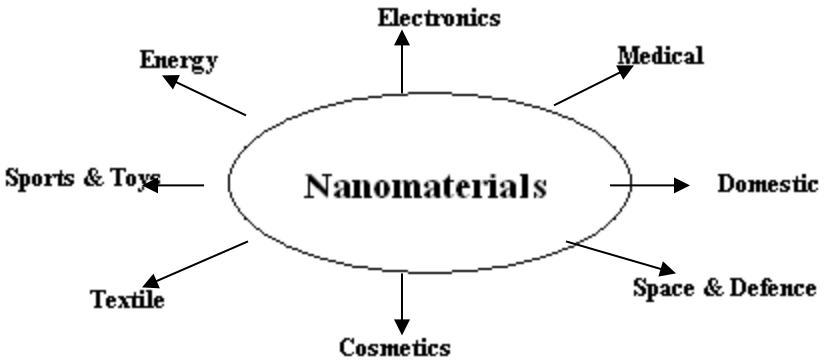
Nanomaterials are used in a variety of fields and devices such as electrical devices, memory devices, fuel cells, defense field, aero-space industry, etc.

1. Electrical and electronic devices:

The nanoparticles are for fabricating nano transistors, multilayer capacitors, quantum computing and display technologies, noise filters and stabilizers. The advantages of these devices are low power loss and smaller in size.

Example: $BaTiO_3$, ZnO etc.

The above materials are used for current controlling devices such as transistors.



Applications of nanomaterials

2. Memory devices:

i. These are used to produce very tiny permanent magnets with high energy product value. The tiny magnets find applications in high density magnetic recording.
ii. Quantum well, quantum dots and quantum wires having quantum confinement are produced from semiconductor nanomaterials which are acting as computer storage materials with high density.
iii. The magnetic devices made of Cu, Fe alloys are used in RAM, Read/Write heads and sensors.

3. Defense industry:

i. The nano particles are used in fuel cells and infrared windows which are effective in defence applications. Eg: Nano ceramic Powders

4. Additives and Pastes:

i. Due to reduction in processing time, cost and higher productivity, the nano phase materials such as $BaTiO_3$, ZnO etc are used for additives and pastes which are used for screen printing.

5. Optical devices:

i. These materials are used for fabrication of semiconductor laser which is smaller in size. These lasers have high efficiency compared to lasers made by conventional materials.

6. Cosmetics and Body care Products:

i. The nano ceramic powders such as CaO, TiO_2, Al_2O_3, SnO_2 etc are used for cosmetic and body care products.

7. Communications:

i. Miniaturized microchip patch antenna which have large bandwidth are manufactured using nano phase materials. The weight of this antenna is very much reduced compared to conventional ones.

8. Energy:

Nanotechnology will play an important role in the field of energy.

i. Hydrogen is a good source of fuel for auto mobile and other transportation purposes. Carbon nanotubes are used as hydrogen storage materials.
ii. Some metal hydride nanoparticles like Ni Hydrides and ultra light weight materials like aerogels are found to be better options than the conventional materials in improved batteries.

9. Automobiles:

i. Very powerful motors using nanoparticles like Ni-Ti alloys perform better and are less power hungry than other motors.
ii. By using nanoparticles clay, better, lightweight and less rubber consuming thinner tyres are possible.

10. Sports and Toys:

i. Good quality tennis rackets are made of carbon.
ii. Nanomaterials based motors are used by toy industry to make them very smooth and soft.

11. Textile:

i. Special threads and dyes used in textile industry are the nanomaterials product. These clothes do not require ironing and frequent cleaning.
ii. Some companies used silver nanoparticles in washing machine which will make clothes germ free.

12. It is used in information storage, refrigeration, computer and solar cells.

13. BIOMEMS are used in biofluidic chips for chemical and biochemical analysis in medical diagnostics and implantable pharmaceutical drug delivery.
14. Nanotube based bio sensors are used for cancer diagnostics.
15. Nanomaterials are used as molecular switches, nanotube flat panel displays, fast logic gates and nanoscopic lasers.
16. Zeolite, a special nanomaterial is used by petroleum industry as catalyst.

Explain about Properties of Nanomaterials.(8 marks)

PROPERTIES OF NANOMATERIALS:

Nanomaterials have the structural features in between of those of atoms and the bulk materials. While most micro structured materials have similar properties to the corresponding bulk materials, the properties of materials with nanometer dimensions are significantly different from those of atoms and bulks materials. This is mainly due to the nanometer size of the materials which render them: (i) large fraction of surface atoms; (ii) high surface energy; (iii) spatial confinement; (iv) reduced imperfections, which do not exist in the corresponding bulk materials.

Due to their small dimensions, nanomaterials have extremely large surface area to volume ratio, which makes a large to be the surface or interfacial atoms, resulting in more "surface" dependent material properties. Especially when the sizes of nanomaterials are comparable to length, the entire material will be affected by the surface properties of nanomaterials. This in turn may enhance or modify the properties of the bulk materials.

OPTICAL PROPERTIES:

One of the most fascinating and useful aspects of nanomaterials is their optical properties. Spectral shift of optical absorption and fluorescence properties, increased quantum efficiency of semiconducting crystal are the optical properties of nanomaterials. Applications based on optical properties of nanomaterials include optical detector, laser, sensor, imaging, phosphor, display, solar cell, photocatalysis, photoelectrochemistry and biomedicine.

ELECTRICAL PROPERTIES:

Electrical Properties of Nanoparticles discuss about fundamentals of electrical conductivity in nanotubes and nanorods, carbon nanotubes, photoconductivity of nanorods, electrical conductivity of nanocomposites.

Resistivity in nanosized materials is in general larger than that in polycrystalline materials. The electrons get scattered at the grain boundaries resulting into increase of resistance. Therefore, resistance of nanosized materials is generally quite large.

One interesting method which can be used to demonstrate the steps in conductance is the mechanical thinning of a nanowire and measurement of the electrical current at a constant applied voltage. The important point here is that, with decreasing diameter of the wire, the number of electron wave modes contributing to the electrical conductivity is becoming increasingly smaller by well-defined quantized steps.

MECHANICAL PROPERTIES:

"Mechanical Properties of Nanoparticles" deals with bulk metallic and ceramic materials, influence of porosity, influence of grain size, superplasticity, filled polymer composites, particle-filled polymers, polymer-based nanocomposites filled with platelets, carbon nanotube-based composites. When the grain size is decreased, the strength of the material is increased. This is because the movement of dislocation is impeded by grain boundary. Hence, nano phase materials with their exceptionally small grain size are found to be highly strong. Similarly the melting temperature of the material is reduced when the grain size is decreased.

MAGNETIC PROPERTIES:

Bulk gold and Pt are non-magnetic, but at the nano size they are magnetic. Surface atoms are not only different to bulk atoms, but they can also be modified by interaction with other chemical species, that is, by capping the nanoparticles. This phenomenon opens the possibility to modify the physical properties of the nanoparticles by capping them with appropriate molecules.

Explain in detail about carbon nanotubes (8 marks)

CARBON NANOTUBES [CNTS]

Introduction

The discovery of carbon leads to form stable, ordered structures other than graphite and diamond, which have stimulated the researchers worldwide to search for other new forms of carbon. The search was given new impetus when it was shown that C_{60} could be produced in a simple arc-evaporation apparatus. This C_{60} is very much useful in the production of carbon nano tubes, so called CNT.

The CNT have remarkable electronic properties and many other unique characteristics. For these reasons they have attracted huge academic and industrial interest.

Definition

Carbon nanotubes (CNT) are molecular-scale tubes of graphitic carbon with outstanding properties. They are among the stiffest and strongest fibres researched till date, with remarkable electronic properties and applications.

Types

Carbon nao-tubes are of two types. viz

(i) Single walled nano-tubes (SWNTs)

(ii) Multi walled nano-tubes (MWNTs)

(iii) Single walled nano-horns (SWNHS).

Types of carbon nanotubes

Three types of nanotube structures are considered by rolling a graphite sheet with different orientations about the axis.

They are

(i) Armchair structure

(ii) Zig-zag structure

(iii) Chiral structure

Armchair structure

When the axis of the tube parallel to C-C bonds of the carbon hexagons, the structure is obtained. It is referred as "armchair" structure.

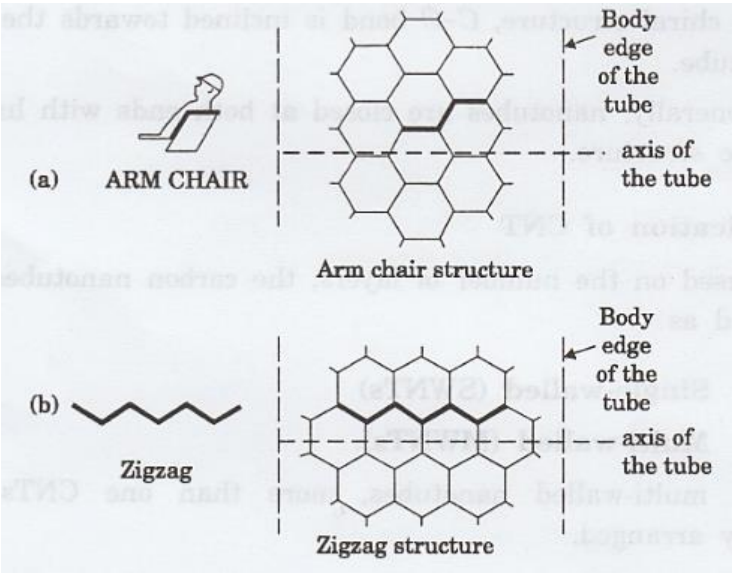
Zig-zag and Chiral structure

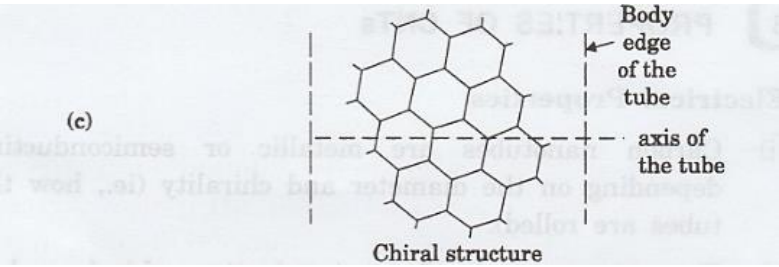
The tubes sketched in the diagram referred as zig-zag and chiral structure. They are formed by rolling graphene sheet such that the axis of the tube is not parallel to C-C bonds.

Zig-zag structure consists of tube axis perpendicular to C-C bonds.

In chiral structure, C-C bond is inclined towards the axis of the tube.

Generally, nanotubes are closed at both ends with half of fullerene structure.





Properties of carbon nanotubes

(i) CNTs have High Electrical Conductivity.
(ii) CNTs have Very High Tensile Strength.
(iii) CNTs are Highly Flexible (can be bent considerably without damage).
(iv) CNTs are Very Elastic (18% elongation to failure).
(v) CNTs have High Thermal Conductivity.
(vi) CNTs have a Low Thermal Expansion Coefficient.
(vii) CNTs are Good Electron Field Emitters.
(viii) CNTs Aspect Ratio.

Applications of carbon nanotubes

(i) The carbon nanotubes are very light in weight, but they are very strong, hence they are used in aerospace.
(ii) They are used in constructing nanoscale electronic devices.
(iii) Carbon nano-tubes (CNTs) are used in battery electrodes, fuel cells, reinforcing fibers etc.
(iv) CNTs are used in the development of flat panel displays for computer monitors and televisions.
(v) Plastic Composite CNTs are used as a light weight shielding materials for protecting electromagnetic radiation.
(vi) Light weight CNTs are also used in military and communication systems, for protecting computers and electronic devices.
(vii) Semiconducting CNTs are used as switching devices.
(viii) Semiconducting CNTs are also used as chemical sensors to detect various gases.
(ix) Nano-tubes can also serve as catalysts for some chemical reactions.
(x) The unique properties of carbon nanotubes will undoubtedly lead to many more applications in future to produce nano-computers, plastic composites etc.

Explain in detail about Electrospinning method. (8 marks)

Electrospinning nanoparticles involves creating a charged solution containing polymer and nanoparticles and then using a strong electrostatic field to draw the liquid into a thin jet, which solidifies into nanofibers as the solvent evaporates. The applied voltage stretches the jet and causes it to whip, with the nanoparticles getting incorporated into the resulting nanofiber structure. This process yields highly porous fibrous materials with high surface areas, which are then collected on a grounded target. The key advantage is the ability to produce nanoscale fibers containing uniformly distributed nanoparticles for applications like drug delivery systems, tissue engineering, and filtration membranes.

The Process of Electrospinning Nanoparticles

1. **Setup:** A solution containing polymer and nanoparticles is placed in a syringe connected to a high-voltage power supply.
2. **Taylor Cone Formation:** A high voltage is applied to the solution at the syringe tip, causing the liquid to form a Taylor cone.
3. **Jet Ejection:** When the repulsive electrical forces overcome the liquid's surface tension, a charged jet is ejected from the apex of the cone.
4. **Jet Thinning and Whipping:** The electric field stretches and thins the jet, while bending instability causes it to whip, elongating the jet and promoting solvent evaporation.

5. **Nanofiber Formation:** As the solvent evaporates or the melt cools, the jet solidifies to form solid or hollow nanofibers. The nanoparticles are simultaneously trapped and incorporated within the polymer matrix.
6. **Fiber Deposition:** The solidified nanofibers are deposited onto a grounded collector, forming a nanofibrous web or scaffold.

Key Factors Influencing the Process

Polymer Properties: Solution viscosity, conductivity, molecular weight, and surface tension all impact the electrospinning process and the resulting fiber morphology.

Process Parameters: Applied voltage, flow rate, distance between the spinneret and collector, and ambient conditions (humidity, temperature) significantly affect fiber diameter and structure.

Nanoparticle Characteristics: The type, size, and concentration of the nanoparticles, as well as their dispersion in the solution, influence how well they are incorporated into the fibers.

Solvent Properties: The choice of solvent impacts the polymer's solubility, solution viscosity, and volatility, which are critical for successful fiber formation.

Advantages of Electrospinning Nanoparticles

High Surface Area & Porosity: The resulting nanofibers have a large specific surface area and highly porous structures.

Tunable Properties: Fiber dimensions, mechanical strength, and morphology can be controlled by adjusting processing parameters and materials.

Versatility: A wide range of materials, including polymers, ceramics, and inorganic compounds, can be used.

Cost-Effective: The technique is relatively simple and affordable.

Nanoparticle Integration: It provides a means to uniformly disperse and embed nanoparticles within a fibrous structure.

Applications

Biomedical Engineering: Used for creating tissue scaffolds for wound healing and drug delivery systems.

Filtration: For producing high-performance filters.

Sensors: Exploited for their high surface area and enhanced sensing capabilities.

Aerospace and Textiles: For developing lightweight, super-strong materials.

List out and explain the applications of nanoparticles (8 marks)

Nanoparticles possess unique electronic properties—such as quantum confinement effects, which alter conductivity, and surface plasmon resonance, affecting optical interactions—leading to applications in advanced electronics like high-efficiency solar cells, flexible circuits, quantum dot displays, and improved data storage technologies. These properties arise from their extremely small size and large surface-area-to-volume ratio, which allows for novel behaviors not seen in bulk materials.

Key Properties of Nanoparticles for Electronics

Quantum Confinement Effects: Reducing a material's size to the nanoscale can restrict the movement of electrons, leading to a change in its electronic properties. This confinement can transform a material's conductivity, making it behave as a semiconductor or insulator where its bulk form was once a conductor.

Surface Plasmon Resonance (LSPR): Noble metal nanoparticles exhibit a strong absorption and scattering of light due to the collective oscillation of their conduction electrons when struck by a photon. This is a size and shape-dependent phenomenon, allowing for precise control of light absorption and emission.

Large Surface Area to Volume Ratio: This fundamental characteristic increases the interaction of nanomaterials with their environment, a crucial factor for developing sensitive sensors and catalysts.

Enhanced Electrical Conductivity: While nanoscale confinement can sometimes reduce

conductivity, certain nanomaterials like carbon nanotubes and graphene possess exceptionally high electrical conductivity, surpassing that of their bulk counterparts, making them ideal for advanced electronics.

Applications in Electronics

Solar Cells: Nanowires and nanoparticles can enhance the light absorption and energy collection efficiency of photovoltaic devices, improving solar power generation.

LEDs and OLEDs: Nanomaterials are used to advance lighting technologies, with quantum dots playing a role in lasers and other light-emitting diodes.

Flexible Electronics: Conductive inks, made with nanoparticles of materials like silver, copper, or graphene, can be printed onto flexible substrates to create circuits for flexible electronic devices.

Advanced Displays: Quantum dots are employed in advanced displays to achieve vibrant and precise light emission and absorption.

Data Storage: Nanotechnology enables the creation of new hard disks, utilizing effects like giant magnetoresistance to enhance data storage capacity and efficiency.

Sensors: The high sensitivity of nanomaterials to their environment makes them suitable for nano-biosensors, chemical sensors, and gas sensors that offer ultra-sensitive, real-time detection capabilities.

Energy Storage: Nanomaterials are being developed for batteries and fuel cells, leading to quicker charging times, higher power density, and improved efficiency.